

MTH6139 Time Series

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Chapter 0

Introduction to time series

This module is an introduction to Time Series Analysis

0.1 What is time series analysis?

Time series analysis is the study of Time Series.

0.2 What is a time series?

A time series is a collection of data sampled through time.

- Examples 0.1.**
1. Price of a stock throughout time. Here we would note by $x_1, x_2, \dots$ the price of this stock taken at times $t_1, t_2, \dots$
 2. Other financial assets such as foreign exchange rates like EUR/USD can also be considered time series.
 3. Temperatures taken at a certain location would give rise to a time series. This might be sampled continuously and give rise to x_t for t in a certain interval of time, $[0, T]$; or sampled discretely $x_1, x_2, \dots$ at certain times $t_1, t_2, \dots$
 4. Fish stocks in a certain area
 5. ...

So typically these are continuous numerical quantities (i.e. real numbers) which exhibit some form of randomness.

There are more sophisticated examples outside of what we will deal with in this module that you might wish to explore as an aside:

- Examples 0.2.**
1. Text. This can be seen as a sequence of words. You could even try to forecast the next word in a sentence. This is what modern AI generative technologies do.
 2. Music. This can be seen as a two time series, one for notes and a sequence of note durations. Again you might try to forecast the next note, and based on these ideas generate some form of AI-music based on a few initial notes.
 3. Sound consists of a continuously varying air pressure measurements which are, on a small time scale, periodic. So that, for example A below middle C on a piano makes air vibrate in this fashion 220 times per second (220Hz). Audio file formats will convert this continuous signal into a discrete set of values, typically sampled 14.4k times a second. All of these constitute time series. And Time Series Analysis can be used to understand and process these signals.
 4. Other collections of data that are sampled in a 'ordered' fashion, like height on a road, can be analysed as time series by replacing time by the relevant quantity, the km on a road in this example.

0.3 How do we analyse Time Series?

We will use a mix of maths and technology, more precisely:

- Probability
- Statistics. In fact Time Series Analysis is considered a branch of statistics.
- Technology

0.3.1 What is the difference between Probability and Statistics?

Simple answer:

- If you see upper case letters, X , Y , ... then you are likely to be dealing with Probability.
- If you see lower case letters, x , y , ... then it is probably Statistics.

Better answer:

- Probability deals with random variables, probability density functions, calculation of probabilities, expectations etc. This is often called **modelling**.

- In statistics you:
 1. Get some real world data (or sometimes, when learning, invented data).
 2. Consider what random variable or probability model best describes this data. This often uses unknown parameters like μ and σ for normal random variables, λ for Poisson or Exponential, p for Binomial/Bernoulli, β_0 and β_1 for linear regression, ...
 3. We then estimate these parameters using statistical techniques like maximum likelihood etc.
 4. Describe how good this parameter fit is using techniques like confidence intervals, p-values, R^2 , hypothesis testing, ...

0.3.2 What technology will we use?

Statistics uses R, we will use R. This module assumes you have some exposure to R.

Other systems are also used in industry:

- Excel. More basic but **universally** used in industry.
- Python. General purpose language. Also has statistical functionality.
- Specialised statistical systems like: SAS, SPSS, ... Have a look at these if you are considering statistics as a professional line of business.

0.3.3 Why should I learn this?

First and foremost, you do not study for the money, you study because you are interested in a field. The main reason is:

- Because it is very interesting.

Other reasons:

- Industry uses time series:
 - Finance deals with stock prices, foreign exchange, inflation rates, etc.
 - Insurance
 - Retail and online retail. Big retail companies study how purchases are made in order to plan stocks, optimize profit, etc.

- Medical sciences. For example, the NHS will study the time series measuring incidence of a particular illness (e.g. flu) to plan resources, vaccines etc.
- Science. Astronomers will measure intensity of sun flares, ecologists the evolving population of a given species like fish stocks, etc.
- Companies like *numerate* candidates.
- Companies like candidates with exposure to technology like R.

0.4 How can we find interesting datasets?

If you are interested in the topic (I hope you are) you could invest some time exploring datasets. We will provide numerous throughout the module. You can also explore:

- [kaggle.com](https://www.kaggle.com). Not all the data are time series but many will be.
- In R, many libraries give you access to time series datasets.
 - Many books have their own libraries with datasets they use. The book *Time Series Analysis and its Applications* by Robert H. Shumway and David S. Stoffer is very interesting. You do not need to buy it as it might be available online or in the QMUL library. We will not use it in the module, but might be interesting. The datasets used in this book are made available in the library **astsa** (which stands for Applied Statistics of Time Series Analysis). If you have never used this you might wish to install it in R as follows

```
# The following line installs the library astsa.
# You only need to do this once
install.packages("astsa")
# You can then 'activate' the library using
library(astsa)
```

Run the following examples to plot some of the data series offered by **astsa**

- * Johnson and Johnson's quarterly earnings per share.

```
library(astsa)

#plotting time series data
#first look at Johnson and Johnson quarterly earnings per share
jj #Johnson and Johnson dataset

#plot time series
#type is plotting symbol, here is o
#ylab is label on the y axis
plot(jj, type="o", ylab="Quarterly Earnings per Share")
```

* Global temperature changes

```
plot(xglobtemp, type="o", ylab="Global Temperature Deviations")
```

* Speech and sound waves

```
#Speech recording of the syllable aaa...hhh  
plot(speech)
```

* El Niño and Fish Population

```
par(mfrow = c(2,1)) #to draw 2 time series in the same plot  
plot(soi, ylab="", xlab="", main="Southern Oscillation Index")  
plot(rec, ylab="", xlab="", main="Estimated new fish")
```

– The library `tseries` allows you to access financial time series from Yahoo Finance. The following code will display Open/Close/High/Low share prices for Apple stock.

```
library(tseries)  
apple = get.hist.quote("AAPL") # this downloads data for the stock  
with ticker AAPL  
plot(apple)
```

You are not expected to memorise the code above, but as you regularly use R you are expected to become familiar with many of these instructions.

0.5 What type of Time Series will we consider

In this introductory course we will consider:

1. Continuous numerical data, i.e. our $x_1, x_2, \dots$ will be in $\mathbb{R}$.
2. Discrete times, i.e. data is sampled on some set times $t_1, t_2, \dots$. This is in contrast with continuously sampled data like temperatures recorded every millisecond.
3. Equispaced times. I.e. our data is sampled at times $t_1, t_2, t_3, \dots$ where $t_2 - t_1 = t_3 - t_2 = \dots$. We even often assume $t_1 = 1, t_2 = 2, t_3 = 3, \dots$ for simplicity.
4. These time series are not simply a deterministic function like $x_t = \cos(1 + 3e^{2t-1})$ but rather there is some *randomness* that we wish to understand and measure.

Example 0.1. A time series as above could be the temperature in a given place sampled every day at 9am.

Often, real world data exhibits quirks and traits that do not fit this scheme. But what we will learn is used to gain the first order understanding of data. Data could exhibit:

- Even in the simple case above, what do we do when time changes twice a year from Greenwich Mean Time (GTM) to British Summer Time (BST) (on the last Sunday in March) and back (on the last Sunday in October)? Then the time interval might not be 24h.

If we are dealing with an international operation these matters can get really messy. For instance in New York, Eastern Standard Time (EST) changes to Eastern Daylight Time (EDT) on the second Sunday in March, and back on the first Sunday in November. This means that London time and New York time are 5 hours apart except for the days between the last Sunday in October and the first Sunday in November, and the days between the second Sunday in March and the last Sunday in March when it will be only 4 hours apart.

- The data might be continuous, such as the temperature. The data analyst will then need to decide when to sample it in order to make it a discrete time series.
- Sometimes you might have missing data (a thermometer broke down a couple of days) or erroneous data (a squirrel nested on the thermometer). The data analyst will need to decide what to do here.

We will not concern ourselves with these complications in this introductory course, but you should be aware of them.

Chapter 1

Additive model, estimating trend

1.1 Aims of TSA

Before getting started let us summarise the aims of Time Series Analysis as this will help frame how the course evolves. When considering a concrete Time Series we often want to:

1. **Understand.** This means have a general *feel* for what the time series does, the units of measurement, any quirk or outlier. This is often supported by charts which in fancy language is called *Visual Analytics*.
2. **Explain.** This refers to using maths to explain why the time series evolves like it does. For example identifying seasonal components, a trend, ...
3. **Predict.** This is the holy grail of TSA. You can often use TSA to come up with an estimate of future values. But this must always be accompanied with confidence intervals and often results might be not conclusive. Do not think that TSA allows you to find the next lottery ticket.
4. **Control.** In many processes, like autopilot systems, or management of a nuclear reaction, you need to decide how to modify your inputs in order to correct deviations in your process. This is called control.

1.2 What are the general features of a TS?

When we look at our first time series we will identify features such as the following:

- **Trend.** This refers to an underlying non-random pattern of evolution. For example a stock might be seen as some error noisiness on top of an exponential growth trend (something like

$x_t = a \exp(bt)$ for some a and b). Or population of a country might seem to be following a linear increase trend (something like $x_t = a + bt$ for some a and b).

- **Seasonality.** This refers to an underlying non-random periodic pattern with a clearly understood period. For example, the temperature taken every minute in the month of August will be seen to go up and down with a 24h period corresponding to the time of the day.
- **Cyclic patterns.** This refers to periodic behaviour with less clearly understood periodicity. For example the boom/bust cycles in the economy are said to occur with period around 56 months on average. We will not look into this in this module but you should be aware of it.
- **Residual noise.** This is what is left of our time series once we have removed all the factors above. It is similar to the residuals in linear regression. However, unlike in regression, in time series we often have some form of correlation between these errors that can greatly aid in understanding the time series.

Additionally we might detect more complex features that require attention, by far the most prevalent one is *heteroscedasticity*

- A time series is called *heteroscedastic* if its variance changes through time. For example it might become more and more variable over time. Many statistical models (like linear regression) do not cope well with *heteroscedastic* data. In these cases we will try to change the data to make it *homoscedastic* which means the variance does not change through time.

1.3 The additive model

The model we will consider for our time series in this module is:

$$X_t = m_t + S_t + Y_t, \quad t = 1, \dots, n. \quad (1.1)$$

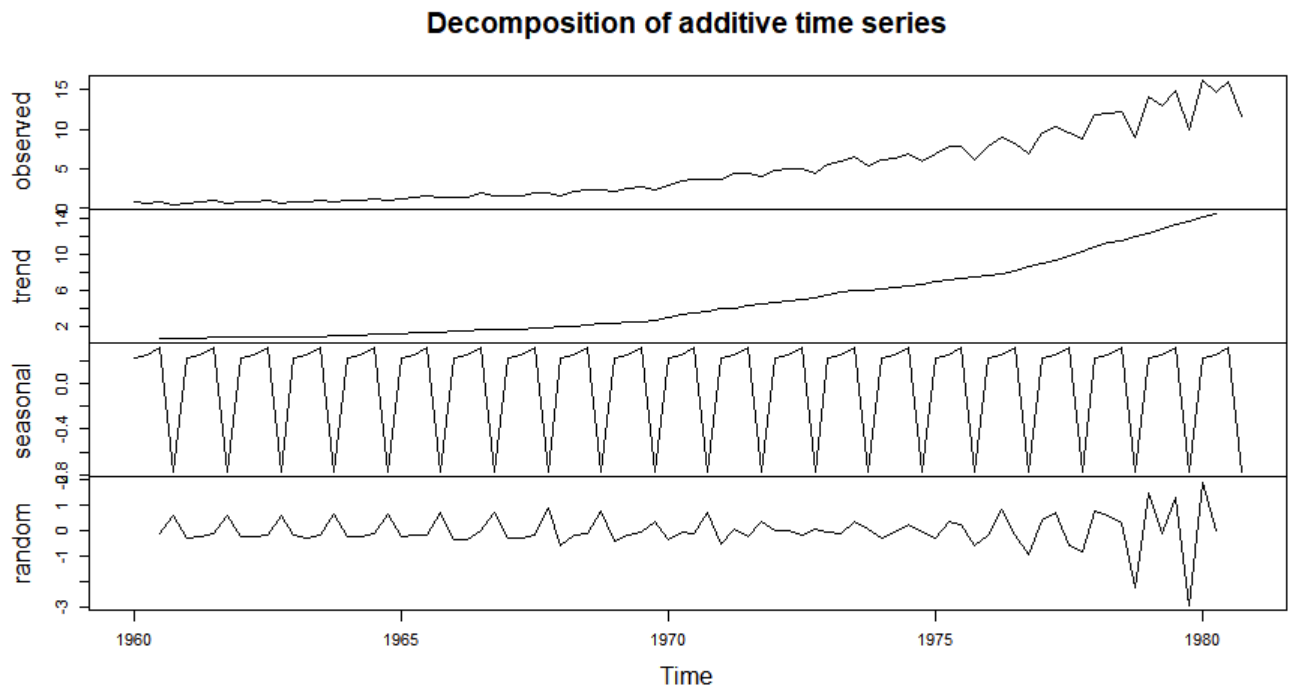
where

- X_t is our original time series.
- m_t is the trend.
- S_t is the seasonal component.
- Y_t is the residual error.

The crux of the matter is to find these m_t , S_t and Y_t given X_t .

A preview in R: The following code draws a chart of these three components for the data `jj` which contains share prices for the company Johnson & Johnson. It uses the R function `decompose` as follows:

```
library(astsa) # this contains the dataset jj
# the following decomposes the time series jj and plots the three resulting
# components as above.
plot(decompose(jj))
```



Try it!.

Note that in the charts above, both S_t and Y_t are centred around zero. This can always be accomplished by changing m_t if necessary. So for example if the average value of the seasonal component is 3.4 then we would redefine $m_t^{new} = m_t + 3.4$ and $S_t^{new} = S_t - 3.4$ and this would yield a new decomposition with seasonal component centred around 0. Same for Y_t .

1.4 Multiplicative Models

We often assume that our time series follow the additive model above because:

1. they are easier to interpret
2. easier to work with mathematically and computationally.

That is why a lot of statistical models are additive.

However, time series where increments tend to be proportional (e.g. like stocks, where the relevant increment is the percentage return rather than the absolute monetary value) are typically better modelled multiplicatively as follows:

$$X_t = m_t S_t Y_t, \quad t = 1, \dots, n. \quad (1.2)$$

Note that this can be converted to an additive model by taking logarithms on both sides:

$$\log X_t = \log m_t + \log S_t + \log Y_t.$$

We would then use the additive model above defining $X_t^{new} = \log(X_t)$, $m_t^{new} = \log(m_t)$, $S_t^{new} = \log(S_t)$, and $Y_t^{new} = \log(Y_t)$ and writing

$$X_t^{new} = m_t^{new} + S_t^{new} + Y_t^{new}, \quad t = 1, \dots, n. \quad (1.3)$$

This is the reason why you will often see logarithm taken of monetary values like house prices, financial assets, etc.

1.5 Trend functions

Let us get started by trying to find the trend function m_t associated to a time series X_t .

For simplicity, here we will consider an additive model with no seasonality:

$$X_t = m_t + Y_t, \quad t = 1, \dots, n, \quad (1.4)$$

As stated above, we assume that the noise, Y_t is centred around zero, $\mathbb{E}(Y_t) = 0$. As otherwise we could redefine m_t to absorb a non-zero $\mathbb{E}(Y_t)$ (see paragraph above, after the chart where this phenomena can be visualised).

In order to estimate what m_t might be, we have two broad models:

- Parametric models. Here we assume m_t has a particular functional form like $m_t = \beta_0 + \beta_1 t$, or $m_t = \beta_0 + \beta_1 t + \dots + \beta_k t^k$, or $m_t = \beta_0 \exp(\beta_1 t)$ or some other function depending on some parameters that will be estimated.
- Non-parametric models are models where we do not need to make assumptions as above. There will be no parameters to be estimated. But still choices will need to be made.

1.5.1 Parametric methods: linear regression

The most basic model for the trend is to assume it is linear, i.e. $m_t = \beta_0 + \beta_1 t$. Let us fit this trend model to our existing data

$$(t_1, x_1), (t_2, x_2), \dots, (t_n, x_n)$$

to do so we pick an arbitrary line $m_t = \beta_0 + \beta_1 t$ and measure how far it is from our data points. This can be made in several ways, but the most standard is by summing the squares of the residual errors:

$$S(\beta_0, \beta_1) = \sum_{i=1}^n (x_i - (\beta_0 + \beta_1 t_i))^2$$

To find the minimum of this function we compute the derivatives and equate to zero:

$$\begin{aligned} \sum_{i=1}^n 2(x_i - (\beta_0 + \beta_1 t_i))(-1) &= 0 \\ \sum_{i=1}^n 2(x_i - (\beta_0 + \beta_1 t_i))(-t_i) &= 0 \end{aligned}$$

We can divide by -2 and add up the terms separately, if we also place the terms with β 's on the RHS we obtain the following 2×2 linear system in the 2 unknowns β_0 and β_1 :

$$\begin{aligned} \sum_{i=1}^n x_i &= \beta_0 n + \beta_1 \sum_{i=1}^n t_i \\ \sum_{i=1}^n t_i x_i &= \beta_0 \sum_{i=1}^n t_i + \beta_1 \sum_{i=1}^n t_i^2 \end{aligned} \tag{1.5}$$

if we now divide the first equation by n and use the standard notation, $\bar{x} = \sum x_i / n$ and $\bar{t} = \sum t_i / n$:

$$\begin{aligned} \bar{x} &= \beta_0 + \beta_1 \bar{t} \\ \sum_{i=1}^n t_i x_i &= \beta_0 n \bar{t} + \beta_1 \sum_{i=1}^n t_i^2 \end{aligned}$$

Next multiply the first equation by $n\bar{t}$ and subtract from the second one:

$$\sum_{i=1}^n t_i x_i - n\bar{t}\bar{x} = \beta_1 \left(\frac{1}{n} \sum_{i=1}^n t_i^2 - n\bar{t}^2 \right)$$

from which we conclude:

$$\beta_1 = \frac{\sum_{i=1}^n t_i x_i - n\bar{t}\bar{x}}{\sum_{i=1}^n t_i^2 - n\bar{t}^2}$$

This is fine, but if desired it can be written in a slightly different fashion. This uses similar maths to that which proves $Cov(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$ and $Var(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2$. Using this we see that an alternative expression¹ for β_1 is

$$\beta_1 = \frac{\sum_{i=1}^n (t_i - \bar{t})(x_i - \bar{x})}{\sum_{i=1}^n (t_i - \bar{t})^2}$$

From equation (1.5) we derive an expression for β_0 :

$$\beta_0 = \bar{x} - \beta_1 \bar{t}$$

Remarks 1.1. 1. Note that the regression line, $x = \beta_1 t + \beta_0$, always passes through the mean value of the x 's and the mean value of the t 's, as the last equation states

$$\bar{x} = \beta_1 \bar{t} + \beta_0$$

2. As these coefficients have been estimated from a sample, it is customary to adorn them with a 'hat' symbol: $\hat{\beta}_0$ and $\hat{\beta}_1$ to emphasize that they are **estimates** of some unknown *population* or *real* unknown parameters β_0 and β_1 .
3. One difference between linear regression and fitting a time series with a trend function as above is that in traditional linear regression we assume that the residual errors $x_t - m_t$ are independent.

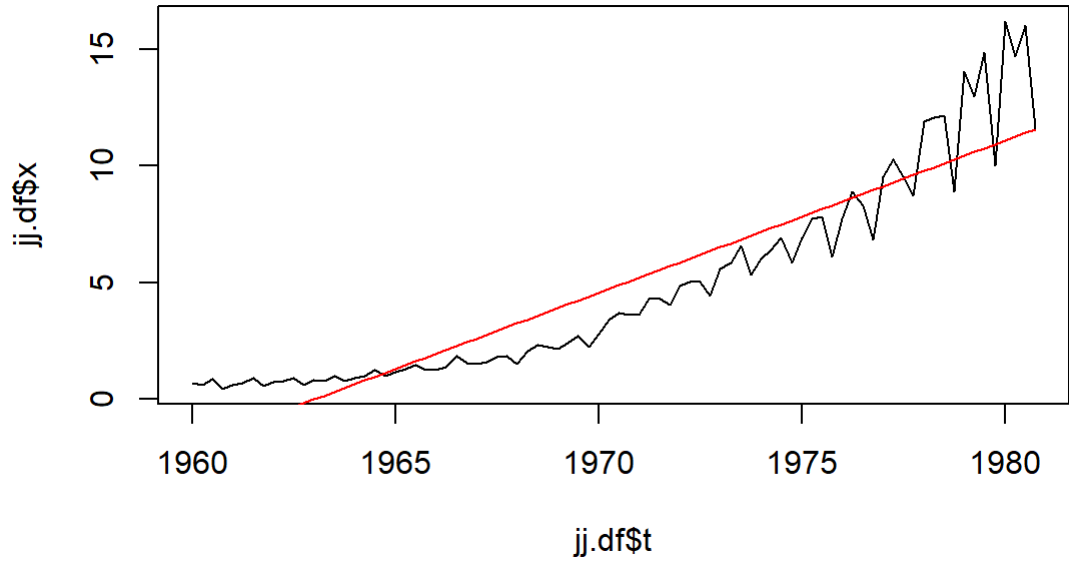
However, in time series analysis, very frequently the error terms $x_t - m_t$ can exhibit interesting correlations that can aid forecasting. For example, it might be that if x_t overshoots m_t for a particular t , then there is a higher chance that x_{t+1} also overshoots m_{t+1} .

4. To fit a linear trend function to a time series we use the `lm` function. See the tutorials for some basic examples. The following example is interesting but a tad more complicated as we need to convert a `ts` object to a dataframe²

```
library(astsa)
# the next line converts the jj time series object to a dataframe
# recall jj contains Johnson & Johnson share prices
jj.df = data.frame(t=as.vector(time(jj)), x=as.vector(jj))
# check first few rows and the structure of the dataframe
head(jj.df)
str(jj.df)
# plot the time series
plot(jj.df$t, jj.df$x, type='l')
# compute linear regression and plot
model = lm(x~t, jj.df)
lines(jj.df$t, fitted(model), type='l', col='red')
```

¹For the denominator: $\sum (t_i - \bar{t})^2 = \sum (t_i^2 - 2t_i\bar{t} + \bar{t}^2) = \sum t_i^2 - 2\bar{t} \sum t_i + n\bar{t}^2 = \sum t_i^2 - 2\bar{t} - 2n\bar{t} + n\bar{t}^2 = \sum t_i^2 - n\bar{t}^2$. For the numerator the derivation is very similar: try it!

²You are not expected to memorise this. You are expected to become familiar with functionality that we use in several examples.



Try it!.

1.5.2 Parametric methods: polynomial regression I

If, after plotting our time series

$$(t_1, x_1), (t_2, x_2), \dots, (t_n, x_n)$$

we see that the general shape does not follow a straight line (as happens in the chart above) but perhaps something resembling a quadratic, or cubic polynomial shape, we might use polynomial regression instead of linear regression.

In this case our trend function will look like

$$m_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \dots + \beta_k t^k.$$

We find the values of $k+1$ unknown parameters $\beta_0, \beta_1, \beta_2, \dots, \beta_k$ by demanding that this polynomial is as close as possible to our data-points. As usual, we will minimise the function

$$S(\beta_0, \beta_1, \dots, \beta_k) = \sum_{i=1}^n (x_i - (\beta_0 + \beta_1 t_i + \beta_2 t_i^2 + \dots + \beta_k t_i^k))^2$$

In other fields, like machine learning, this function is often called the **loss function**. In these other fields often one uses different loss functions. We will stick to the one above called, **sum of squares** loss function.

In order to minimise the sum of squares we need to differentiate with respect to the variables $\beta_0, \beta_1, \beta_2, \dots, \beta_k$ and equate these derivatives to zero. This is easy:

$$\begin{aligned} 2 \sum_{i=1}^n (x_i - (\beta_0 + \beta_1 t_i + \beta_2 t_i^2 + \dots + \beta_k t_i^k)) (-1) &= 0 \\ 2 \sum_{i=1}^n (x_i - (\beta_0 + \beta_1 t_i + \beta_2 t_i^2 + \dots + \beta_k t_i^k)) (-t_i) &= 0 \\ &\vdots \\ 2 \sum_{i=1}^n (x_i - (\beta_0 + \beta_1 t_i + \beta_2 t_i^2 + \dots + \beta_k t_i^k)) (-t_i^k) &= 0 \end{aligned}$$

Solving this seems a bit complicated but it is doable. However, we will explore some interesting alternative techniques described as *matrix calculus*. On the positive side note that the system above is $k+1 \times k+1$ linear system in the $k+1$ unknowns $\beta_0, \beta_1, \beta_2, \dots, \beta_k$ so it seems reasonable. Even though polynomial regression might have sounded like we would have to deal with some crazy non-linear equations, at the end of the day we just need to solve a $k+1 \times k+1$ linear system.

The first thing to try to do is write the loss function in a more compact form using vector multiplications. If you think of it:

$$S(\beta_0, \beta_1, \dots, \beta_k) = \sum_{i=1}^n (x_i - (\beta_0 + \beta_1 t_i + \beta_2 t_i^2 + \dots + \beta_k t_i^k))^2$$

is just the square of the norm of the residual errors vector

$$\begin{pmatrix} x_1 - (\beta_0 + \beta_1 t_1 + \beta_2 t_1^2 + \dots + \beta_k t_1^k) \\ x_2 - (\beta_0 + \beta_1 t_2 + \beta_2 t_2^2 + \dots + \beta_k t_2^k) \\ \vdots \\ x_n - (\beta_0 + \beta_1 t_n + \beta_2 t_n^2 + \dots + \beta_k t_n^k) \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} - \begin{pmatrix} \beta_0 + \beta_1 t_1 + \beta_2 t_1^2 + \dots + \beta_k t_1^k \\ \beta_0 + \beta_1 t_2 + \beta_2 t_2^2 + \dots + \beta_k t_2^k \\ \vdots \\ \beta_0 + \beta_1 t_n + \beta_2 t_n^2 + \dots + \beta_k t_n^k \end{pmatrix}$$

Just to make sure we are tracking everything: there are n known data points, $(t_1, x_1), \dots, (t_n, x_n)$, and $k+1$ unknown coefficients $\beta_0, \dots, \beta_k$. The vector above has size n . Recall vectors are *vertical*³.

To save us writing, let us use the notation

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}, \quad \text{and} \quad W = \begin{pmatrix} 1 & t_1 & t_1^2 & t_1^3 & \dots & t_1^k \\ 1 & t_2 & t_2^2 & t_2^3 & \dots & t_2^k \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & t_n & t_n^2 & t_n^3 & \dots & t_n^k \end{pmatrix}$$

Then the vector above is just $X - W\beta$ and the loss function is its squared norm⁴:

$$S(\beta) = (X - W\beta)^T (X - W\beta)$$

³horizontal vectors will be called *co-vectors*.

⁴We are using the fact that if v is a vector (vectors are vertical), then its square norm is the matrix product $v^T v$. For example, in dimension 2, if $v = (v_1, v_2)^T$, then $v^T v = (v_1, v_2)(v_1, v_2)^T = v_1^2 + v_2^2$.

To minimise this function we will need to learn how to deal with matrix calculus. This is used extensively in machine learning and other fields.

1.5.3 Matrix calculus

In the discussion below, for sake of simplicity we use concrete number of variables and dimensions. For example, vectors of size 3, and 2 variables, β_0 and β_1 . It is trivial to modify the arguments to extend everything to arbitrary dimensions and number of variables.

Derivative of a vector

Let us consider a vector

$$f = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

where a , b and c are all functions of β_0 and β_1 .

Then all the possible derivatives of these functions with respect to β_0 and β_1 are arranged in a matrix called the Jacobian and noted $J(f)$ or $\partial f / \partial \beta$:

$$J(f) = \frac{\partial f}{\partial \beta} = \begin{pmatrix} \frac{\partial a}{\partial \beta_0} & \frac{\partial a}{\partial \beta_1} \\ \frac{\partial b}{\partial \beta_0} & \frac{\partial b}{\partial \beta_1} \\ \frac{\partial c}{\partial \beta_0} & \frac{\partial c}{\partial \beta_1} \end{pmatrix}$$

Case of a vector with one component

If the vector above only has one component,

$$f = a$$

Then this is just a single function of β_0 and β_1 .

The Jacobian then only has one row and we call it the *gradient* and also note it by ∇f .

$$\nabla f = \frac{\partial f}{\partial \beta} = \left(\frac{\partial a}{\partial \beta_0}, \frac{\partial a}{\partial \beta_1} \right)$$

Note that this is a horizontal vector, i.e. a *co-vector*.

Derivative of a linear function

In Calculus I you have seen that if α is a constant number, then $(\alpha f)' = \alpha f'$. With vectors we can consider multiplying a vector by a constant matrix. Let

$$f = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

where a , b and c are all functions of β_0 and β_1 . And

$$A = \begin{pmatrix} \alpha_{11} & \alpha_{21} & \alpha_{31} \\ \alpha_{12} & \alpha_{22} & \alpha_{32} \end{pmatrix}$$

where α_{11} , α_{21} , α_{31} , α_{12} , α_{22} , and α_{32} are all constants (or quantities that do not depend on β_0 and β_1).

We want to calculate the Jacobian of

$$Af = \begin{pmatrix} \alpha_{11} & \alpha_{21} & \alpha_{31} \\ \alpha_{12} & \alpha_{22} & \alpha_{32} \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} \alpha_{11}a + \alpha_{21}b + \alpha_{31}c \\ \alpha_{12}a + \alpha_{22}b + \alpha_{32}c \end{pmatrix}$$

By applying the definition:

$$\begin{aligned} J(Af) &= \begin{pmatrix} \alpha_{11} \frac{\partial a}{\partial \beta_0} + \alpha_{21} \frac{\partial b}{\partial \beta_0} + \alpha_{31} \frac{\partial c}{\partial \beta_0} & \alpha_{11} \frac{\partial a}{\partial \beta_1} + \alpha_{21} \frac{\partial b}{\partial \beta_1} + \alpha_{31} \frac{\partial c}{\partial \beta_1} \\ \alpha_{12} \frac{\partial a}{\partial \beta_0} + \alpha_{22} \frac{\partial b}{\partial \beta_0} + \alpha_{32} \frac{\partial c}{\partial \beta_0} & \alpha_{12} \frac{\partial a}{\partial \beta_1} + \alpha_{22} \frac{\partial b}{\partial \beta_1} + \alpha_{32} \frac{\partial c}{\partial \beta_1} \end{pmatrix} = \begin{pmatrix} \alpha_{11} & \alpha_{21} & \alpha_{31} \\ \alpha_{12} & \alpha_{22} & \alpha_{32} \end{pmatrix} \begin{pmatrix} \frac{\partial a}{\partial \beta_0} & \frac{\partial a}{\partial \beta_1} \\ \frac{\partial b}{\partial \beta_0} & \frac{\partial b}{\partial \beta_1} \\ \frac{\partial c}{\partial \beta_0} & \frac{\partial c}{\partial \beta_1} \end{pmatrix} \\ &= AJ(f) \end{aligned}$$

This is the generalisation of $(\alpha f)' = \alpha f'$. It can similarly be seen that $J(fA) = J(f)A$.

Derivative of a quadratic function

In Calculus I you have seen that $(f^2)' = 2f'f$ and also that $(fg)' = f'g + fg'$. We will next see how this generalises to matrices. The functions f and g will be replaced by vectors of the same dimension, and the product will be the scalar product.

Recall the scalar product is defined in high school as $(a, b) \cdot (c, d) = ac + bd$. After learning some linear algebra you prefer to write all your vectors vertically (horizontal vectors are called co-vectors) and so the scalar product is just a matrix multiplication:

$$(a, b)(c, d)^T = (a, b) \begin{pmatrix} c \\ d \end{pmatrix} = ac + bd.$$

Now assume all these variables a , b , c , and d are functions of β_0 and β_1 and call f the scalar product

$$f = (a, b)(c, d)^T = (a, b) \begin{pmatrix} c \\ d \end{pmatrix} = ac + bd$$

All of these things are functions of β_0 and β_1 . How can we calculate the derivative of f , i.e. the gradient $\partial f / \partial \beta$?

There will be a formula similar to the formula for the derivative of a product from Calculus I; let us discover it.

We start calculating the derivative by hand:

$$\frac{\partial f}{\partial \beta} = \left(\frac{\partial(ac + bd)}{\partial \beta_0}, \frac{\partial(ac + bd)}{\partial \beta_1} \right) = \left(\frac{\partial a}{\partial \beta_0}c + a \frac{\partial c}{\partial \beta_0} + \frac{\partial b}{\partial \beta_0}d + b \frac{\partial d}{\partial \beta_0}, \frac{\partial a}{\partial \beta_1}c + a \frac{\partial c}{\partial \beta_1} + \frac{\partial b}{\partial \beta_1}d + b \frac{\partial d}{\partial \beta_1} \right)$$

This is a bit of a pain to write and will get out of hand in higher dimensions. Can we write it more compactly?

We can split it in the sum of two bits: one with factors of c and d , and the other with a and b . For the first bit

$$\left(\frac{\partial a}{\partial \beta_0}c + \frac{\partial b}{\partial \beta_0}d, \frac{\partial a}{\partial \beta_1}c + \frac{\partial b}{\partial \beta_1}d \right) = (c, d) \begin{pmatrix} \frac{\partial a}{\partial \beta_0} & \frac{\partial a}{\partial \beta_1} \\ \frac{\partial b}{\partial \beta_0} & \frac{\partial b}{\partial \beta_1} \end{pmatrix}$$

and the second bit, the ones with a and b as factors, we get:

$$\left(a \frac{\partial c}{\partial \beta_0} + b \frac{\partial d}{\partial \beta_0}, a \frac{\partial c}{\partial \beta_1} + b \frac{\partial d}{\partial \beta_1} \right) = (a, b) \begin{pmatrix} \frac{\partial c}{\partial \beta_0} & \frac{\partial c}{\partial \beta_1} \\ \frac{\partial d}{\partial \beta_0} & \frac{\partial d}{\partial \beta_1} \end{pmatrix}$$

If we denote $u = (a, b)^T$ and $v = (c, d)^T$ then we get that the gradient of the scalar product $u^T v$ is

$$\frac{\partial}{\partial \beta} (u^T v) = u^T \frac{\partial v}{\partial \beta} + v^T \frac{\partial u}{\partial \beta}$$

In the case of a square norm, $u = v$, we have

$$\frac{\partial}{\partial \beta} \|u\|^2 = 2u^T \frac{\partial u}{\partial \beta}$$

Remarks 1.2. 1. These are very interesting formulæ, they generalise to vectors the well known formulæ $(fg)' = f'g + fg'$ and $(f^2)' = 2ff'$.

2. We have performed the calculations above in the case the vectors u and v have dimension 2, and when these depend on two variables β_0 and β_1 . Obviously all these calculations also work for vectors of any dimensions, and any number of variables.
3. Just to keep track of things: if u is a vector of dimension n depending on m variables. Then the gradient above will be a co-vector (horizontal vector) with m components. and $\partial u / \partial \beta$ will be a matrix with m columns and n rows. It makes sense to multiply a co-vector of size n by such a matrix, and the result will be a co-vector of size m . Check this.

1.5.4 Parametric methods: polynomial regression II

The matrix calculus techniques developed above allow us to compactly perform the derivative of the loss function

$$S(\beta) = \|X - W\beta\|^2$$

We get:

$$\frac{\partial S}{\partial \beta} = 2(X - W\beta)^T(-W)$$

If we equate this gradient to zero, and transpose the resulting matrix, we obtain the so called *normal equations*:

$$W^T X = W^T W \beta$$

The matrix W^T is called the Vandermonde matrix and it can be seen that it has rank $k + 1$ if $t_1, \dots, t_n$ are all different and $k + 1 \leq n$. This implies that the square $(k + 1) \times (k + 1)$ matrix $W^T W$ also has rank $k + 1$ and is therefore invertible. So the solution is:

$$\beta = (W^T W)^{-1} W^T X$$

This resolves the polynomial regression problem.

Remarks 1.3. 1. Polynomial **interpolation** refers to the case in which the number of unknown parameters is equal to the number of points. For example, linear regression in which you have only two data points and where you have to find two parameters (fitting a line to two points), or fitting a quadratic function to 3 points, ...

Interpolation problems are quite different from **regression** in that you can fit to the data exactly and so the loss function actually reaches zero. In this module we will typically have many more data points than parameters and so will not consider interpolation. But interpolation is very important in many other fields.

In the case of interpolation the matrix W above is actually invertible and the normal equation simplifies to $\beta = W^{-1}X$.

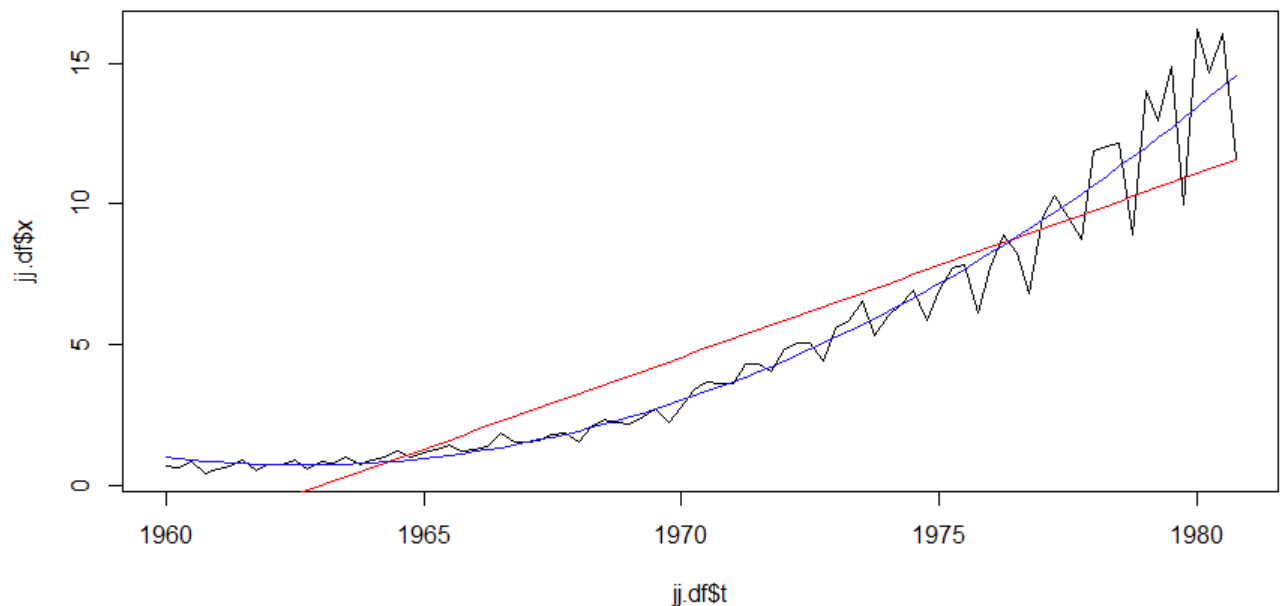
2. It is generally not advised to use models with too many parameters as you can encounter the phenomenon of *overfitting*. This means that your model fits the given data very well, but does not perform with new data.
3. It is generally advised to only use complex models if there is some rational reason to consider the world follows that model. So if you use cubic regression, then ideally there should be some logical reason why that is to be used. Otherwise you risk over-fitting the data.
4. To fit a polynomial trend function to a time series in R we use the `lm` function but you now need to add powers of the explanatory variables. If `x` and `t` are R vectors of the same size describing a data series, then you can find the degree 3 polynomial regression model by running `lm(x~t+I(t^2)+I(t^3))`. See the tutorials for some basic examples.

The following example is interesting but a tad more complicated as we need to convert a `ts` object to a dataframe

```

library(astsa)
# the next line converts the jj time series object to a dataframe
# recall jj contains Johnson & Johnson share prices
jj.df = data.frame(t=as.vector(time(jj)), x=as.vector(jj))
# check first few rows and the structure of the dataframe
head(jj.df)
str(jj.df)
# plot the time series
plot(jj.df$t, jj.df$x, type='l')
# compute linear regression and plot
model = lm(x~t, jj.df)
lines(jj.df$t, fitted(model), type='l', col='red')
# compute cubic regression and plot
model2 = lm(x~t+I(t^2)+I(t^3), jj.df)
lines(jj.df$t, fitted(model2), type='l', col='blue')

```



Try it!.

Other parametric families

Other forms of m_t are also used in practice. For example, the **exponential trend** model with $m_t = \beta_0 e^{\beta_1 t}$ is used to model exponential growth in variables that grow on a percentage basis.

On the other hand, the **generalized logistic trend** or Richards' growth model⁵

$$m_t = \frac{\theta_1}{[1 + \xi \exp \{-\theta_2(t - \theta_3)\}]^{-1/\xi}},$$

⁵See https://en.wikipedia.org/wiki/Generalised_logistic_function

This curve is appropriate to model populations or quantities that cannot grow indefinitely (this is described as *saturated growth*) as it results in an S-shaped curve called **sigmoid** function.

Remarks 1.4. 1. To find the suitable values of the parameters above (β_0 and β_1 in the first case, and θ_1 , θ_2 , θ_3 , and ξ in the second) we have to define a *loss function* that measures the error and find its minimum and then find its minimum.

Unfortunately, in these two cases the resulting minimisation problems do not result in equations that can be resolved analytically. To find the optimal values we must use numerical approximations. As an aside, note that most real-world mathematical problems cannot be solved analytically and need numerical approximation. The set of techniques to find the best numerical approximations is a very interesting field called *Numerical Analysis* or *Numerical Methods*.

2. Both exponential trend and Richard's growth curve were used in modelling [COVID-19](#) infection trajectories (cumulative cases across time).
3. The loss function may be different than the one we considered for linear and polynomial regression (sum of squared residuals). Another loss functions often considered is likelihood (this is maximized rather than minimised).

Chapter 2

Non-parametric Trend Methods I

A downside of parametric models is that they are very restrictive as we impose a specific form for the trend function. Non-parametric models, in some sense are more *data driven* in that the shape of the trend function, m_t , depends more on the data.

Even though we use the word non-parametric, these methods still use some form of parameters. The expression ‘*non-parametric*’ just refers to the fact that we do not use functional forms with unknown parameters such as $m_t = \beta_0 + \beta_1 t + \cdots \beta_k t^k$, $m_t = \beta_0 \exp(\beta_1 t)$, or the like.

2.1 Moving Average Trend estimates

The general idea of the decomposition $X_t = m_t + Y_t$, is that m_t should capture the *smooth* moves of X_t whereas Y_t will be the residual erratic move around the trend.

A simple idea to reduce the variability in a set of numbers is averaging¹.

According to this we could take an average of values of X_t , for instance

$$m_t = \frac{X_{t-1} + X_t + X_{t+1}}{3}$$

and this would be less variable, more like a trend.

In statistics, whenever we use sampled data to make an estimation, we often distinguish between the ‘*real*’ number and that obtained from our particular sample by adorning the sampled estimate with a hat symbol. Accordingly we note the moving average above as $\hat{m}_t$

¹This is explained in basic statistics as: if the variance of a random variable X is σ^2 , then the variance of the average of n iid samples $(X_1 + \cdots + X_n)/n$ is σ^2/n . In business this is called diversification of risk: ‘*do not place all your eggs in one basket*’

Remarks 2.1. 1. Note that, if our time series is finite

$$x_0, x_1, \dots, x_n$$

then $\hat{m}_0$ is not defined, because the formula above would read $\hat{m}_0 = (\textcolor{red}{x}_{-1} + x_0 + x_1)/3$ which is not defined, as x_{-1} does not exist. And likewise with $\hat{m}_n = (x_{n-1} + x_n + \textcolor{red}{x}_{n+1})/3$ is not defined either.

2. Below we extend this to averages of more terms. Note however that we need to pick an odd number of terms, $2q + 1$. If we were to average an even number of terms, for example two, we would end up with expressions like $(x_0 + x_1)/2$ which would be placed in time 0.5, halfway between the time at which x_0 and x_1 happen.

Definition 2.1. Let

$$x_0, x_1, \dots, x_n$$

be a time series and q a positive integer, then we define the $2q + 1$ -point moving average as the time series

$$m_t = \frac{x_{t-q} + \dots + x_t + \dots + x_{t+q}}{2q + 1}$$

Remarks 2.2. 1. If you get mixed up with the indices: m_t is the average of the following $2q + 1$ numbers:

- q samples before t : $x_{t-q}, \dots, x_{t-1}$.
- x_t
- q samples after t : $x_{t+1}, \dots, x_{t+q}$.

2. As in the case of the 3-point moving average, the first q terms ($m_0, \dots, m_{q-1}$) and the last q terms ($m_{n-(q-1)}, \dots, m_n$) are not defined.

Example 2.1. Let us consider the time series

$$(x_0, x_1, \dots, x_9) = (5, 3, 2, 4, 5, 6, 7, 8, 3, 9)$$

and let us compute its 3-point moving average:

$$\begin{aligned} \hat{m}_0 &\text{ is not defined} \\ \hat{m}_1 &= \frac{1}{3}(5 + 3 + 2) = \frac{10}{3} \\ \hat{m}_2 &= \frac{1}{3}(3 + 2 + 4) = \frac{9}{3} \\ &\vdots \\ \hat{m}_8 &= \frac{1}{3}(8 + 3 + 9) = \frac{20}{3} \\ \hat{m}_9 &\text{ is not defined} \end{aligned}$$

Remarks 2.3. 1. R can calculate moving averages using the function `filter`. Its first argument is the R vector containing the time series, and the second argument the coefficients involved in the averaging operation, for a 3-point average these are: $1/3$, $1/3$, and $1/3$.

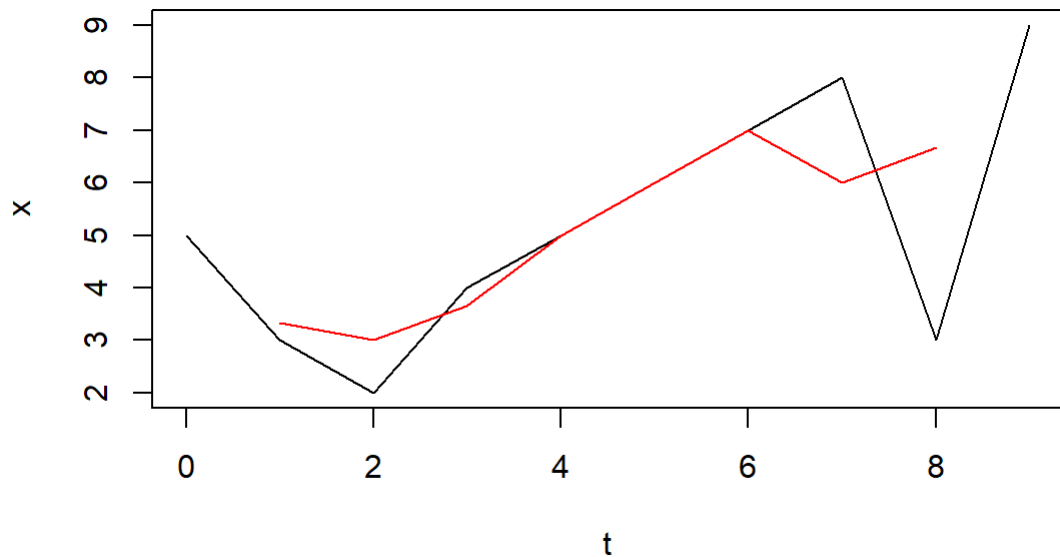
2. We can perform the calculations in the previous example in R as follows

```
> t = 0:9
> x = c(5,3,2,4,5,6,7,8,3,9)
> m = filter(x, c(1/3, 1/3, 1/3))
> x
[1] 5 3 2 4 5 6 7 8 3 9
> m
Time Series:
Start = 1
End = 10
Frequency = 1
[1] NA 3.333333 3.000000 3.666667 5.000000 6.000000 7.000000 6.000000
[9] 6.666667 NA
```

Note that the first and last element of `m` are undefined (in R this is displayed as `NA`).

3. We can also plot this and check the *smoothing* effect of taking averages.

```
plot(t, x, type="l")
lines(t, m, type="l", col="red")
```



The graph is a bit basic as the time series only has 10 elements, but it should be visible that the red line (m_t) is less zig-zaggy than the black line (x_t).

As stated above, when applying the $2q + 1$ -moving average algorithm to a finite time series

$$x_0, x_1, \dots, x_n$$

the first q values $m_0, \dots, m_{q-1}$ are undefined, as the last q values $m_{n-(q-1)}, \dots, m_n$. There are several ways we can deal with this.

1. Leave these values, $m_0, \dots, m_q$ and $m_{n-(q-1)}, \dots, x_n$. undefined. As seen above, this is the default approach that R takes: it places **NA** in these positions.
2. Set anything below x_0 to be 0 and anything above x_n as 0. This is the easiest mathematically but it leads to silly values for the first and last m_t . It is called *zero padding*.
3. Set anything below x_0 to be x_0 and anything above x_n to be x_n . It leads to more realistic values of m_t than zero padding. This is called *censoring*.
4. Make time circular: $x_{10} = x_0$, $x_{11} = x_1, \dots$. In modular arithmetic, $x_a = x_b$ when $a \equiv b \pmod{10}$. This could be appropriate if data was known to be seasonal.
5. Modify the moving average procedure for data near the boundary so that it is one-sided or asymmetric, while the unmodified moving average is used for interior data.

Using the third convention above (censoring):

$$\begin{aligned}\widehat{m}_1 &= \frac{1}{3}(5 + 5 + 3) = \frac{13}{3} \\ \widehat{m}_{10} &= \frac{1}{3}(3 + 9 + 9) = \frac{21}{3}.\end{aligned}$$

2.2 Linear filters and convolutions

In the previous section we averaged the time series, x_t , by calculating $m_t = (x_{t-q} + \dots + x_{t+q})/(2q+1)$ which can be seen as multiplying $(x_{t-q}, \dots, x_{t+1})$ by the weights $(1/(2q+1), \dots, 1/(2q+1))$ and adding the resulting numbers. We could consider averaging using different weights, for example, instead of $(1/3, 1/3, 1/3)$ we could use $(1/6, 2/3, 1/6)$. This would result in another estimate for the trend: $m_t = x_{t-1}/6 + 2x_t/3 + x_{t+1}/6$. The reason for using unequal weights might be to give values closer to x_t more weight and those further away less.

To explore this, we need to introduce the concept of linear filters.

Consider a sequence of coefficients $\mathbf{a} = (a_{-q}, a_{-q+1}, \dots, a_0, \dots, a_{q-1}, a_q)$ and a time series $\mathbf{x} = (x_1, x_2, \dots, x_n)$, define their convolution $\mathbf{w}$ to be time series

$$w_t = \sum_{i=-q}^q a_i x_{t-i}.$$

We write the convolution operator as $*$ and we can write the above compactly as $\mathbf{w} = \mathbf{a} * \mathbf{x}$.

Remarks 2.4. 1. You can think of the convolution as some form of average using the weights $a_{-q}, \dots, a_q$. For instance, if you take all a_i to be $1/(2q+1)$ then this reduces to the moving average in the previous section.

2. Note that the vector of weights is used *in reverse*, in that to calculate w_t we start with $a_q x_{t-q}$ which involves the last element of $\mathbf{a}$.

This is so that the sub-indices add up nicely. The sum of the sub-indices of each of the terms, $a_i x_{t-i}$, is the sub-index of the sum, w_t .

3. As happened with the moving average in the previous section, w_t is not defined for $t \in \{0, \dots, q-1\} \cup \{n-(q-1), \dots, n\}$ because the formula would require you to use values of the time series prior to x_0 or after x_n .

To simplify the mathematical treatment, **we will assume in the rest of this section that the time series has been extended by zero values before x_0 and after x_n**

$$\dots 0, 0, x_0, x_1, \dots, x_n, 0, 0, \dots$$

(this is called *zero padding*). With this convention w_t is defined for all values of t .

Example 2.2. What is $(1/3, 1/3, 1/3) * (1/2, 1/2)$?

Let us represent $(1/3, 1/3, 1/3)$ as (a_{-1}, a_0, a_1) and $(1/2, 1/2)$ as (x_1, x_2) . The idea is for each w_t , we sum the products $a_i x_j$ such that $i + j = t$:

$$\begin{aligned} w_0 &= a_{-1} x_1 = 1/6 \\ w_1 &= a_{-1} x_2 + a_0 x_1 = 1/3 \\ w_2 &= a_1 x_1 + a_0 x_2 = 1/3 \\ w_3 &= a_1 x_2 = 1/6. \end{aligned}$$

Hence $(1/3, 1/3, 1/3) * (1/2, 1/2) = (1/6, 1/3, 1/3, 1/6)$.

A **linear filter** is the transformation that takes a time series $\{x_t\}$ and produces another time series $\{w_t\}$ by convolving with filter coefficients $\mathbf{a} = (a_{-q}, \dots, a_q)^T$:

$$w_t = \sum_{j=-q}^q a_j x_{t-j} = \mathbf{a} * \{x_t\}. \quad (2.1)$$

In the case that these coefficients are the weights in an averaging operation, we call this a **low-pass filter**. Intuitively, low-pass filters remove high frequency (rapidly fluctuating) part of x_t and allow low frequency component such as m_t to pass through undistorted. By definition, this is a **two-sided** linear filter. In applications to signal processing in sound systems, low-pass filters are used to remove noise or hissing sounds.

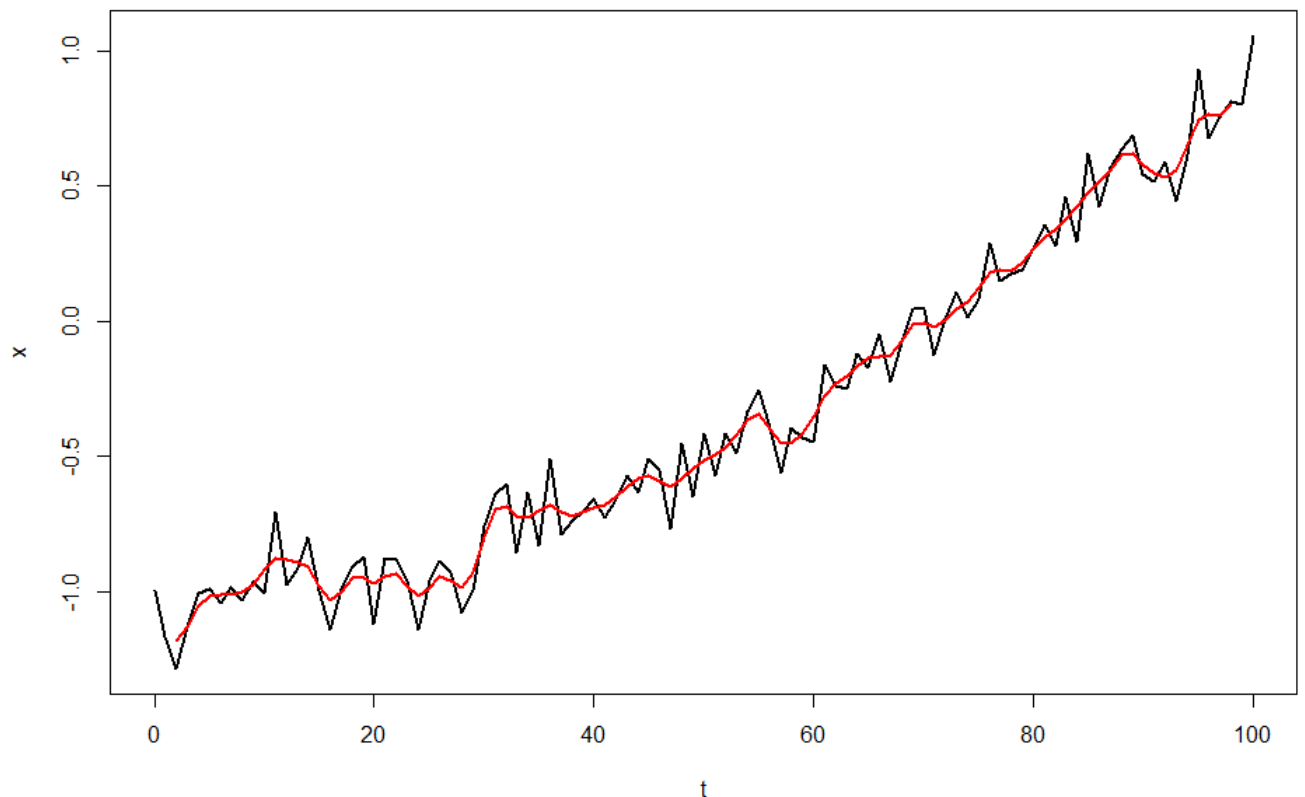
Example 2.3. In the R example below we create a time series, x_t , by adding random numbers to the function $2(t/100)^2 - 1$. We then use the low pass linear filter $\mathbf{a} = (1/16, 4/16, 6/16, 4/16, 1/16)$ to produce a smooth trend $m_t = \mathbf{a} * x$.

```

# values for time: 0, 1, ..., 100
# notice we have 101 time points
t = seq(0,100)
# just a nice quadratic function
# divide t by 100 as otherwise numbers get very big
x = 2*(t/100)^2-1
plot(t, x, type="l", lwd=2)
# add a bit of noise
# remember rnorm is standard normal, N(0,1), random number
# divide by 10 to make it a bit smaller
x = x + rnorm(101)/10
plot(t, x, type="l", lwd=2)
# the filter, more weight in center, less on sides.
a = c(1/16, 4/16, 6/16, 4/16, 1/16)
# we estimate the trend by the convolution a*x
m = filter(x, a)
lines(t, m, type="l", col="red", lwd=2)

```

Notice we have used the parameter `lwd=2` to make the lines in the chart thicker than the default `lwd=1`.



If $a_j = a_{-j}$ for all j , we call $\mathbf{a}$ a **symmetric filter**. If a_j 's are all the same, we say $\mathbf{a}$ is **simple**. Hence the $(2q + 1)$ -point moving average is a low-pass simple two-sided symmetric filter, with filter coefficients $a_j = (2q + 1)^{-1}$ for $|j| \leq q$ and 0 otherwise.

So how do we choose $\mathbf{a}$ to achieve the desired smoothing of our time series data?

2.3 Filter design

Another way of looking at the (2.1) is that the linear filter distorts the incoming time series $\{x_t\}$ to produce an outgoing distorted series $\{w_t\}$. For a trend function m_t , we say m_t passes through the linear filter given by (2.1) **without distortion** when

$$m_t = \sum_{j=-q}^q a_j m_{t-j}. \quad (2.2)$$

Another way to say this is that the linear filter **reproduces** m_t , i.e. when we apply the filter to the sequence m_t , the result is m_t again.

We design a filter by choosing a_j 's so that:

1. it is effective in suppressing noise of the time series data;
2. it allows a large class of trend functions to pass undistorted through the filter.

What trends does our basic moving $2q + 1$ -step moving average filter allow to pass undistorted?

Let us see that a linear trend $m_t = \beta_0 + \beta_1 t$ passes undistorted through a $(2q + 1)$ -point simple moving average filter. Recall this has coefficients $a_j = (2q + 1)^{-1}$ for $-q \leq j \leq q$, so that

$$\begin{aligned} \sum_{j=-q}^q \frac{1}{2q+1} m_{t-j} &= \frac{1}{2q+1} \sum_{j=-q}^q [\beta_0 + \beta_1(t-j)] \\ &= \frac{1}{2q+1} \sum_{j=-q}^q (\beta_0 + \beta_1 t) - \frac{1}{2q+1} \sum_{j=-q}^q \beta_1 j \\ &= \beta_0 + \beta_1 t = m_t, \quad \text{since } \sum_{j=-q}^q j = 0. \end{aligned}$$

2.3.1 Filter design for linear trend

If we choose a_j 's such that

$$\sum_{j=-q}^q a_j = 1, \quad \sum_{j=-q}^q j a_j = 0,$$

a linear trend $m_t = \beta_0 + \beta_1 t$ will pass through the filter undistorted.

The proof is almost the same as the case of the $2q + 1$ -step moving average above.

2.3.2 Filter design for quadratic trend

If we choose a_j 's such that

$$\sum_{j=-q}^q a_j = 1, \quad \sum_{j=-q}^q j a_j = 0, \quad \sum_{j=-q}^q j^2 a_j = 0,$$

a quadratic trend $m_t = \beta_0 + \beta_1 t + \beta_2 t^2$ will pass through the filter undistorted.

2.3.3 Filter design for cubic trend

If we choose a_j 's such that

$$\sum_{j=-q}^q a_j = 1, \quad \sum_{j=-q}^q j a_j = 0, \quad \sum_{j=-q}^q j^2 a_j = 0, \quad \sum_{j=-q}^q j^3 a_j = 0,$$

a cubic trend $m_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3$ will pass through the filter undistorted.

2.4 Well known low-pass filters

Example 2.4. A filter that satisfies the first three summation conditions is the **Spencer's 15-point moving average**:

$$a_0 = \frac{74}{320}, a_1 = \frac{67}{320}, a_2 = \frac{46}{320}, a_3 = \frac{21}{320}, a_4 = \frac{3}{320}, a_5 = -\frac{5}{320}, a_6 = -\frac{6}{320}, a_7 = -\frac{3}{320}$$

with $a_j = a_{-j}$ for $|j| \leq 7$ (symmetric). In other words, a quadratic trend passes through the Spencer's 15-point moving average without distortion. Note that this filter is not simple.

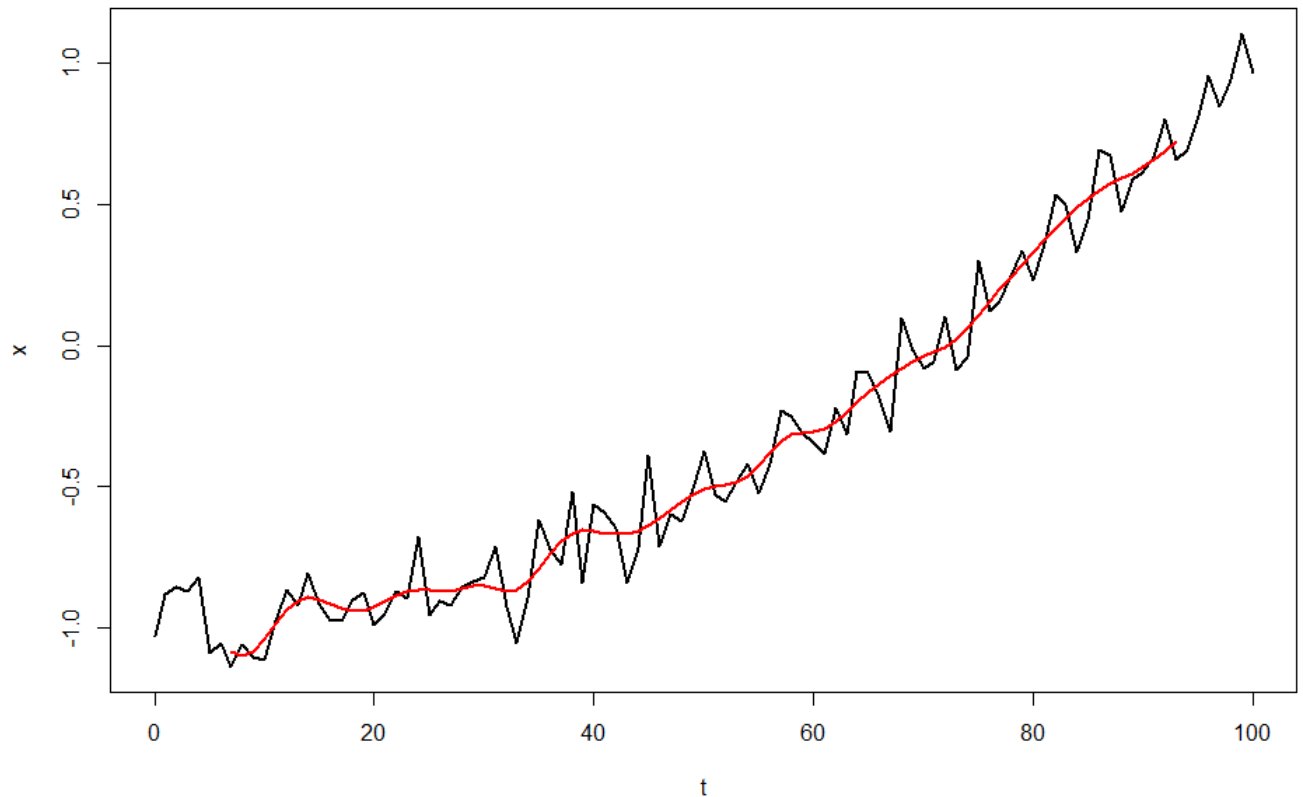
In R you can access the Spencer filter with the function `spencerFilter()` in the library `signal`. Try out the following code:

```
# values for time: 0, 1, ..., 100
# notice we have 101 time points
t = seq(0, 100)
# just a nice quadratic function
# divide t by 100 as otherwise numbers get very big
x = 2*(t/100)^2-1
plot(t, x, type="l", lwd=2)
# add a bit of noise
# remember rnorm is standard normal, N(0,1), random number
# divide by 10 to make it a bit smaller
x = x + rnorm(101)/10
plot(t, x, type="l", lwd=2)
```

```

# the filter, more weight in center, less on sides.
a = signal::spencerFilter()
a # let us view these values
# we estimate the trend by the convolution a*x
m = filter(x, a)
lines(t, m, type="l", col="red", lwd=2)

```



Example 2.5. Another widely used filter is **Henderson’s filter**. For example, the coefficients for its 13-point version are

$$a_0 = 0.240, a_1 = 0.214, a_2 = 0.147, a_3 = 0.066, a_4 = 0, a_5 = -0.028, a_6 = -0.019$$

with $a_j = a_{-j}$ for $|j| \leq 6$. This filter satisfies the four summation conditions and hence it allows a cubic trend to pass through undistorted.

These filter coefficients are found using (weighted) least squares around a neighbourhood of x_t .

2.5 High-pass filters

Note that not all filters are low-pass, i.e. they do not all eliminate noise. A silly example is obtained by taking the difference between x_t and it’s smoother moving average $m_t = (x_{t-1} + x_t + x_{t+1})/3$

which will be some form of residual error

$$y_t = x_t - (x_{t-1} + x_t + x_{t+1})/3 = -x_{t-1}/3 + 2x_t/3 - x_{t+1}/3$$

and is the result of applying the filter $(-1/3, 2/3, -1/3)$ to our time series. Try out the following code to check this:

```
# same as in previous examples
t = seq(0,100)
x = 2*(t/100)^2-1 + rnorm(101)/10
# this filter will not smoothen things
a = c(-1/3, 2/3, -1/3)
# y is the convolution a*x
y = filter(x, a)
plot(t, y, type="l", lwd=2)
```

You should see a series y_t that is not smooth at all.

Chapter 3

Non-parametric Trend Methods II

3.1 Exponential Smoothing

Recall that our current task is to identify the trend m_t underlying a time series x_t . This is often referred to as *smoothing* as the trend is meant to be a smooth function capturing the general move of x_t . The residual error $y_t = x_t - m_t$ is expected to capture the noisiness or *zig-zaggyness* of the time series.

3.1.1 Definition of Exponential Smoothing

In the previous chapter we considered techniques that averaged the time series around the value at time t by taking q values before and q values ahead. This was referred to as *two-sided* moving average, and as it uses future values it is not applicable to forecasting.

In this chapter we will consider a technique that only uses past values, it can therefore be described as *one-sided* and can be applied to forecasting.

Take a number between 0 and 1, α . For instance α could be 30%. Then construct m_t as a weighted average of past values of x_t as follows:

- Add $\alpha = 30\%$ times x_t : αx_t
- Plus $1 - \alpha = 70\%$ of past values of the time series as follows:
 - $\alpha = 30\%$ of that $1 - \alpha = 70\%$, times x_{t-1} . So $\alpha(1 - \alpha)x_{t-1}$
 - $1 - \alpha = 70\%$ of that $1 - \alpha = 70\%$, to the past values as follows:
 - * $\alpha = 30\%$ of that $(1 - \alpha)^2$, times x_{t-2} . So $\alpha(1 - \alpha)^2 x_{t-2}$
 - * $1 - \alpha = 70\%$ of that $(1 - \alpha)^2$, to the past values as follows:

• ...

So in maths

$$m_t = \alpha x_t + \alpha(1 - \alpha)x_{t-1} + \alpha(1 - \alpha)^2 x_{t-2} + \dots$$

If there are infinitely many past values then this is a series, and if the values of x_t are not too large in the past then it will converge.

Most frequently the time series starts at some point in the past, say x_0 and the procedure above has to stop there:

$$m_t = \alpha x_t + \alpha(1 - \alpha)x_{t-1} + \alpha(1 - \alpha)^2 x_{t-2} + \dots + \alpha(1 - \alpha)^{t-1} x_1 + (1 - \alpha)^t x_0$$

(the last term is right! it only has $1 - \alpha$).

Note that this is a weighted average of all past values of the time series and that the further in the past, the smaller the weights. For example, for $\alpha = 30\%$ we get the values

α	$\alpha(1 - \alpha)$	$\alpha(1 - \alpha)^2$	$\alpha(1 - \alpha)^3$	$\alpha(1 - \alpha)^4$	$\alpha(1 - \alpha)^5$	$\alpha(1 - \alpha)^6$	$\alpha(1 - \alpha)^7$	$\alpha(1 - \alpha)^8$	$\alpha(1 - \alpha)^9$	$\alpha(1 - \alpha)^{10}$
0.3000	0.2100	0.1470	0.1029	0.0720	0.0504	0.0353	0.0247	0.0173	0.0121	0.0085

The method is called *exponential* because these weights decay like powers of $(1 - \alpha)$.

Homework 3.1. 1. When using the expression *weighted average* with some weights $w_1, w_2, \dots$ one usually assumes the sum of the weights is 1=100%.

- In the infinite case above, where the weights are $\alpha, \alpha(1 - \alpha), \alpha(1 - \alpha)^2, \dots$, check that their sum is indeed 1.¹
- Check that this is also the case for the finite case², where the weights are $\alpha, \alpha(1 - \alpha), \alpha(1 - \alpha)^2, \dots, \alpha(1 - \alpha)^{t-1}, (1 - \alpha)^t$.

You will need to use the formula for a geometric series $1 + r + r^2 + r^3 \dots = 1/(1 - r)$ (valid if $|r| < 1$), and the formula for a geometric sum $r^a + r^{a+1} + \dots + r^b = (r^a - r^{b+1})/(1 - r)$.

- By differentiating the formula $\sum_{i=0}^{\infty} r^n = 1/(1 - r)$ we get $\sum_{i=0}^{\infty} nr^{n-1} = 1/(1 - r)^2$. Multiplying by r you get

$$\sum_{i=0}^{\infty} nr^n = \frac{r}{(1 - r)^2}.$$

This can be interpreted as some form of *weighted average value* of n weighted by the weights r^n .

Consider the exponential average

$$\alpha x_t + \alpha(1 - \alpha)x_{t-1} + \alpha(1 - \alpha)^2 x_{t-2} + \alpha(1 - \alpha)^3 x_{t-3} + \dots$$

¹. $1 = ((\alpha - 1) - 1)/1 \dots \alpha = \sum_{i=0}^{\infty} \alpha(1 - \alpha)^i$ The sum is equal to 1

². $1 = \alpha(1 - \alpha)^0 + \alpha(1 - \alpha)^1 + \dots + \alpha(1 - \alpha)^{t-1} + (1 - \alpha)^t$ Answer: $\sum_{i=0}^{t-1} \alpha(1 - \alpha)^i + (1 - \alpha)^t = 1$

If you want to provide some kind of *average relevant time*, t , you could calculate the weighted average of the times:

$$T = \alpha t + \alpha(1 - \alpha)(t - 1) + \alpha(1 - \alpha)^2(t - 2) + \alpha(1 - \alpha)^3(t - 3) + \dots$$

Show that this is equal to

$$T = t - \alpha \sum_{i=0}^{\infty} (1 - \alpha)^i i = t - \alpha \frac{1 - \alpha}{(1 - (1 - \alpha))^2} = t - \frac{1 - \alpha}{\alpha}$$

We denote $(1 - \alpha)/\alpha$ by N . This number measures some kind of *average time back* for the exponential moving average and is called the *span*.

If t is measured in days, and you want this *span* to be thirty days back, say $N = 30$, then we just need to solve:

$$30 = \frac{1 - \alpha}{\alpha}$$

which has solution $\alpha = 1/(30 + 1)$, or in general $\alpha = 1/(N + 1)$.

This is what in finance is referred to as the 30-day exponential moving average, an exponential moving average with $\alpha = 1/(30 + 1)$. Sometimes a factor of 2 is added $\alpha = 2/(30 + 1)$ for reasons we will not discuss here.

There is a field called, *technical analysis*, where some people construct trading signals out of these exponential moving average. There is not much scientific rationale for technical analysis, but in finance, if sufficiently many players believe something is on the rise, it might go up for a while as a result of the buying activity that results from this belief. Check

<https://en.wikipedia.org/wiki/MACD>

if you want to learn about one of these indicators³

3.1.2 Definition by recurrence

Note that

$$\begin{aligned} m_t &= \alpha x_t + \alpha(1 - \alpha)x_{t-1} + \alpha(1 - \alpha)^2x_{t-2} + \dots \\ &= \alpha x_t + (1 - \alpha)(\alpha x_{t-1} + \alpha(1 - \alpha)x_{t-2} + \dots) \\ &= \alpha x_t + (1 - \alpha)m_{t-1} \end{aligned}$$

which is another way of defining exponential smoothing. We also need the initial value $m_0 = x_0$.

³I recommend you do not get into trading with your own money.

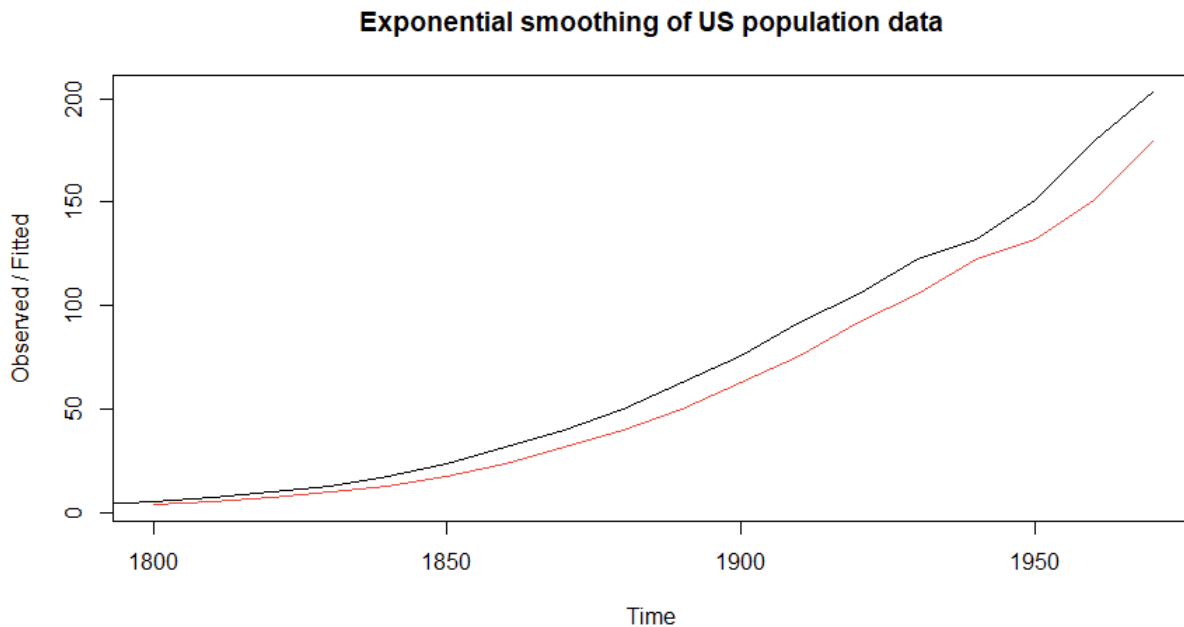
3.1.3 Use for forecasting

In forecasting one can take the exponential average m_t as a forecast for x_{t+1} .

This forecast will be different for different values of α . One can find the optimal parameter α by minimising the sum of squares of the prediction errors $x_{t+1} - m_t$ for known values in our time series.

In R this is accomplished by using the function `HoltWinters` with parameters `beta` and `gamma` set to `FALSE` as follows

```
# uspop is a dataset in the library astsa containing US population numbers from  
# 1800 to 1960  
# it is saved as a ts (time series) object  
library(astsa)  
m <- HoltWinters(uspop, beta=FALSE, gamma = FALSE)  
plot(m, main="Exponential smoothing of US population data")
```



This is not a very good fit as exponential smoothing is just an average of past values, and so for data that is growing, it will be always below the actual data. Exponential smoothing is more appropriate for data that does not exhibit a clear trend.

3.1.4 Holt Method

As seen above, exponential smoothing is not very good at forecasting when there is some form of grow trend. This is because the predicted value for x_t is just some weighted average of past values and this will not capture any larger values.

In the 1950's Charles Holt⁴ devised a method that captures the general upward or downward trend of the time series in another time series called b_t which is itself an exponential average:

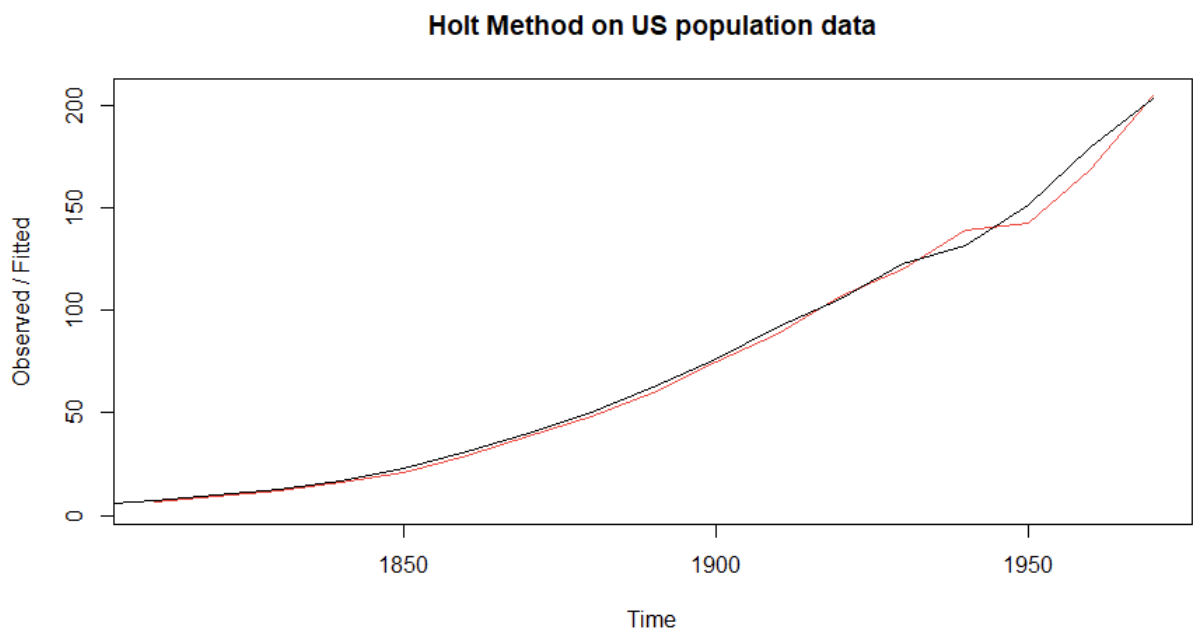
$$m_t = \alpha x_t + (1 - \alpha)(m_{t-1} + b_{t-1})$$

$$b_t = \beta(m_t - m_{t-1}) + (1 - \beta)b_{t-1}$$

For some parameters $\alpha, \beta \in [0, 1]$.

- Remarks 3.1.**
1. b_t measures the slope or trend, i.e. how much x_t changes from t to $t + 1$. To avoid noisiness this is done using the trend $m_t - m_{t-1}$. And for further smoothing, this is mixed with past values as in exponential smoothing. This explains the equation for b_t .
 2. The equation for m_t is the same than for exponential smoothing but now includes b_{t-1} which measures the trend. This way m_t will grow if there is a positive trend.
 3. When the method is used for forecasting, we set $\hat{x}_{t+h} = m_t + hb_t$, i.e. we forecast x_t to keep growing with trend b_t .
 4. As with exponential smoothing, the parameters α, β are found by optimizing past forecasting performance.
 5. In R you can use the Holt Method by using the function `HoltWinters` with parameter `gamma` set to `FALSE` as follows.

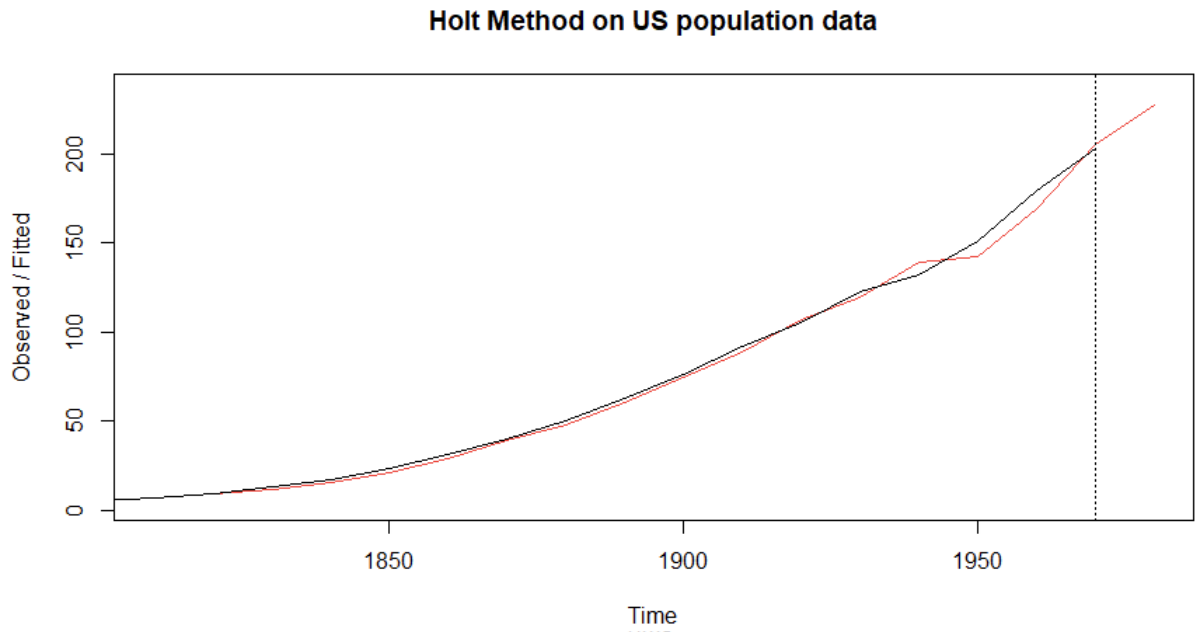
```
## Holt method
m2 <- HoltWinters(uspop, gamma = FALSE)
plot(m2, main="Holt Method on US population data")
```



⁴https://en.wikipedia.org/wiki/Charles_C._Holt

As you can see, in this case the trend follows the time series much more accurately and so will be more suitable for forecasting. You can do this in R with the function `predict` as follows.

```
## Holt method with forecasting
m2 <- HoltWinters(uspop, gamma = FALSE)
p <- predict(m2, 1, prediction.interval = TRUE)
plot(m2, p, main="Holt Method on US population data")
```



The uspop time series has `frequency=0.1`

```
> uspop
Time Series:
Start = 1790
End = 1970
Frequency = 0.1
 [1]  3.93  5.31  7.24  9.64 12.90 17.10 23.20 31.40 39.80 50.20
    62.90 76.00 92.00 105.70 122.80 131.70
[17] 151.30 179.30 203.20
```

This means there is one datapoint every 10 years. We have set the second parameter of `predict` to 1, which will forecast the population one time step ahead, to 1980. You could try other values.

6. As an aside, Excel provides functionality for Exponential Smoothing and the Holt method via the function `FORECAST.ETS`. You can also try the function `FORECAST.LINEAR` which calculates linear regression.

3.2 Trend differencing

3.2.1 Definition of ∇ and B

The estimated trend $\hat{m}_t$ gives you an idea how the mean of the time series changes over time. However in some practical applications, m_t is not the main object of interest, and our main aim is to understand the dependence structure of the noise $\{y_t\}$. To this end, we could remove the effect of trend from our time series by computing $x_t - \hat{m}_t$. This operation is called **detrending** or **trend adjustment**. We then use the detrended data for our dependence analysis.

The following method, called trend differencing, detrends a time series without having to compute $\hat{m}_t$. To do this, we transform time series data using the differencing operator. The **trend differencing operator**, written as ∇ , is defined as⁵

$$\nabla x_t = x_t - x_{t-1}. \quad (3.1)$$

Since we are differencing once, this is also called a **first order difference**. In contrast to moving average filters discussed previously, differencing is a **high-pass filter** that allows high frequency signals to pass through while removing low frequency components.

To study higher order differences, we define the **backshift operator** B such that⁶

$$Bx_t = x_{t-1}. \quad (3.2)$$

That is, B moves time backwards by one step.

Then, (3.1) can be rewritten as

$$\nabla x_t = (I - B)x_t, \quad (3.3)$$

where I is the identity operator such that $Ix_t = x_t$ (i.e. just like 1 in the number system).

It is easy to get mixed up with indices, so for clarity here is a table for illustration

t	$\dots$	0	1	2	$\dots$	$t-1$	t	$t+1$	$\dots$
x	$\dots$	x_0	x_1	x_2	$\dots$	x_{t-1}	x_t	x_{t+1}	$\dots$
Bx	$\dots$	x_{-1}	x_0	x_1	$\dots$	x_{t-2}	x_{t-1}	x_t	$\dots$
∇x	$\dots$	$x_0 - x_{-1}$	$x_1 - x_0$	$x_2 - x_1$	$\dots$	$x_{t-1} - x_{t-2}$	$x_t - x_{t-1}$	$x_{t+1} - x_t$	$\dots$

Here you see that Bx as a vector has shifted upwards, which can cause confusion with the terminology *backshift*.

⁵Note, that if we are rigorous, then if x is a time series, with terms $\dots, x_1, x_2, x_3, \dots$, then ∇x is the time series with terms $(\nabla x)_1 = x_1 - x_0, (\nabla x)_2 = x_2 - x_1, \dots$. So strictly speaking ∇x_t really means $(\nabla x)_t$. You only need concern yourself with these details if something does not work or you are confused.

⁶Same comment applies to B : strictly speaking if x is a time series, then Bx is another time series defined by $(Bx)_t = x_{t-1}$.

3.2.2 Higher order differencing

To move time back 2 steps, we can apply B twice to get $B^2x_t = x_{t-2}$. In general to move back d steps, we apply B d -times to get $B^dx_t = x_{t-d}$.

We can also iterate ∇ twice, the result is called the **second order difference** and can be calculated as follows:

$$\nabla^2x_t = \nabla(\nabla x_t),$$

This just means

$$\begin{aligned}\nabla^2x_t &= \nabla(x_t - x_{t-1}) = (x_t - x_{t-1}) - (x_{t-1} - x_{t-2}) \\ &= x_t - 2x_{t-1} + x_{t-2}.\end{aligned}$$

In view of (3.3), this can also be computed using the language of operators:

$$\begin{aligned}\nabla^2x_t &= \nabla[(I - B)x_t] = (I - B)(I - B)x_t = (I - 2B + B^2)x_t \\ &= x_t - 2x_{t-1} + x_{t-2}.\end{aligned}$$

Generally, we define the d th **order difference** to be

$$\nabla^dx_t = \nabla(\cdots \nabla(\nabla x_t)) = (I - B)^dx_t. \quad (3.4)$$

To write this out explicitly, we can use the binomial expansion to expand $(I - B)^d$.

3.2.3 Detrending

The idea is to use the operation ∇ or ∇^d to isolate the noise component of x_t in order to study it. As stated in the introduction often the study of ∇x can be very informative in our understanding of x_t . Here the order of differencing d will be a tuning parameter.

Differencing cancels a linear trend

To see how differencing works, consider again our additive model $x_t = m_t + y_t$ with linear trend $m_t = \beta_0 + \beta_1 t$. Then

$$\begin{aligned}\nabla x_t &= x_t - x_{t-1} \\ &= m_t + y_t - (m_{t-1} + y_{t-1}) \\ &= \beta_0 + \beta_1 t - [\beta_0 + \beta_1(t-1)] + \nabla y_t \\ &= \beta_1 + \nabla y_t.\end{aligned}$$

Hence the first order difference ∇x_t removes linear trend.

∇^2 cancels a quadratic trend

If $m_t = \beta_0 + \beta_1 t + \beta_2 t^2$ is quadratic, then

$$\begin{aligned}\nabla^2 x_t &= x_t - 2x_{t-1} + x_{t-2} \\ &= m_t + y_t - 2(m_{t-1} + y_{t-1}) + m_{t-2} + y_{t-2} \\ &= \beta_0 + \beta_1 t + \beta_2 t^2 - 2[\beta_0 + \beta_1(t-1) + \beta_2(t-1)^2] + \beta_0 + \beta_1(t-2) + \beta_2(t-2)^2 + \nabla^2 y_t \\ &= 2\beta_2 + \nabla^2 y_t.\end{aligned}$$

Therefore, the second order difference $\nabla^2 X_t$ removes quadratic trend.

∇^d cancels a polynomial trend of degree d

In general if $m_t = \beta_0 + \beta_1 t + \dots + \beta_d t^d$, then we have

$$\nabla^d x_t = d! \beta_d + \nabla^d y_t.$$

Hence the d th order differencing removes a polynomial trend of degree d . In practice, $d = 1$ or $d = 2$ suffices for most time series applications.

3.2.4 Closing comments on detrending

Remarks 3.2. 1. Advantages of differencing:

- (a) Estimation of fewer parameters, only have to worry about d , the order of differencing.
- (b) Can remove a wide variety of trends, and does not assume that trend remains fixed.

2. Disadvantage of differencing:

- (a) Does not give an explicit estimate of the trend, we need to use other methods if we want to study trends. This method is used when we are only interested in modelling the time dependence of $\{y_t\}$.

3. In R you can difference a vector as follows:

```
> x = c(1,3,4,5)
> y = diff(x)
> y
[1] 2 1 1
```

one disadvantage of doing things this way is that R does not properly account for the position of these elements. The vector x has values $x[1]=1$, $x[2]=3$, $x[3]=4$, and $x[4]=5$. And y has values $y[1]=2=x[2]-x[1]$, $y[2]=1=x[3]-x[2]$, $y[3]=1=x[4]-x[3]$. But this does not correspond to our definition of $\nabla x_t = x_t - x_{t-1}$.

A better way of dealing with this is using time series objects in R (`ts`) which have additional time data. The example above would be

```
> x = ts(c(1,3,4,5))
> x
Time Series:
Start = 1
End = 4
Frequency = 1
[1] 1 3 4 5
```

This is how the time series x is encoded in R. It has a start time of 1, and end of 4, frequency of 1 (you can ignore this now) and its elements are on the last line: $x[1]=1$, $x[2]=3$, $x[3]=4$, and $x[4]=5$.

Let us calculate $\nabla = \text{diff}$ of this time series:

```
> diff(x)
Time Series:
Start = 2
End = 4
Frequency = 1
[1] 2 1 1
```

Here the output is the same as before (the 2, 1, 1 in the last line) but we also see that this time series now starts at time 2 which is what we expect from the mathematical definition: the first difference is $\nabla x_2 = x_2 - x_1$.

In R there is also a `lag` function, that performs the operation B described above. You need to provide it the lag parameter `-1`.

```
> lag(x, -1) # this is the same as B(x) in the lesson
Time Series:
Start = 2
End = 5
Frequency = 1
[1] 1 3 4 5
> x-lag(x,-1)
Time Series:
Start = 2
End = 4
Frequency = 1
[1] 2 1 1
```

Notice that $Bx = \text{lag}(x, -1)$ starts at time 2 with the value $1 = x[1]$ as expected. We have computed $x - \text{lag}(x, -1)$ and see, as expected, that it coincides with `diff(x)`.

4. The technical term used to refer to operations like B , and ∇ is *operators*.

Operators are functions that map functions to functions. We can view a time series x_t or $x(t)$ as a function that assigns to each time the value of the time series:

$$\begin{aligned} x : \mathbb{Z} &\longrightarrow \mathbb{R} \\ t &\longmapsto x_t \end{aligned}$$

(so x sends t to x_t). The backshift operator B maps this function x to the function Bx defined as

$$\begin{aligned} Bx : \mathbb{Z} &\longrightarrow \mathbb{R} \\ t &\longmapsto x_{t-1} \end{aligned}$$

Likewise ∇ , B^d , ∇^d and I are operators that map the function x as above to another function.

These operators are all linear operators meaning that if, x and y are time series and a , b numbers, then $B(ax + by) = aB(x) + bB(y)$ (prove it!). In most cases, you can work with operators like you would with numbers, e.g. $(I - B)(I - B) = I - 2B + B^2$. Many mathematical operations you learned previously are linear operators, e.g. matrices, differentiation, integration etc.

Chapter 4

Seasonality

Recall at the start we described the additive model for a time series as a decomposition of a time series into trend, seasonal component and residual error:

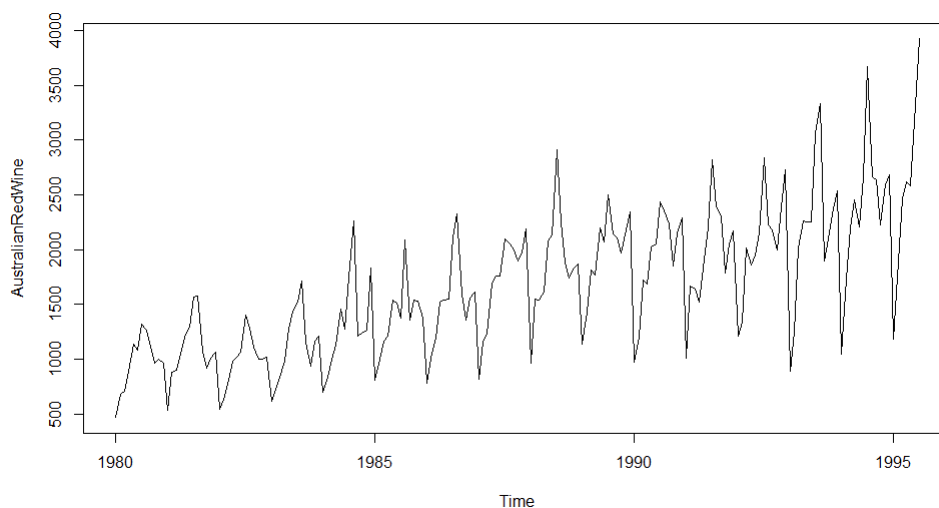
$$x_t = m_t + s_t + y_t.$$

(remember: m_t and s_t are not assumed to be random, y_t is meant to capture the residual randomness)

So far we have assumed our data did not exhibit seasonal effects and hence set $s_t = 0$ in our discussion. We will now consider what to do if our time series exhibits seasonal effects.

4.1 An example: Australian wine sales

Example 4.1. The following is a chart displaying monthly sales of Australian red wine in thousands of litres. A clear seasonal pattern is visible. When is most wine purchased?



This data is available in the R library `Rssa` and the chart was produced with the code below. Run it and try variations (for example display time series for Australian fortified wine instead)

```
library(Rssa)
data(AustralianWine)
?AustralianWine
class(AustralianWine)
head(AustralianWine)
AustralianRedWine = AustralianWine[, "Red"]
plot(AustralianRedWine)
```

In case the library `Rssa` is not installed you will get an error message. You can install it with `install.packages("Rssa")`.

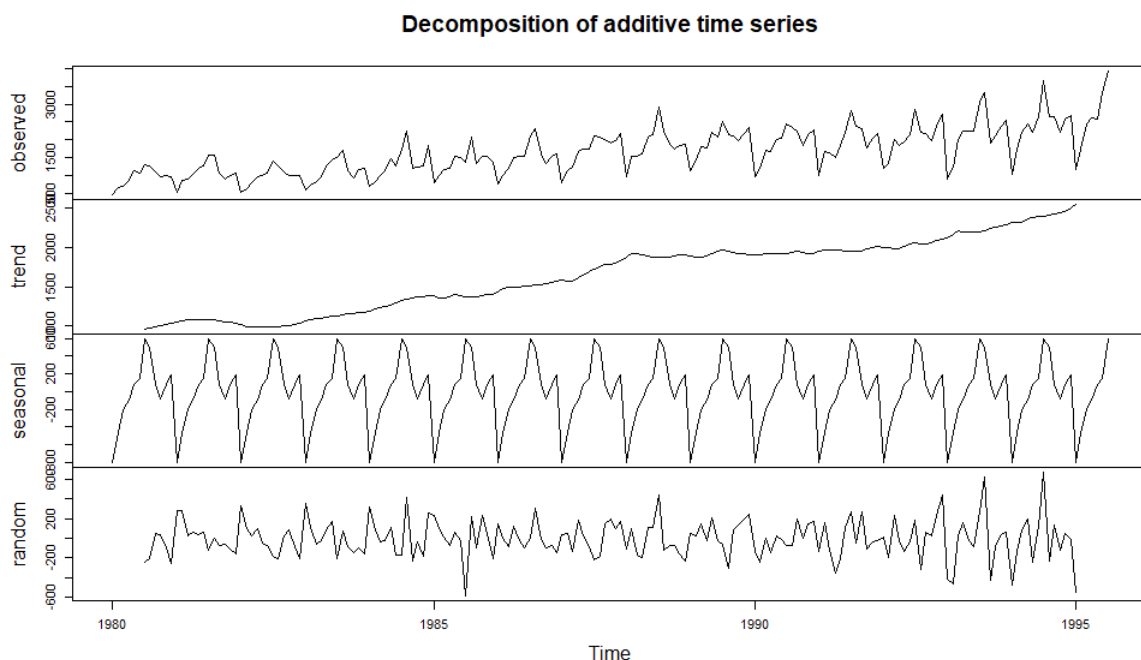
One interesting side observation in the chart above is that the variations seem to be getting bigger. Our current model assumes that data is *homoscedastic*, i.e. variability is constant with time. The time series above exhibits *heteroscedasticity* as with time the time series variability increases. We will see later how this can be addressed.

As an extra, to identify the months where most wine is purchased, you can proceed as follows

```
library(xts)
AustralianRedWine2 = as.xts(AustralianRedWine)
plot(AustralianRedWine2)
```

Try it!

The aim of this chapter is to explore ways in which we can understand the third of the following four charts



which you can produce with the following R instruction:

```
plot(stats::decompose(AustralianRedWine))
```

(we need to add `stats::` to access the usual `decompose` function, as the library `Rssa` redefines the function `decompose`).

4.2 Definition and terminology

It will now be important to know the units in which time is measured. For example t could represent years, months, weeks, days, hours...

Definition 4.1. The *period* of the seasonal component is the number of time units, S , such that

$$s_{t+S} = s_t$$

In other words, s_t will be periodic with period S .

Remarks 4.1. 1. In most cases we will consider t measured in months and $S = 12$. I.e. we will look at monthly data with yearly seasonality. The wine data above is an example of this type of seasonality. Many other time series related to commerce, weather, retail, health ... exhibit yearly seasonality.

2. Weekly seasonal effects are also interesting. For instance, Amazon will be interested in exploring how purchasing habits change with days of the week in order to maximise the impact of any marketing campaign.
3. We will often assume that the average of all the seasonal components is zero: $(s_1 + \dots + s_S)/S = 0$. If this was not the case and the average is, say 13.5, then we would redefine the seasonal component and the trend as follows $s_t^{new} = s_t - 13.5$ and $m_t^{new} = m_t + 13.5$, the time series would still add up to these:

$$\begin{aligned} x_t &= m_t + s_t + y_t = m_t + 13.5 + s_t - 13.5 + y_t = \\ &= m_t^{new} + s_t^{new} + y_t \end{aligned}$$

and now we would have that the average seasonal effect is zero

$$\frac{s_1^{new} + \dots + s_S^{new}}{S} = \frac{s_1 - 13.5 + \dots + s_S - 13.5}{S} = \frac{s_1 + \dots + s_S}{S} - 13.5 = 0$$

You can see in the chart above that the seasonal effects are centred around zero on the y -axis.

4. We might also, to simplify notation, assume m_t is zero. If not, then just consider a new time series $x_t^{new} = x_t - m_t$ and we will work with x_t^{new} instead.

4.3 R terminology

In R time series are often codified as a **ts** (time series) object. Time in a **ts** object is measured in years. The two following functions describe the time between observations:

- **frequency()** returns the number of observations per year. For monthly data this would be 12.
- **deltat()** returns the time (in years) between observations, i.e. $1/\text{frequency}()$. For monthly data this number will be $1/12$.

This has nothing to do with seasonality, it just refers to how many data-points are recorded per year.

For instance, the dataset **birth** in the library **astsa** contains monthly births in the US.

```
> library(astsa)
> frequency(birth)
[1] 12
> deltat(birth)
[1] 0.08333333
> head(birth)
[1] 295 286 300 278 272 268
> head(time(birth))
[1] 1948.000 1948.083 1948.167 1948.250 1948.333 1948.417
```

4.4 Seasonal indicator and the Brockwell Davies decomposition

4.4.1 The simplest possible seasonality model

Recall that the seasonal component of our time series is a periodic time series

$$s_1, s_2, \dots \text{ with } s_t = s_{t+S}.$$

For instance, in monthly data with yearly seasonality, $S = 12$; and if t is a time in January:

$$\begin{array}{ll} s_1 = s_{1+12} = s_{1+24} = \dots & \text{seasonal component for January} \\ s_2 = s_{2+12} = s_{2+24} = \dots & \text{seasonal component for February} \\ \vdots & \\ s_{12} = s_{12+12} = s_{12+24} = \dots & \text{seasonal component for December} \end{array}$$

So we only need to find these 12 numbers.

A simple way of estimating such an s_t is as follows:

- Estimate the trend, m_t , and remove it from x_t to find a de-trended time series, $x_t - m_t$.
- Set
 - s_1 to be the average of all values $x_1 - m_1, x_{1+12} - m_{1+12}, \dots$ I.e. the average of all January values of $x_t - m_t$.
 - s_2 to be the average of all values $x_2 - m_2, x_{2+12} - m_{2+12}, \dots$ I.e. the average of all February values of $x_t - m_t$.
 - $\dots$
 - s_{12} to be the average of all values $x_{12} - m_{12}, x_{12+12} - m_{12+12}, \dots$ I.e. the average of all December values of $x_t - m_t$.
- If the average of $s_1, \dots, s_{12}, \bar{s}$, is not zero, then subtract it from all $s_1, \dots, s_{12}$ so that the average is now zero. And add this subtracted average, $\bar{s}$, to m_t .
- Extend the definition of s_t to other values of t using periodicity, $s_{t+12} = s_t$. For example, $s_{13} = s_1$ or $s_{-1} = s_{11}, \dots$

Remarks 4.2. 1. For clarity we have restricted the discussion above to monthly data and yearly seasonality, $S = 12$. It can easily be written in general. Write it for quarterly data with yearly seasonality, in which every year has 4 quarters.

2. The approach above is less suitable if S is a larger number. For example, for daily data we would have to calculate the 365 numbers, $s_1, \dots, s_{365}$. This could lead to *overfitting*, i.e. using too many parameters to fit data resulting in a loss of predictive power. In this case we would prefer a model for seasonality that explains these 365 numbers using less parameters.

4.4.2 Even moving averages

We previously estimated the trend, m_t , using the average of an odd number of points, $2q + 1$, around each value x_t . For example, for $q = 1$, we had

$$\begin{array}{ccccccc} \dots, & x_{t-2}, & x_{t-1}, & x_t, & x_{t+1}, & x_{t+2}, & \dots \\ \dots, & m_{t-2}, & m_{t-1}, & m_t = \frac{x_{t-1} + x_t + x_{t+1}}{3}, & m_{t+1}, & m_{t+2}, & \dots \end{array}$$

It will be useful to extend this to an even number of points, $2q$. How can we do this? Say $q = 2$, so we are averaging $2q = 4$ points around x_t . What points do we choose? We could do:

- $(x_{t-1} + x_t + x_{t+1} + x_{t+2})/4$, or
- $(x_{t-2} + x_{t-1} + x_t + x_{t+1})/4$

But both of these seem strange in that they have more points to one side or the other of x_t . Let us instead take the average of both:

$$m_t = \frac{x_{t-2}/2 + x_{t-1} + x_t + x_{t+1} + x_{t+2}/2}{4}$$

Definition 4.2. Let

$$x_0, x_1, \dots, x_n$$

be a time series and q a positive integer, then we define the $2q$ -point moving average as the time series:

$$m_t = \frac{x_{t-q}/2 + x_{t-(q-1)} + \dots + x_t + \dots + x_{t+q-1} + x_{t+q}/2}{2q}$$

Note that it is a bit cheating in that there are $2q + 1$ numbers above and not $2q$, but the two at the extremes (x_{t-q} and x_{t+q}) are counted only half; so you can say that in some sense we are averaging $2q$ numbers.

4.4.3 The Brockwell and Davies algorithm

The Brockwell and Davies algorithm is a method of decomposing a time series x_t into trend, seasonal component and residual error. It uses the steps described above.

Let $x_1, \dots, x_n$ be a time series with n samples. We assume that n is a multiple of S . This means that if the time series is monthly then n covers an integer number of full years (so $n = 25$ months would not be allowed).

The steps in the algorithm are:

1. Estimate the trend using moving averages.

What size of window (the $2q + 1$ or the $2q$) should we use? Take for example, $S = 12$ for monthly data with yearly seasonality. Then if the size of the window is smaller than S , like 3 months, then we will still see seasonal oscillations in the trend. We want to keep these seasonal effects to s_t and not in m_t . It seems therefore clear that we should use a window of size S , or 12 months in this case.

So we estimate the trend, m_t , as the S -point moving average of x_t .

- If S is even, $S = 2q$, this means

$$\hat{m}_t = \frac{1}{S} (x_{t-q}/2 + x_{t-q+1} + \dots + x_{t+q-1} + x_{t+q}/2), \quad q + 1 \leq t \leq n - q.$$

- If S is odd, $S = 2q + 1$, then

$$\hat{m}_t = \frac{1}{S} (x_{t-q} + x_{t-q+1} + \dots + x_{t+q-1} + x_{t+q}), \quad q + 1 \leq t \leq n - q.$$

Averaging over S elements, has the effect of removing (by averaging) any seasonal effect from m_t .

Also, note that as usual with averaging, $m_1, \dots, m_q$ and $m_{n-(q-1)}, \dots, m_n$ are not defined. The trend, m_t , is only defined for the $n - 2q$ values $t = q + 1, \dots, n - q$.

2. Estimate the seasonal component as explained in section 4.4.1. I.e. let w_t be the average of $\{(x_{t+jS} - \hat{m}_{t+jS}), q < t + jS \leq n - q\}$:

$$w_t = \frac{1}{\frac{n-2q}{S}} \sum_{\frac{q-t}{S} < j \leq \frac{n-q-t}{S}} (x_{t+jS} - \hat{m}_{t+jS}).$$

(in the monthly case this fancy formula just means average all elements of the time series corresponding to January, etc.)

However these average deviations do not necessarily average to zero, we instead estimate the seasonal component s_t by

$$\hat{s}_t = w_t - \frac{1}{S} \sum_{i=1}^S w_i, \quad k = 1, \dots, S,$$

and extend to other values of k $\hat{s}_k = \hat{s}_{k-S}$, $k > S$.

3. Deseasonalize $d_t = x_t - \hat{s}_t$, $t = 1, \dots, n$.
4. Re-estimate the trend $\tilde{m}_t$ using your favourite method based on the deseasonalized series $\{d_t\}$. Note that if you use the same moving average as in step 1, the effect will be to add onto $\hat{m}_t$ the constant number $\frac{1}{S} \sum_{i=1}^S w_i$.
5. Calculate the residuals $\hat{y}_t = x_t - \tilde{m}_t - \hat{s}_t$ (detrend and deseasonalize).

This algorithm is implemented in R as the function `decompose` used above.

Exercise. In order to understand the procedure above, it is very useful to get your hands dirty and do the calculations yourself. Consider the following time series consisting of quarterly observations for 8 full years:

```
> # a vector with 32 numbers
> x <- c(-50, 175, 149, 214, 247, 237, 225, 329,
        729, 809, 530, 489, 540, 457, 195, 176,
        337, 239, 128, 102, 232, 429, 3, 98,
        43, -141, -77, -13, 125, 361, -45, 184)
> # make it into a time series of quarterly data from Q1 2016 to Q4 2023
> # S=4, n= 32, n/S=32/4=8 full years
> x <- ts(x, start = c(2016, 1), frequency = 4)
> x
      Qtr1 Qtr2 Qtr3 Qtr4
2016  -50  175  149  214
2017  247  237  225  329
```

2018	729	809	530	489
2019	540	457	195	176
2020	337	239	128	102
2021	232	429	3	98
2022	43	-141	-77	-13
2023	125	361	-45	184

Can you calculate some of the numbers in the Brockwell Davis algorithm by hand? For example, as $S = 4$ we have

$$m_t = \frac{x_{t-2}/2 + x_{t-1} + x_t + x_{t+1} + x_{t+2}/2}{4}$$

and the first such number is

$$m_3 = \frac{x_1/2 + x_2 + x_3 + x_4 + x_5/2}{4} = \frac{-50/2 + 175 + 149 + 214 + 247/2}{4} = 159.125.$$

Calculate m_4 . Can you set up an Excel spreadsheet to calculate the trend? You can also compute these m_t by running `filter(x, c(0.5,1,1,1,0.5)/4)`, do you understand why?

Calculate and display the time series $x_t - m_t$ and average each of the quarter columns (these are the number w_1, w_2, w_3 , and w_4 above).

Run the R code below and check that `m$trend` agrees with the numbers you just calculated.

```
m <- stats::decompose(x)
m$seasonal
m$trend
m$random
```

Try to calculate all these numbers in Excel. Plot the Brockwell Davis decomposition for this time series as we did at the start of the chapter for the Australian red wine time series.

4.5 Holt Winters

A point of terminology: In the Holt Winters model we use a time series, b_t , to track the increases of the time series, x_t or m_t . This b_t is normally called the **trend**, as it describes how much x_t and m_t are expected to increase. For this reason, in this model, we rename m_t and call it **level** instead of the usual word *trend*.

We have already seen the version of Holt-Winters without seasonal adjustment in which both m_t and b_t followed an exponential average process.

$$\hat{x}_{t+h} = \hat{m}_t + hb_t \quad h = 0 \text{ for smoothing, } h \geq 1 \text{ for prediction}$$

$$\begin{aligned} \hat{m}_t &= \alpha x_t + (1 - \alpha)(\hat{m}_{t-1} + b_{t-1}) && \text{Level smoothing} \\ b_t &= \beta(\hat{m}_t - \hat{m}_{t-1}) + (1 - \beta)b_{t-1} && \text{Trend smoothing} \end{aligned}$$

We now add seasonality effects through an additional process s_t :

$$\hat{x}_{t+h} = \hat{m}_t + hb_t + \hat{s}_{t-S+1+\lfloor(h-1) \bmod S\rfloor} \quad h = 0 \text{ for smoothing, } h \geq 1 \text{ for prediction}$$

$$\hat{m}_t = \alpha(x_t - \hat{s}_{t-S}) + (1 - \alpha)(\hat{m}_{t-1} + b_{t-1}) \quad \text{Level smoothing}$$

$$b_t = \beta(\hat{m}_t - \hat{m}_{t-1}) + (1 - \beta)b_{t-1} \quad \text{Trend smoothing}$$

$$\hat{s}_t = \gamma(x_t - \hat{m}_{t-1} - b_{t-1}) + (1 - \gamma)\hat{s}_{t-S} \quad \text{Seasonal smoothing}$$

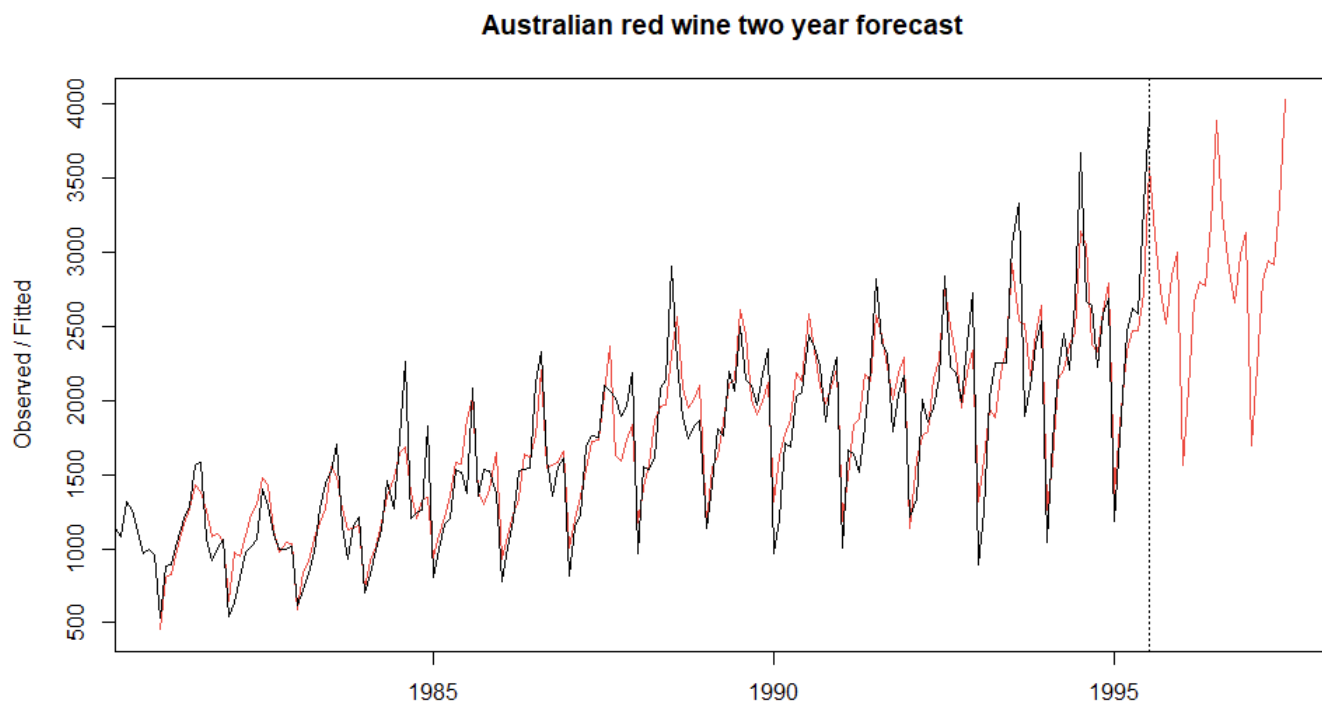
This method has three tuning parameters, each in the range $[0, 1]$:

- α controls the degree of level smoothing;
- β controls the trend smoothing;
- and γ controls the seasonal smoothing.

α, β and γ would be chosen to minimize the one-step ahead prediction error. The values for $\hat{m}_t$, b_t and $\hat{s}_t$ at the start of the series may be chosen using moving averages on some subset of the data at the start, for example the first two full periods of data.

Let us use it to predict Australian red wine sales two years ahead:

```
model = HoltWinters(AustralianRedWine)
p = predict(model, 24)
plot(model, p, main="Australian red wine two year forecast")
```



4.6 Seasonal differencing

Recall that in analysing the trend of a time series, one option was to *kill* the trend by differencing the time series. This differenced time series, ∇x_t , would often have no trend. Sometimes we need to difference more than once, $\nabla^d x_t$. Often $d = 1$ or 2 suffice.

The same idea applies to seasonal effects. In some problems it can be useful, rather than to model seasonality, to *kill* seasonality. This is achieved by an operator similar to ∇ .

4.6.1 The seasonal differencing operator

Definition 4.3. The *lag- S seasonal differencing operator* is defined as:

$$\nabla_S x_t = x_t - x_{t-S} = (I - B^S)x_t.$$

Recall that the usual differencing operator, ∇ as well as its iterates, ∇^d , are linear. The same happens with the seasonal differencing operator.

Proposition 4.1. The seasonal differencing operators is a linear operator meaning that if x_t and y_t are time series and a and b are numbers, then

$$\nabla_S (ax_t + by_t) = a\nabla_S x_t + b\nabla_S y_t$$

Proof. The proof is a straightforward application of the definition:

$$\begin{aligned}\nabla_S (ax_t + by_t) &= (ax_t + by_t) - (ax_{t-S} + by_{t-S}) = a(x_t - x_{t-S}) - b(y_t - y_{t-S}) \\ &= a\nabla_S x_t + b\nabla_S y_t\end{aligned}$$

□

Remarks 4.3. 1. For example, with monthly data with yearly seasonality, $S = 12$, $\nabla_S x_t$ is the difference between the value this month and the value 12 months ago.

2. Note that this is different from applying the trend differencing S times, $\nabla^S = (I - B)^S$.

3. As a trivial remark, note that for lag 1, $S = 1$, this operator is the plain difference operator: $\nabla_1 = \nabla$.

4. If we apply the ∇_S operator to a periodic time series with period S the result is zero:

$$\nabla_S s_t = s_t - s_{t-S} = 0$$

So, you can say that the operator ∇_S *kills* any seasonal effect.

5. If our time series, x_t , is the sum of a seasonal trend with period S , and residual noise:

$$x_t = s_t + y_t$$

Then, by linearity

$$\nabla_S x_t = \nabla_S s_t + \nabla_S y_t = \nabla_S y_t \quad (4.1)$$

And so, again, we see that ∇_S *kills* the seasonal component, or in more precise language it *deseasonalizes* the time series.

6. In R you know that the operator ∇ corresponds to the function `diff`. In fact this function has additional parameters to specify lag (this is S above) or number of iterates (the d in ∇^d). To see these you can run the help command in R:

```
?diff
```

If you want to calculate a lagged difference (the ∇_S above) you just need to run something like `diff(x, lag=12)`

```
> # A random time series
> x = c(4, 9, 8, 12, 6, -2, -4, -7, -9, -9, -2, 1, 4, 4, 12, 9, 5, 2, -3,
      -7, -8, -7, -5, -4, 6, 9, 10, 6, 4, 1, -2, -9, -9, -9, -8, -1)
> # S=12 lagged difference
> diff(x, lag=12)
[1] 0 -5 4 -3 -1 4 1 0 1 2 -3 -5 2 5 -2 -3 -1 -1 1 -2 -1 -2 -3 3
> # Check
> x[13]-x[1] ; x[14]-x[2] ; x[15]-x[3] ; x[16]-x[4]
[1] 0
[1] -5
[1] 4
[1] -3
```

Try this. Invent your own time series, try with `lag=4`.

4.6.2 Properties of ∇ and ∇_S

We can both detrend and deseasonalize, by applying both the trend differencing operator ∇^d and the seasonal differencing operator ∇_S to $\{x_t\}$.

In which order should we do this?

- First ∇ and after that ∇_S . I.e. apply the operator $\nabla_S \nabla$ to x_t ? or
- First ∇_S and after that ∇ . I.e. apply the operator $\nabla \nabla_S$ to x_t ?

In general linear operators (which are basically like matrices) do not commute¹, so the order matters. However we have the following result.

¹Two matrices, A , and B , are said to *commute* if $AB = BA$. This is not generally the case: most matrices do not commute. Understanding which matrices commute with a given one is an interesting pure maths problem.

Proposition 4.2. The operators ∇ and ∇_S commute. That is, for a time series x_t we have that

$$\nabla \nabla_S x_t = \nabla_S \nabla x_t$$

Proof. The proof consists on just writing what each of these things mean:

$$\begin{aligned}\nabla \nabla_S x_t &= \nabla (x_t - x_{t-S}) = (x_t - x_{t-S}) - (x_{t-1} - x_{t-1-S}) \\ \nabla_S \nabla x_t &= \nabla_S (x_t - x_{t-1}) = (x_t - x_{t-1}) - (x_{t-S} - x_{t-S-1})\end{aligned}$$

And it is trivial to check both these quantities are the same. □

Corollary 4.1. For any d , we have $\nabla^d \nabla_S = \nabla_S \nabla^d$.

Proof. By the previous proposition we can commute ∇ and ∇_S , so

$$\nabla^d \nabla_S = \nabla^{d-1} \nabla \nabla_S = \nabla^{d-1} \nabla_S \nabla$$

If we continue doing the same then we can move all the ∇ s to the other side (for more rigour apply induction). □

4.6.3 Trend and seasonal differencing

So now assume we have a time series

$$x_t = m_t + s_t + y_t$$

and we wish to remove both trend and seasonal effect.

If the seasonal effect has a period of S and we assume the trend is linear $m_t = \beta_0 + \beta_1 t$ then by applying $\nabla \nabla_S$ we will kill both the trend and the seasonal effect.

Let us see this. By linearity

$$\nabla \nabla_S x_t = \nabla \nabla_S m_t + \nabla \nabla_S s_t + \nabla \nabla_S y_t$$

Next, since s_t is periodic with period S

$$\nabla \nabla_S s_t = \nabla 0 = 0$$

and since m_t is linear, and $\nabla \nabla_S = \nabla_S \nabla$ we have

$$\nabla \nabla_S m_t = \nabla_S \nabla m_t = \nabla_S \beta_0 = 0$$

So in total we have

$$\nabla \nabla_S x_t = \nabla \nabla_S y_t$$

which means we have removed both the trend and the seasonal effects.

Homework 4.1. What should we do to remove seasonal effects and trend if the trend is assumed to be polynomial $m_t = \beta_0 + \beta_1 t + \dots + \beta_d t^d$?

4.7 Harmonic decomposition

The discussion above is based on the premise that there is a single source of seasonality. For instance, you can have **one of the following**:

- Daily data with some form of weekly seasonality. For instance, purchases on an online platform might have some form of weekly pattern.
- Monthly data with yearly seasonality. For example, sales of ice-creams are presumably higher in summer and lower in winter, so we expect some form of seasonal effect here.
- Hourly data with daily seasonality. For instance temperatures in a few days in August are expected to be lower at night and higher during the day.

Both the Brockwell Davis and the Holt Winters methods will work fine here.

However, often we have data with several seasonal components of different periodicity. For instance, electricity consumption has a yearly seasonal pattern (more electricity is used in winter), but also a weekly seasonality (more electricity is used in weekdays than in weekends) and then also a daily seasonal pattern (more electricity is used around 4pm than 4am).

This requires to consider periodic functions in which several periods are added together. This is the content of a fascinating area of analysis called *Fourier Series* or more generally *Harmonic Analysis*. The main idea is that all periodic functions can be written, in some sense, as sums of sin and cos functions of different periods.

The outcome is that the seasonal component will be described as:

$$s_t = a_0 + \sum_{j=1}^k \left(a_j \cos \left(\frac{2\pi jt}{S} \right) + b_j \sin \left(\frac{2\pi jt}{S} \right) \right),$$

where $a_0, a_1, \dots, a_k$ and $b_1, \dots, b_k$ are unknown parameters.

The right hand side is a **Fourier series** truncated at k . We can view this model is a linear regression problem and we can estimate these parameters using least squares. Here k the series truncation point is our tuning parameter.

Chapter 5

Analysis of the residual error

We have explored several ways in which we can extract the trend and seasonal component of a time series.

$$x_t = m_t + s_t + y_t.$$

Let us next turn our attention to the residual error term, y_t .

The first question we must address is: have we done a good job at extracting m_t and s_t ? What do we expect y_t to look like?

There will be occasions when the resulting y_t does not have the characteristics appropriate for our subsequent analysis. How can we detect this?

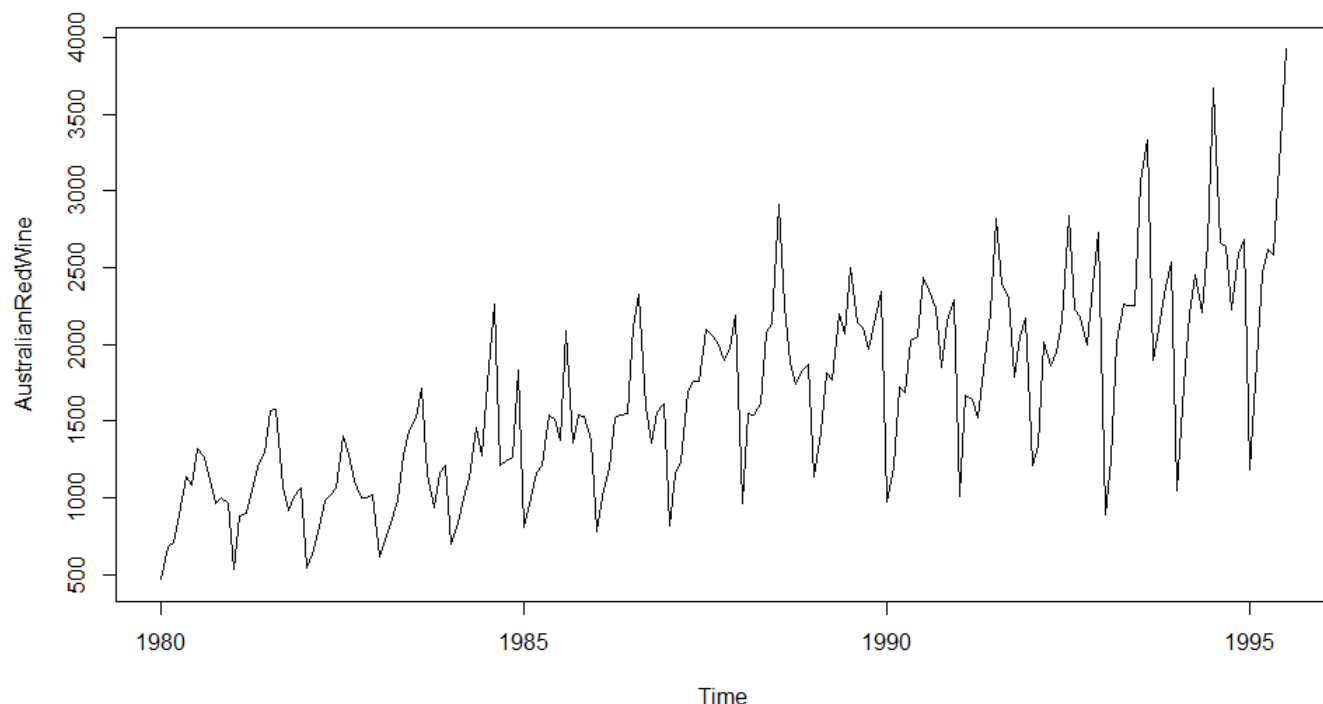
The general response is that we need y_t to be *stationary*. We will look into this later. For now, let us focus on one typical problem and its remedy.

5.1 Heteroskedasticity

5.1.1 Definition of heteroskedasticity and homoskedasticity

The word *heteroskedasticity* literally means *different* (ἕτερο) *scattered* (σχεδάννυμι) and refers to cases such as the one displayed below, where we see that the variance of the data varies through time (in the case below it increases).

Mathematically this can be expressed by saying that $\text{Var}(X_t)$ depends on t .



Our techniques to deal with residual error require time series to be stationary (to be defined later). A stationary time series is, in particular, *homoskedastic* which comes from *same* (ὁμός) *scattered* (σχεδάννυμι).

How can we fix this?

5.1.2 The Box-Cox transformations

To fix the problem of heteroskedasticity the idea is to *transform* our data using a function, i.e. instead of considering x_t , look at $x_t^{new} = f(x_t)$ for some suitable function.

Box-Cox considered the following family of functions

$$\text{BoxCox}(x_t) = \begin{cases} (x_t^\lambda - 1)/\lambda & \lambda \neq 0, \\ \log x_t & \lambda = 0. \end{cases}$$

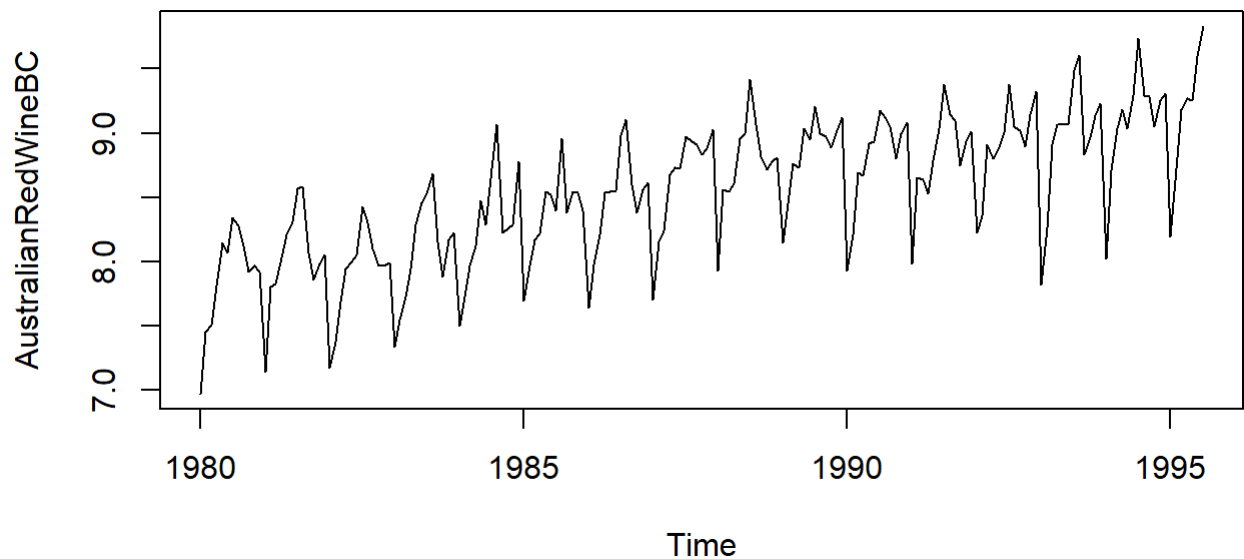
These transformations are parameterized by λ and we choose the best λ by maximum likelihood using past data.

In practice, data scientists often simplify this by just doing a **log transformation**. BoxCox belongs to the family of **power transforms** and hence you can also use other members of this family to stabilize the time series variance.

5.1.3 Box-Cox in R

In R you can use the function `BoxCox.lambda` function from the `forecast` package to find the optimal λ , and the function `BoxCox` to calculate the transform as described in the previous subsection. If we plot the resulting time series, we see that the increasing variance has disappeared.

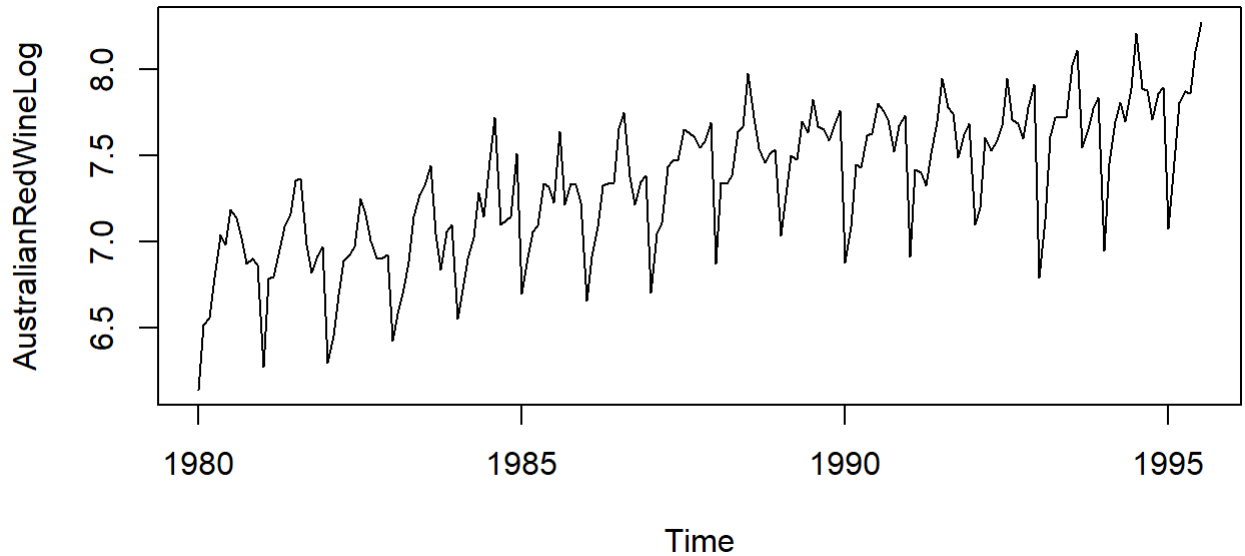
```
library(forecast)
lambda = BoxCox.lambda(AustralianRedWine)
lambda
AustralianRedWineBC = BoxCox(AustralianRedWine, lambda)
plot(AustralianRedWineBC)
```



As stated above, motivated by *parsimony*¹ most people use a simple logarithm to stabilise the variance.

```
AustralianRedWineLog = log(AustralianRedWine)
plot(AustralianRedWineLog)
```

¹Parsimony means using the simplest possible methods to explain a phenomenon. It is also referred to as the rule of Ockham's razor. Google it!



5.2 Revision of expectations, covariances and correlations

We are going to *model* the error term y_t . Modelling means finding random variables from which our data is a sample. So we will revise the main properties of random variables.

5.2.1 One-dimensional random variables

Discrete case

In reality discrete data is not one-dimensional, but rather zero-dimensional. We consider here discrete random variables whose set of values lives in $\mathbb{R}$ and so in some sense are one-dimensional. For simplicity of exposition we consider only the finite case.

A finite discrete random variable, X , can attain a finite number of values $x_1, \dots, x_n$ with probabilities $p_1, \dots, p_n$. These numbers are called the pmf and are often displayed in tabular format:

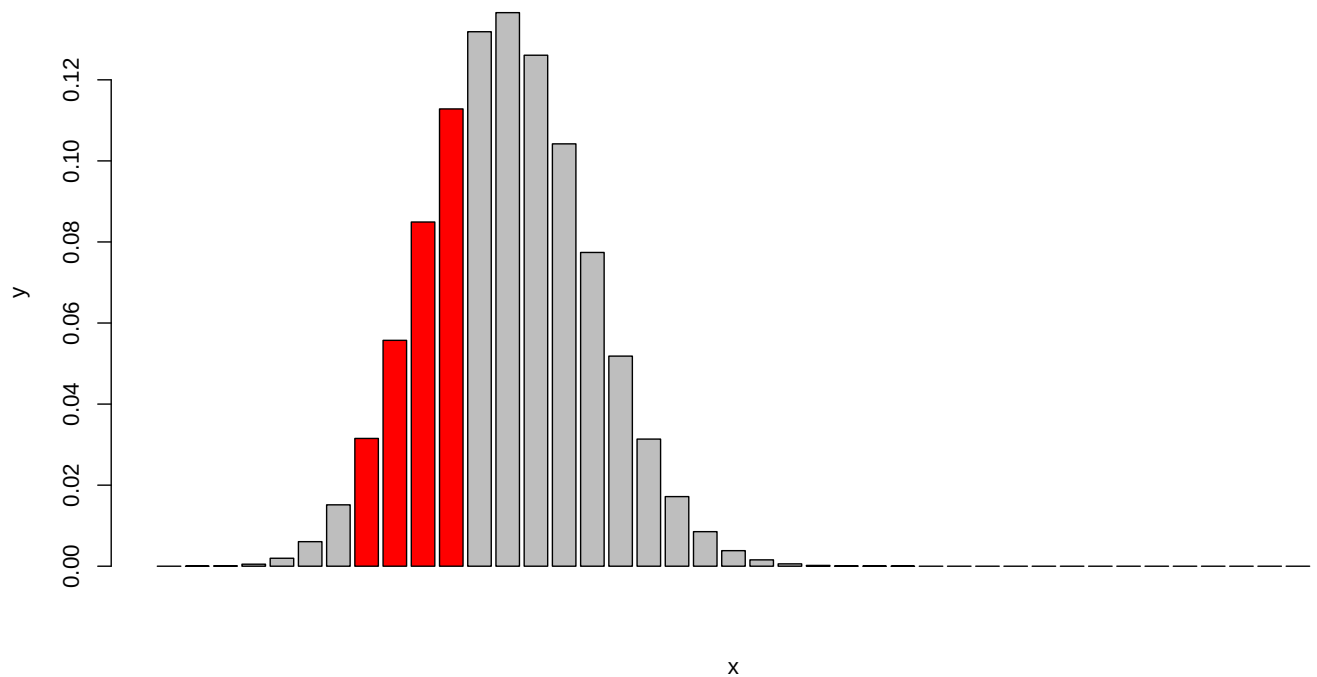
x_1	x_2	$\dots$	x_n
p_1	p_2	$\dots$	p_n

Or with a barplot²

```

2n = 40
p = 0.3
x = 0:n

```



You can then calculate probabilities like $\mathbb{P}(a < X \leq b)$ by adding the relevant p_i 's as in the barplot coloured in red.

The expected value of a function, g , of X is defined as its average, weighted by probability:

$$\mathbb{E}(g(X)) = \sum_{i=1}^n g(x_i)p_i$$

In particular, with the function $g(X) = X$ we get the expected value of X

$$\mathbb{E}(X) = \sum_{i=1}^n x_i p_i$$

And with $g(X) = (X - \mathbb{E}(X))^2$ (notice here $\mathbb{E}(X)$ is a number, not random) we have the variance of X which is the average of the square of deviations of X away from its mean:

```
y = dbinom(x, n, p)
colours = rep("grey", n+1)
colours[8:11]= "red"
barplot(y x, col=colours, axisnames=FALSE )
```

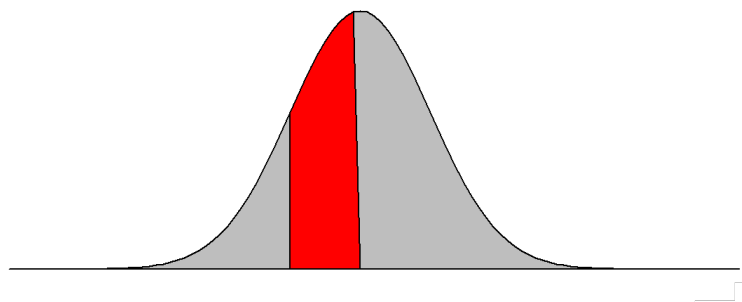
$$\text{Var}(X) = \mathbb{E}((X - \mathbb{E}(X))^2) = \sum_{i=1}^n (x_i - \mathbb{E}(X))^2 p_i = \sum_{i=1}^n x_i^2 p_i - \mathbb{E}(X)^2$$

There are many known discrete random variables for which probabilities,, p_k , are given by certain formulæ: Poisson, Bernoulli, binomial, geometric, negative binomial, ...

Continuous case

A continuous random variable is defined by a function called *probability density function*, f_X which is positive and with total area one. Probabilities are then defined by³

$$\mathbb{P}(a < X < b) = \int_a^b f_X(x) dx$$



```

3mu = 5
sigma = 1
x = seq(0,10, 0.1)
y = dnorm(x, mu, sigma)
plot(x,y, type="l", axes=FALSE)
# Fill area under curve with grey colour
# area defined by polygon (x, y)
polygon(x, y, col = "grey")
# Fill between values with red
Lvalue = 4
Uvalue = 5
polygon(c(Lvalue, x[(x >= Lvalue) & (x < Uvalue)], Uvalue),
c(0, y[(x >= Lvalue) & (x < Uvalue)], 0),
col = "red")
polygon(c(Lvalue, x[(x >= Lvalue) & (x < Uvalue)], Uvalue),
c(0, y[(x >= Lvalue) & (x < Uvalue)], 0),
col = "red", angle=45, density=40, fillOddEven = TRUE)
polygon(c(Lvalue, x[(x >= Lvalue) & (x < Uvalue)], Uvalue),
c(0, y[(x >= Lvalue) & (x < Uvalue)], 0),
col = "orange", angle=45, density=20, fillOddEven = TRUE)

```

The expectation of a function, g , of X , is defined as in the discrete case, as weighted average of the values that $g(X)$ can take as follows

$$\mathbb{E}(g(X)) = \int_{-\infty}^{+\infty} g(x)f_X(x)dx.$$

With $g(X) = X$ and $g(X) = (X - \mathbb{E}(X))^2$ we get the usual expectation and variance

$$\begin{aligned}\mathbb{E}(X) &= \int_{-\infty}^{+\infty} xf_X(x)dx. \\ \text{Var}(X) &= \int_{-\infty}^{+\infty} (x - \mathbb{E}(X))^2 f_X(x)dx.\end{aligned}$$

For the purposes of calculation of probabilities such as $\mathbb{P}(a < X < b)$, to avoid calculating integrals, we often use the *cumulative distribution function* defined as

$$F_X(x) = \int_{-\infty}^x f_X(t)dt$$

Then, the probability above can be written as

$$\mathbb{P}(a < X < b) = F_X(b) - F_X(a)$$

R can calculate F_X for a variety of distributions with the functions `pnorm`, `pexp`, `punif`, `...`

5.2.2 Two dimensional random variables

Discrete case

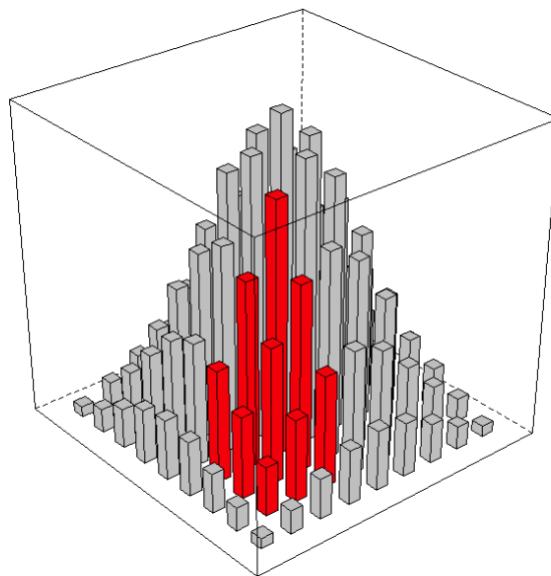
We consider here discrete random variables (which in strict sense should be called zero-dimensional) whose set of values is in $\mathbb{R}^2$ and for simplicity only consider the finite case.

As in the one-dimensional case, a two-dimensional discrete random variable, (X, Y) is defined by giving the set of values it can attain, and the probability of each of these values. This is often arranged in a table as follows:

	x_1	x_2	$\dots$	x_n
y_1	$\mathbb{P}((X, Y) = (x_1, y_1)) = p_{11}$	$\mathbb{P}((X, Y) = (x_2, y_1)) = p_{21}$	$\dots$	$\mathbb{P}((X, Y) = (x_n, y_1)) = p_{n1}$
y_2	$\mathbb{P}((X, Y) = (x_1, y_2)) = p_{12}$	$\mathbb{P}((X, Y) = (x_2, y_2)) = p_{22}$	$\dots$	$\mathbb{P}((X, Y) = (x_n, y_2)) = p_{n2}$
$\vdots$				
y_m	$\mathbb{P}((X, Y) = (x_1, y_m)) = p_{1m}$	$\mathbb{P}((X, Y) = (x_2, y_m)) = p_{2m}$	$\dots$	$\mathbb{P}((X, Y) = (x_n, y_m)) = p_{nm}$

Where all the numbers p_{ij} must be positive and add to one.

You could also display these numbers in a 3D barplot, though this is less common⁴



If you wanted to calculate something like $\mathbb{P}(a < X \leq b, c < Y \leq d)$ you would add all the cases as in the red bars above.

The expectation of a numerical quantity, such as $g(X, Y) = 2X^2 + XY - 1$ is defined, as usual, as the average of all values weighted by probability.

$$\mathbb{E}(g(X, Y)) = \sum_{i=1}^n \sum_{j=1}^m g(x_i, y_j) \mathbb{P}((X, Y) = (x_i, y_j)) = \sum_{i=1}^n \sum_{j=1}^m g(x_i, y_j) p_{ij}$$

```

4library(latticeExtra)
x = seq(-5, 5, 1)
y = seq(-5, 5, 1)
mu = c(0, 0)
sigma = matrix(c(2, -1, -1, 2), nrow = 2)
f = function(x, y) exp(-(x2+y2)/10)
z = outer(x, y, f)
zcol = z
for(i in 1:length(x))
{for(j in 1:length(y))
{if( (x[i] < 0) & (x[i] > -4) & (y[j] < 0) & (y[j] > -4))
{zcol[i,j]="red"}
else
{zcol[i,j]="grey"}}}}
cloud(z, panel.3d.cloud=panel.3dbars, col.facet=zcol, xbase=0.4, ybase=0.4,scales=list(arrows=FALSE,
draw=FALSE), xlab=NULL, ylab=NULL, zlab=NULL)

```

We often write the probabilities of the one dimensional random variables X and Y on the sides. Note that these are just obtained by adding up cases, for example $\mathbb{P}(X = x_1) = \mathbb{P}[(X, Y) = (x_1, y_1)] + \mathbb{P}[(X, Y) = (x_1, y_2)] + \cdots + \mathbb{P}[(X, Y) = (x_1, y_m)]$.

	x_1	x_2	$\cdots$	x_n	
y_1	p_{11}	p_{12}	$\cdots$	p_{1n}	$\mathbb{P}(Y = y_1) = p_{11} + p_{12} + \cdots + p_{1n}$
y_2	p_{21}	p_{22}	$\cdots$	p_{2n}	$\mathbb{P}(Y = y_2) = p_{21} + p_{22} + \cdots + p_{2n}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
y_m	p_{m1}	p_{m2}	$\cdots$	p_{mn}	$\mathbb{P}(Y = y_m) = p_{m1} + p_{m2} + \cdots + p_{mn}$
	$\mathbb{P}(X = x_1)$	$\mathbb{P}(X = x_2)$	$\cdots$	$\mathbb{P}(X = x_n)$	
	$=$	$=$	$\cdots$	$=$	
	$p_{11} + \cdots + p_{m1}$	$p_{12} + \cdots + p_{m2}$	$\cdots$	$p_{1n} + \cdots + p_{mn}$	

These probabilities (which are just the probabilities of X and Y separately) are sometimes called *marginal* probabilities as they live in the margins of the table.

- Remarks 5.1.** 1. Note that by knowing the pmf of each of the random variables X and Y , we do not know the joint behaviour. In the table above, knowing the sums on the margins is not enough to know all the numbers inside the table.
2. However, if X and Y are independent, then $\mathbb{P}((X, Y) = (x_i, y_j)) = \mathbb{P}(X = x_i) \cdot \mathbb{P}(Y = y_j)$. This means the numbers inside the table are just the product of the numbers on the margins. Therefore in this case, it is sufficient to know the pmf of X and Y to understand the random vector (X, Y) .

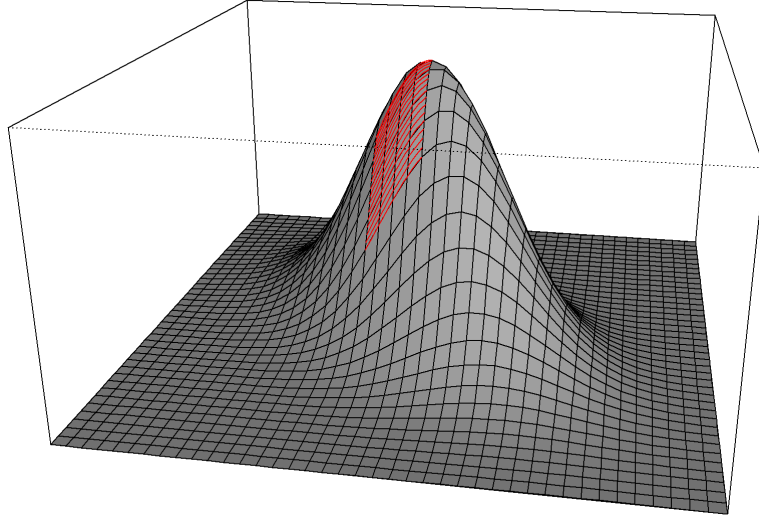
Continuous case

A continuous two-dimensional random variable is defined by its density function $f_{(X,Y)}(x, y)$, which is positive and of total volume 1⁵.

```

5library(mnormt)
x = seq(-5, 5, 0.25)
y = seq(-5, 5, 0.25)
mu = c(0, 0)
sigma = matrix(c(2, -1, -1, 2), nrow = 2)
f = function(x, y) dmnorm(cbind(x, y), mu, sigma)
z = outer(x, y, f)
persp(x, y, z, theta = 10, phi = 20, shade = 0.75, col = "grey", expand = 0.5, r = 2, ltheta =
25, ticktype = "detailed", axes=FALSE)-> res
xselect = seq(-1,0,0.25/4)
yselect = seq(-1,0,0.25/4)
for(i in 1:length(yselect))
points(trans3d(xselect,yselect[i], f(xselect,yselect[i]), res), col = "red", type = "l",lwd = 1)

```



Probabilities are then defined by

$$\mathbb{P}((X, Y) \in [a, b] \times [c, d]) = \int_a^b \int_c^d f_{(X,Y)}(x, y) dx dy$$

I.e. as the volume below the red region in the chart

We will rarely have to calculate any of these integrals as R will take care of this. But it is a good idea to understand as much as possible the abstract bits.

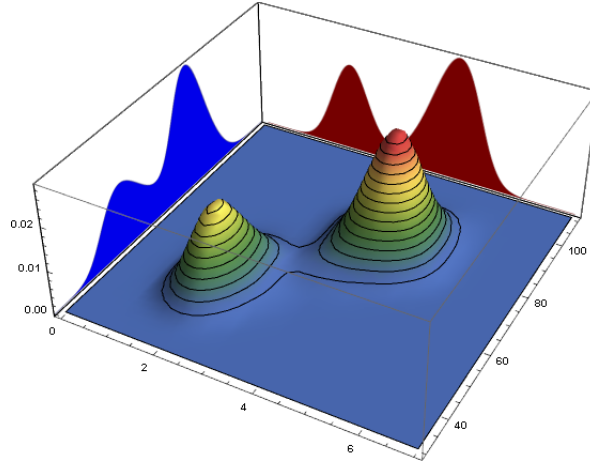
To simplify calculations we use the cumulative distribution function defined as

$$F_{(X,Y)}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{(X,Y)}(t, s) dt ds$$

As in the discrete case we can define the marginals (which are just the pdf's of X and Y) by integrating out one of the variables (analogous to adding up a row or column before)

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{+\infty} f_{(X,Y)}(x, y) dy \\ f_Y(y) &= \int_{-\infty}^{+\infty} f_{(X,Y)}(x, y) dx \end{aligned}$$

The chart below should help your intuition of what marginal pdf means. The surface is a graph of $f_{(X,Y)}(x, y)$ and the functions on either side are the marginals.



As before, if you have a numerical function, g , of (X, Y) , you can calculate its expectation as its average value, weighted by the pdf:

$$\mathbb{E}(g(X, Y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{(X, Y)}(x, y) dx dy$$

- Remarks 5.2.**
1. We rarely have to deal with these integrals as we will use ready-made distributions such as the normal for which R can perform the calculations. However, it is important to understand the theory as it will clarify what you are doing.
 2. As happened with the discrete case, knowing X and Y separately, i.e. knowing the pdfs $f_X(x)$ and $f_Y(y)$ is not sufficient to understand the random vector (X, Y) . The function $f_{(X, Y)}(x, y)$ can be plotted as a surface and it has much more information than the two single variable functions $f_X(x)$ and $f_Y(y)$.
 3. However, when X and Y are independent random variables we have that $f_{(X, Y)}(x, y) = f_X(x)f_Y(y)$.

5.2.3 Higher dimensional distributions

The same kind of definition and calculations can be performed for higher dimensional random vectors. A random vector, $(X_1, \dots, X_n)$, is defined by its pdf which is a function of n variables $f_{(X_1, \dots, X_n)}(x_1, \dots, x_n)$.

In the same fashion we can define cdf, expectation, and the marginal distributions, $f_{X_i}(x_i)$, which are just the pdf's of the one dimensional variables X_i .

Again, note that knowing each of the random variables $X_1, \dots, X_n$ separately, i.e. knowing each of their pdf's, $f_{X_1}(x_1), \dots, f_{X_n}(x_n)$ is much less than knowing their *joint* pdf $f_{(X_1, \dots, X_n)}(x_1, \dots, x_n)$.

5.2.4 Normal random variables

Dimension 1

The Central Limit Theorem states that the sum and the average of many independent identically distributed random variables converges to a normal random variable. As a consequence, many measurements in the real world, that are the sum of many different influences (such as the height of a person) are normally distributed.

A one dimensional random variable, X , is said to be normally distributed if its pdf has the form

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

for certain numbers, μ , and σ called the mean and standard deviation.

As usual, we normally use the cdf, $F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x \exp(-(t - \mu)^2/(2\sigma^2))dt/\sqrt{2\pi\sigma^2}$ which in R is computed as `pnorm(x, mu, sigma)`.

Dimension 2

A two dimensional random variable, (X, Y) , is said to be normal if its pdf is of the form

$$f_{(X,Y)}(x, y) = \frac{1}{\sqrt{(2\pi)^2 \det(\Sigma)}} e^{-\frac{1}{2}(x - \mu_X, y - \mu_Y)\Sigma^{-1}\begin{pmatrix} x - \mu_X \\ y - \mu_Y \end{pmatrix}}$$

For a certain vector⁶ (μ_X, μ_Y) called the mean, and a 2×2 matrix, Σ , called the matrix of variances and covariances.

Sometimes you will read the expression *X and Y are jointly normal*, this just means that the vector (X, Y) is normal as above.

Example 5.1. If the matrix Σ is diagonal

$$\Sigma = \begin{pmatrix} \sigma_X^2 & 0 \\ 0 & \sigma_Y^2 \end{pmatrix}$$

⁶We sometimes write vectors horizontally for ease of typesetting. Recall vectors are vertical. Horizontal vectors are called co-vectors. Vector quiz: if $u = \begin{pmatrix} a \\ b \end{pmatrix}$ and $v = \begin{pmatrix} c \\ d \end{pmatrix}$, calculate $u^T v$ and uv^T . They have different size but same trace. The fact that the trace of AB is the same as that of BA means that the trace is a central function.

Then it is easy to see that the pdf above decomposes as the product of two one-dimensional normal pdf's

$$\begin{aligned} f_{(X,Y)}(x,y) &= \frac{1}{\sqrt{(2\pi)^2 \det(\Sigma)}} e^{-\frac{1}{2}(x-\mu_X, y-\mu_Y) \begin{pmatrix} \sigma_X^{-2} & 0 \\ 0 & \sigma_Y^{-2} \end{pmatrix} \begin{pmatrix} x-\mu_X \\ y-\mu_Y \end{pmatrix}} \\ &= \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}} \cdot \frac{1}{\sqrt{2\pi\sigma_Y^2}} e^{-\frac{(y-\mu_Y)^2}{2\sigma_Y^2}} \end{aligned}$$

Which just means that X and Y are independent and normally distributed.

Higher dimensions

The definition above extends to higher dimensions in a more or less trivial fashion. A random vector $(X_1, \dots, X_n)$ is said to be normal if there exists a vector $(\mu_1, \dots, \mu_n)$, and an $n \times n$ matrix Σ , such that its pdf is

$$f_{(X_1, \dots, X_n)}(x_1, \dots, x_n) = \frac{1}{\sqrt{(2\pi)^n \det(\Sigma)}} e^{-\frac{1}{2}(x_1 - \mu_1, \dots, x_n - \mu_n) \Sigma^{-1} (x_1 - \mu_1, \dots, x_n - \mu_n)^T}$$

5.2.5 Covariances and correlations

For a 2-dimensional random vector (X, Y) with a given pdf $f_{(X,Y)}(x, y)$, its covariance is defined to be

$$\text{Cov}(X, Y) = \mathbb{E}((X - \mathbb{E}(X))(Y - \mathbb{E}(Y)))$$

Remarks 5.3. 1. A trivial consequence of the definition is that $\text{Cov}(X, X) = \text{Var}(X)$.

2. Another trivial property is symmetry: $\text{Cov}(X, Y) = \text{Cov}(Y, X)$.
3. Another standard fact is that you can calculate covariance as $\mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$. Prove it!⁷
4. As a consequence of the last point, if X and Y are independent, then $\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) = 0$.
5. It is also easy to see that Cov is linear in both sides. Let us see it is linear in the second parameter:

$$\begin{aligned} \text{Cov}(X, aY_1 + bY_2) &= \mathbb{E}(X(aY_1 + bY_2)) - \mathbb{E}(X)\mathbb{E}(aY_1 + bY_2) \\ &= a\mathbb{E}(XY_1) - +b\mathbb{E}(XY_2) - a\mathbb{E}(X)\mathbb{E}(Y_1) - b\mathbb{E}(X)\mathbb{E}(Y_2) \\ &= a\text{Cov}(X, Y_1) + b\text{Cov}(X, Y_2) \end{aligned}$$

Prove it for the first parameter, $\text{Cov}(aX_1 + bX_2, Y)$. And for both at the same time $\text{Cov}(aX_1 + bX_2, cY_1 + dY_2)$.

⁷... = $[(\lambda)\mathbb{E}(X)\mathbb{E}] + \lambda(X)\mathbb{E} - (\lambda)\mathbb{E}X - \lambda X] \mathbb{E} = [((\lambda)\mathbb{E} - \lambda)((X)\mathbb{E} - X)] \mathbb{E}$

6. A particular consequence of the properties above is the well known formula:

$$\begin{aligned}\text{Var}(X + Y) &= \text{Cov}(X + Y, X + Y) = \text{Cov}(X, X) + \text{Cov}(X, Y) + \text{Cov}(X, Y) + \text{Cov}(Y, Y) \\ &= \text{Var}(X) + 2\text{Cov}(X, Y) + \text{Var}(Y)\end{aligned}$$

7. The following provides some intuition for the expression above.

(a) If our random vector (X, Y) is such that

- when X ends up being above average, then also Y tends being above its average, and
- when X ends up being below average, then also Y tends being below its average, and

Then the random number $(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))$ will tend to be a product of positive times positive, or negative times negative. Therefore, in this case the covariance will be positive.

(b) Conversely, if (X, Y) is such that

- when X ends up being above average, then Y tends to be below its average, and
- when X ends up being below average, then Y tends to be above its average,

then the random number $(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))$ will tend to be a product of positive times negative, or negative times positive. Therefore, in this case the covariance will be negative.

8. It makes no sense to calculate the covariance if you only have the single pdf's, f_X and f_Y , and not the joint probability density, $f_{(X,Y)}$.

This can be understood with a statistical example.

- If you record all heights and weights of a bunch of students in a school as (x_i, y_i) then you can draw a two-dimensional scatter plot and compute the covariance and correlation between heights and weights.
- However, if you are just given a list of heights of students in one school $x_1, \dots, x_n$ and a list of weights of students in another school $y_1, \dots, y_n$. Then you cannot calculate any covariance as you only have statistics of these two separate quantities.

9. Covariances are numbers roughly in the same scale as XY so if your numbers are big, then $\text{Cov}(X, Y)$ will also be big. If X and Y are measured in cm then Cov will be in cm^2 . It is useful to come up with a rescaled number that always has the same range of magnitudes and no unit. We define the correlation as

$$\text{Cor}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}.$$

Correlation is also noted ρ_{XY} .

10. It can be proved that $\text{Cor}(X, Y) \in [-1, +1]$. To see this it is enough to check that the square of this number is below 1:

$$\text{Cor}(X, Y)^2 = \frac{\text{Cov}(X, Y)^2}{\text{Var}(X)\text{Var}(Y)} \leq 1$$

This is proved with the following trick. We know that variances are always positive, so for any value of z , $\text{Var}(X + zY) \geq 0$. But if we expand this we get a quadratic in z that must always be positive

$$\text{Var}(X) + 2z\text{Cov}(X, Y) + z^2\text{Var}(Y) \geq 0$$

But this means that the quadratic above has no zeroes. And from high school we know this happens when the discriminant (the expression inside the square root in the solution to the quadratic $(-b \pm \sqrt{b^2 - 4ac})/(2a)$) is less than zero⁸. In our case this means

$$4\text{Cov}(X, Y)^2 - 4\text{Var}(X)\text{Var}(Y) \leq 0$$

If we now divide by 4 and by $\text{Var}(X)\text{Var}(Y)$ we get

$$\rho_{XY}^2 = \frac{\text{Cov}(X, Y)^2}{\text{Var}(X)\text{Var}(Y)} \leq 1$$

as we wished to prove.

11. If (X, Y) is bivariate normal with mean (μ_X, μ_Y) and covariance matrix Σ it can be seen that this covariance matrix actually contains the variances and covariances of X and Y . So

$$\Sigma = \begin{pmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(Y, X) & \text{Var}(Y) \end{pmatrix}$$

To see this, one must calculate some integrals such as

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy \frac{1}{\sqrt{(2\pi)^2 \det(\Sigma)}} e^{-\frac{1}{2}(x - \mu_X, y - \mu_Y)\Sigma^{-1}\begin{pmatrix} x - \mu_X \\ y - \mu_Y \end{pmatrix}}$$

we will not do this here.

12. As a consequence of the last point. If (X, Y) is bivariate normal and $\text{Cov}(X, Y) = 0$ then we are in the diagonal case described in Example 5.1 and so X and Y are independent.

This is often phrased as ‘*uncorrelated normal random variables are independent*’, though this is a bit imprecise.

13. It is not true that uncorrelated random variables are independent. The easiest example is as follows. Take $X \sim N(0, 1)$ and Z an independent random variable with $\mathbb{P}(Z = +1) = 1/2$ and $\mathbb{P}(Z = -1) = 1/2$. Then take $Y = ZX$. It is easy to see that X and Y are not independent because when X is, say 3.2, then Y can only be +3.2 or -3.2. But their covariance and correlation is zero:

$$\begin{aligned} \text{Cov}(X, Y) &= \text{Cov}(X, ZX) = \mathbb{E}(X^2Z) - \mathbb{E}(X)\mathbb{E}(XZ) \\ &= \mathbb{E}(X^2)\mathbb{E}(Z) - \mathbb{E}(X)^2\mathbb{E}(Z) = 0 \end{aligned}$$

Where the last line is a result of independence and $\mathbb{E}(Z) = 0$.

⁸If $b^2 - 4ac > 0$ you can then find two different roots of the quadratic equation $az^2 + bz + c = 0$. Therefore, the graph of the parabola $y = az^2 + bz + c$ will change from positive to negative when z crosses these two roots. If $az^2 + bz + c$ is never negative then it must be that there are no two roots, i.e. $b^2 - 4ac \leq 0$.

14. It is not even true that uncorrelated normal random variables are independent. In the example above the random variable Y is also normal, so both X and Y are normal, with zero correlation but not independent. To see Y is normal we check its cdf is the cdf of a normal random variable:

$$\begin{aligned}\mathbb{P}(Y \leq y) &= \mathbb{P}(XZ \leq y) = \frac{1}{2}\mathbb{P}(X \leq y) + \frac{1}{2}\mathbb{P}(-X \leq y) \\ &= \frac{1}{2}\mathbb{P}(X \leq y) + \frac{1}{2}\mathbb{P}(X \geq -y) = \mathbb{P}(X \leq y) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^y e^{-t^2/2} dt\end{aligned}$$

Where the penultimate equality results from the symmetry of the $N(0, 1)$ pdf.

15. What is true is that for (X, Y) bivariate normal uncorrelated is the same as independent.

5.2.6 Partial correlation

Consider the following situation:

- X is a random variable measuring the reading ability of a random individual within a population.
- Y is a random variable modelling the height in cm of a random individual.
- Z stands for the age.

Depending on our population, it could easily happen that there is a substantial correlation between X and Y . This most likely has nothing to do with an influence of height on reading ability, but rather than, our population might include children, so that there will be a positive correlation between both height and reading ability with age, Z .

So, $\rho_{XY} = \text{Cor}(X, Y)$ somehow will include positive correlation that is simply a consequence of both of them being positively correlated with Z , and nothing to do with a direct connection between x and Y .

A way of removing the effect of Z is using *partial correlation*, noted $\rho_{XY \cdot Z}$. In the example above, this could be thought as computing the correlation between reading ability (X) and height (Y) keeping age (Z) fixed.

Homework 5.1. The following steps will guide you through the definition and calculation of the partial correlation.

1. Find the number a such that $X' = X - aZ$ has zero covariance with Z . (Hint: just equate $\text{Cov}(X - aZ, Z) = 0$, a will be a number depending on quantities like $\text{Var}(Z)$, $\text{Cov}(X, Z)$...)

2. Find the number b such that $Y' = Y - aZ$ has zero covariance with Z .
3. These new random variables (with the values of a and b found above) are understood as versions of X and Y with the effects of Z subtracted away. The partial correlation, $\rho_{XY \cdot Z}$, is defined to be the correlation between X' and Y' .
4. Calculate the variance of X' .
5. Calculate the variance of Y' .
6. Calculate the covariance of X' and Y' .
7. Recall that the partial correlation of X and Y with respect to Z , $\rho_{XY \cdot Z}$, is defined as the correlation of X' and Y' . Calculate this and simplify it to an expression involving the correlations of the initial variables: ρ_{XY} , ρ_{XZ} , and ρ_{YZ} .

5.3 Weakly stationary process

We will now return to dealing with the noise component of our time series.

5.3.1 Random processes

In order to describe its properties we will model it using random processes.

Definition 5.1. A *random process* is a sequence of random variables

$$X_1, X_2, \dots, X_n$$

or alternatively a random vector $(X_1, X_2, \dots, X_n)$. It is therefore defined by its pdf $f_{(X_1, X_2, \dots, X_n)}$ or cdf

$$F_{(X_1, X_2, \dots, X_n)}(x_1, \dots, x_n) = \mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n).$$

Remarks 5.4. 1. We will also consider the case of infinite random processes

$$X_1, X_2, \dots, X_n, \dots$$

The maths can become a bit more complicated. For example the pdf would become a function in infinitely many variables. We will not deal with these technicalities.

2. A time series will be a sample of a random process. We encounter here the typical paradigm in statistics:
 - we construct certain random variables that reflect reality (probability),
 - and then take real world data and try to infer properties or parameters of these random variables (statistics).

As usual, we will be using the mean and covariance structure of a random vector. In this context we will use the following notation.

Definition 5.2. Let X_t be a random process The *mean function* of X_t is

$$\mu(t) = \mathbb{E}(X_t).$$

The *covariance function* of X_t is

$$\gamma(r, s) = \text{Cov}(X_r, X_s) = \mathbb{E}((X_r - \mu(r))(X_s - \mu(s))),$$

for all integers r and s .

5.3.2 Stationary processes

The X_t above is meant to model our de-trended and de-seasonalized time series. So, we would not expect the mean or variance of X_t to change with time as this should have been captured by m_t and s_t . This idea gives rise to the notion of a stationary process as a process that is *time invariant*. If you were to look at the random variables

$$(X_a, \dots, X_b)$$

then you should not notice any difference some time ahead

$$(X_{a+t}, \dots, X_{b+t}).$$

Definition 5.3. A random process X_t is said to be *stationary* if the pdf of the random vector

$$(X_a, \dots, X_b)$$

is the same as that of the vector

$$(X_{a+t}, \dots, X_{b+t})$$

for all a, b and t such that the random variables are defined.

This definition is fine but it is a bit difficult to work with in practice because it requires us to work with the pdf's of these random vectors which will be complicated functions.

A weaker but more practical definition is as follows:

Definition 5.4. A random process X_t is *weakly stationary* if

- $\mu(t) = \mathbb{E}(X_t)$ is independent of t , and
- $\gamma(t+h, t) = \text{Cov}(X_{t+h}, X_t)$ is independent of t for each h

Chapter 6

Properties of stationary processes

6.1 A stationary process is weakly stationary

Recall that a *stationary process* is a process for which the random vector $(X_{t_1}, \dots, X_{t_n})$ has the same distribution as the random vector $(X_{t_1+s}, \dots, X_{t_n+s})$ for any value of $n, t_1, \dots, t_n, s$.

Remarks 6.1. 1. As a particular case, the pdf of X_t will be the same as the pdf of X_{t+s} for any t and s . And in particular $\mathbb{E}(X_t) = \mathbb{E}(X_{t+s})$. Recall that we had defined the mean function of the process to be $\mu_X(t) = \mathbb{E}(X_t)$. So that for a stationary process:

$$\mu_X(t) \text{ is constant}$$

For simplicity we usually assume $\mu_X = 0$.

2. If a process is stationary, then the pdf of the vector (X_{t_1}, X_{t_2}) will be the same as the pdf of (X_{t_1+s}, X_{t_2+s}) for any t_1, t_2 and s . In particular $\text{Cov}(X_{t_1}, X_{t_2}) = \text{Cov}(X_{t_1+s}, X_{t_2+s})$. Recall that we defined the covariance function of the process to be $\gamma_X(t_1, t_2) = \text{Cov}(X_{t_1}, X_{t_2})$. So that for a stationary process:

$$\gamma_X(t_1, t_2) = \gamma_X(t_1 + s, t_2 + s) \text{ for all } t_1, t_2, s.$$

In particular, taking $s = t_2 - t_1$ this means that

$$\gamma_X(t_1, t_2) = \gamma_X(0, t_2 - t_1)$$

$t_2 - t_1$ is called the *lag* and we will often note it by h . We will also simplify $\gamma_X(0, h)$ to $\gamma_X(h)$. So the autocovariance function of a stationary process is defined by $\gamma_X(h)$.

3. The two points above prove that a stationary process is also a weakly stationary process. However, to be fully rigorous, this is only the case if the expectations and covariances are actually defined. There are random variables without expectation or variance, for instance the

Cauchy distribution with pdf $f_X(x) = \frac{1}{\pi(1+x^2)}$ does not have expectation or variance because its tails are *fat* (i.e. it decays too slowly at infinity)

$$\begin{aligned}\mathbb{E}(X) &= \int_{-\infty}^{+\infty} x \frac{1}{\pi(1+x^2)} dx = \text{undefined} \\ \mathbb{E}(X^2) &= \int_{-\infty}^{+\infty} x^2 \frac{1}{\pi(1+x^2)} dx = \infty\end{aligned}$$

The first integral is undefined as the integrand behaves like $1/x$ which is not integrable at infinity (its integral is $\ln(x)$ which does not go to zero at infinity). The integrand of the second integral behaves asymptotically as $1/\pi$ which does not go to zero and therefore the integral is infinite.

With this, we could cook up a stationary process made of Cauchy random variables, for which $\mu_X(t)$ and $\gamma_X(t_1, t_2)$ are not defined, and so the process would not be weakly stationary.

So in full rigour: a stationary process with first and second moments is weakly stationary.

4. We will mostly focus on weakly stationary processes and will just use the word *stationary* to refer to a weakly stationary process if it is clear from the context. We sometimes refer to stationary processes as *strictly stationary* if we want to emphasize the difference with weakly stationary.
5. A trivial observation is that for a weakly stationary process

$$\gamma(0) = \text{Var}(X_t)$$

is independent of t . In particular a weakly stationary process is homoskedastic.

6. We will often omit the subindex X from $\mu_X(t)$ and $\gamma_X(h)$ and just write $\mu(t)$ and $\gamma(h)$ if there is no risk of confusion.

6.2 White noise

Our first example of stationary process is white noise.

Definition 6.1. White noise is the process X_t for which X_t are independent normal random variable with mean zero and constant variance σ^2 .

Proposition 6.1. White noise is stationary and weakly stationary with $\mu(t) = 0$, $\gamma(0) = \sigma^2$ and $\gamma(h) = 0$ for lags, h , greater than zero.

Proof. The pdf of X_t is the pdf of a $N(0, \sigma^2)$ random variable:

$$f_{X_t}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}}$$

and since X_t for different t are independent, the pdf of the random vector $(X_{t_1}, \dots, X_{t_n})$ is just the product of such functions

$$f_{(X_{t_1}, \dots, X_{t_n})}(x_1, \dots, x_n) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x_1^2}{2\sigma^2}} \dots \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x_n^2}{2\sigma^2}}$$

Since none of these functions depend on time the process is stationary.

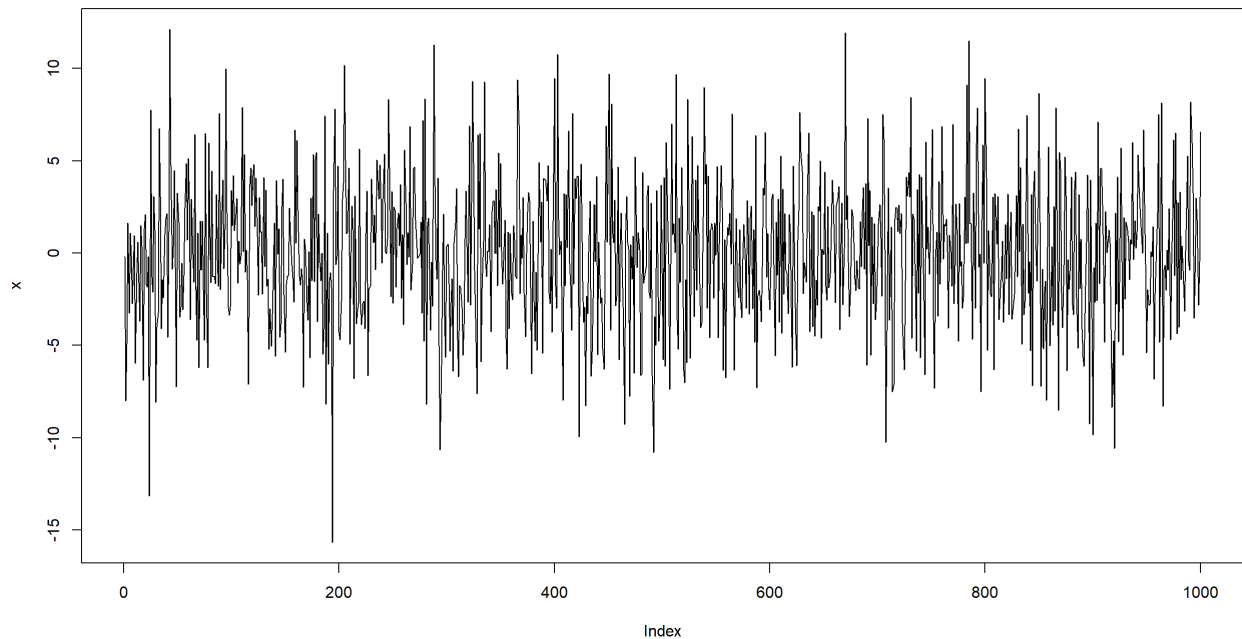
The mean of $X_t \sim N(0, \sigma^2)$ is zero and the covariance function is

$$\gamma(h) = \text{Cov}(X_t, X_{t+h})$$

which is zero for $h > 0$ as then X_t and X_{t+h} are independent. And $\gamma(0) = \text{Var}(X_t) = \sigma^2$. □

It is easy to simulate white noise in R:

```
sigmaSquared = 4
x = rnorm(1000, 0, sigmaSquared)
plot(x, type="l")
```



Remarks 6.2. 1. On the term *white noise*:

- *Noise* refers to the fact that a sound wave with that shape sounds like noise, see [White Noise Black Screen — Sleep, Study, Focus — 10 Hours](#).

- *White* alludes to white light, as in a mathematical sense (Fourier Analysis) white noise contains all the frequencies. The colour white also contains a mix of all the different electromagnetic frequencies corresponding to all colours hence the name *white*.

2. Other than as an example, white noise is important for the following reasons (amongst others):

- It is used as a building block for other processes. For example, random walk and moving average processes which we will shortly discuss are based on white noise.
- It represents the time series with the *least predictability possible*. If something is white noise then the next observation is independent from all the previous so we have no idea what it might be.

When we build models for forecasting time series we will be left with residuals. Once the residual is white noise then there is no more forecasting to be done, we have *squeezed* all predictability out of the time series.

6.3 Random walk

Another interesting process is random walk. It can be constructed in the following fashion.

Definition 6.2. Take

$$Z_0, Z_1, Z_2, \dots$$

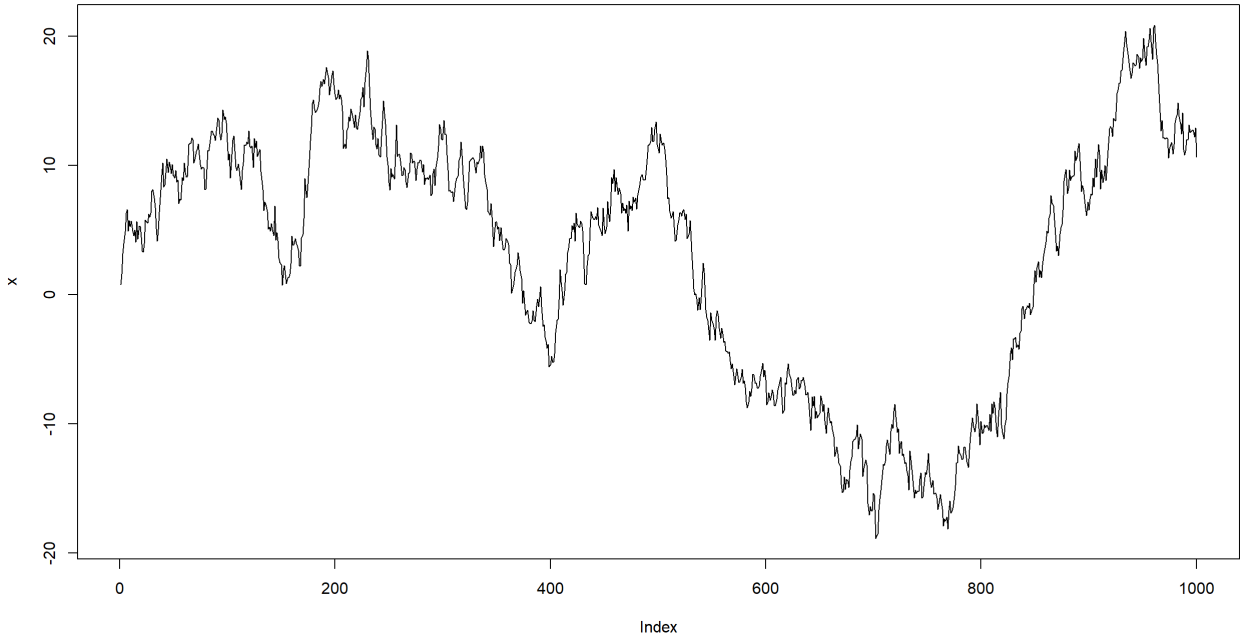
to be independent $N(0, 1)$ random variables and define random walk as the following process:

$$\begin{aligned} X_0 &= Z_0 \\ X_1 &= Z_0 + Z_1 \\ X_2 &= Z_0 + Z_1 + Z_2 \\ X_3 &= Z_0 + Z_1 + Z_2 + Z_3 \\ &\vdots \end{aligned}$$

Remarks 6.3. 1. There are other, similar definitions of random walk. Sometimes Z_i are taken to have values in $\{+1, -1\}$ with probability $1/2$. Sometimes a drift component can be added.

2. It is easy to simulate this in R using the function `cumsum` which stands for cumulative sum.

```
z = rnorm(1000)
x = cumsum(z)
plot(x, type="l")
```



3. Random walk is also called *Brownian motion*, after the botanist Robert Brown who first observed similar motions in pollen particles suspended in water¹.

Is random walk stationary? A simple calculation disproves this:

$$\text{Var}(X_t) = \text{Var}(Z_1 + Z_2 + \cdots + Z_t) = \text{Var}(Z_1) + \text{Var}(Z_2) + \cdots + \text{Var}(Z_t) = t$$

and this is not independent of t .

6.4 Properties of the autocovariance function

Recall that the autocovariance function (sometimes noted ACF) of a stationary process, X_t , is defined as

$$\gamma_X(h) = \text{Cov}(X_t, X_{t+h})$$

where h is called the lag and the function above does not depend on t as the process is stationary.

Definition 6.3. The autocorrelation function (ACF) of a weakly stationary process, X_t , is defined as

$$\rho_X(h) = \frac{\gamma(h)}{\gamma(0)}.$$

¹Brownian motion was subsequently explained as the result of random hits by water molecules on the pollen particle, similar to the way new information can 'hit' the price of an asset.

Remark 6.1. Since $\gamma(h) = \text{Cov}(X_t, X_{t+h})$ and $\gamma(0) = \text{Var}(X_t) = \text{Var}(X_{t+h})$, $\rho_X(h)$ is just the correlation between X_t and X_{t+h} , $\text{Cov}(X_t, X_{t+h})/\sqrt{\text{Var}(X_t)\text{Var}(X_{t+h})}$.

Proposition 6.2. We have the following properties of the autocovariance and autocorrelation functions:

1. $\gamma(0) \geq 0$.
2. $\rho(0) = 1$ and $\rho(h) \in [-1, +1]$.
3. $\gamma(-h) = \gamma(h)$ and $\rho(-h) = \rho(h)$.

As the proposition above states that both γ_X and ρ_X are even functions, we will typically only consider positive values of the lag h .

Proof. 1. $\gamma_X(0) = \text{Var}(X_t)$ and this is greater than or equal to zero because the variance of a random variable is the expected value of something positive $\mathbb{E}((X_t - \mathbb{E}(X_t))^2)$.

2. $\rho(0) = 1$ because it is defined as $\gamma(0)/\gamma(0)$ or alternatively because it is the correlation between the random variable, X_t , and itself.

$\rho(h) \in [-1, +1]$ because is the correlation between X_t and X_{t+h} and correlations are always between -1 and +1².

3. This follows from the symmetry property of the covariance remarked in the previous chapter: $\text{Cov}(X, Y) = \text{Cov}(Y, X)$, together with stationarity as follows:

$$\begin{aligned}
 \gamma(-h) &= \text{Cov}(X_t, X_{t-h}) && \text{definition} \\
 &= \text{Cov}(X_{t+h}, X_t) && \text{stationarity} \\
 &= \text{Cov}(X_t, X_{t+h}) && \text{symmetry of Cov} \\
 &= \gamma(h) && \text{definition}
 \end{aligned}$$

□

6.5 Sample autocorrelation and autocovariance functions

In statistical problems, we do not start with a model but with the observed data. We use these data to compute the sample ACF. The precise definition of the sample ACF is as follows:

²This was proved in the previous chapter.

Definition 6.4. Let $x_1, x_2, \dots, x_n$ be our time series. The *sample mean* is

$$\bar{x} = \frac{1}{n} \sum_{t=1}^n x_t.$$

The *sample autocovariance function (ACVF)*, for $0 \leq h < n$, is

$$\hat{\gamma}(h) = \frac{1}{n} \sum_{t=1}^{n-h} (x_{t+h} - \bar{x})(x_t - \bar{x})$$

The *sample autocorrelation function (ACF)* is

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}.$$

Remarks 6.4. 1. Note that the definition of sample autocovariance at lag h is similar to the sample correlation between

$$(x_1, \dots, x_{n-h}) \text{ and } (x_{1+h}, \dots, x_n),$$

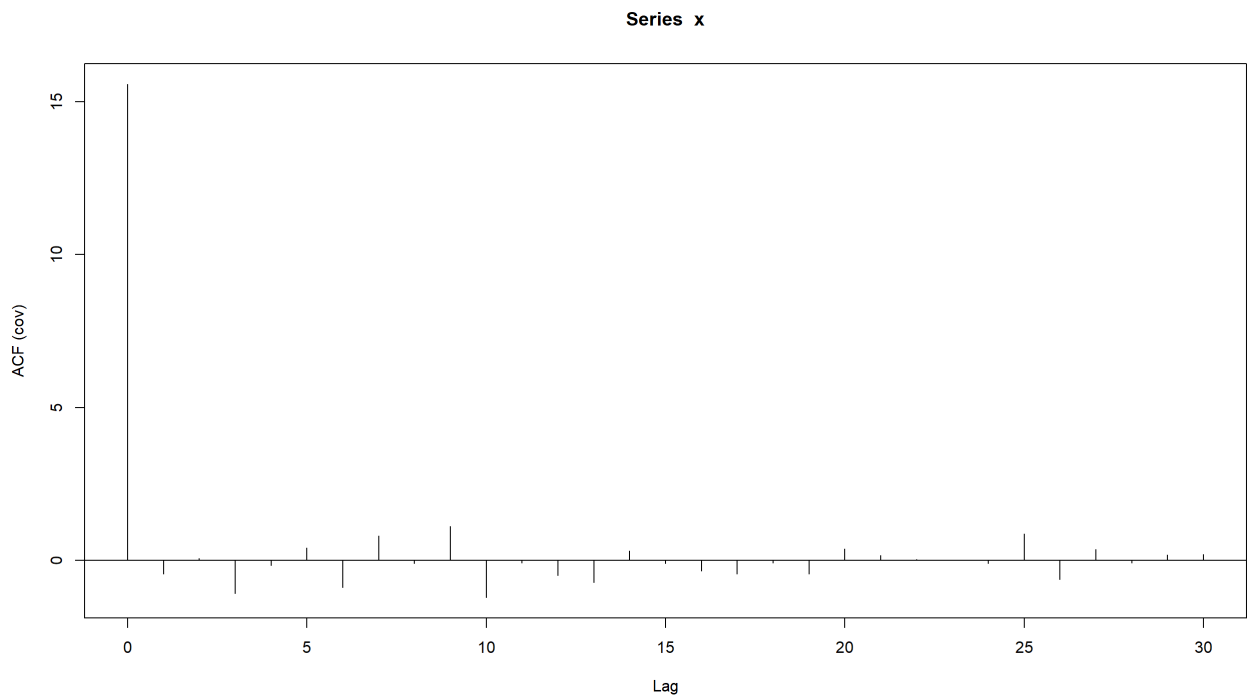
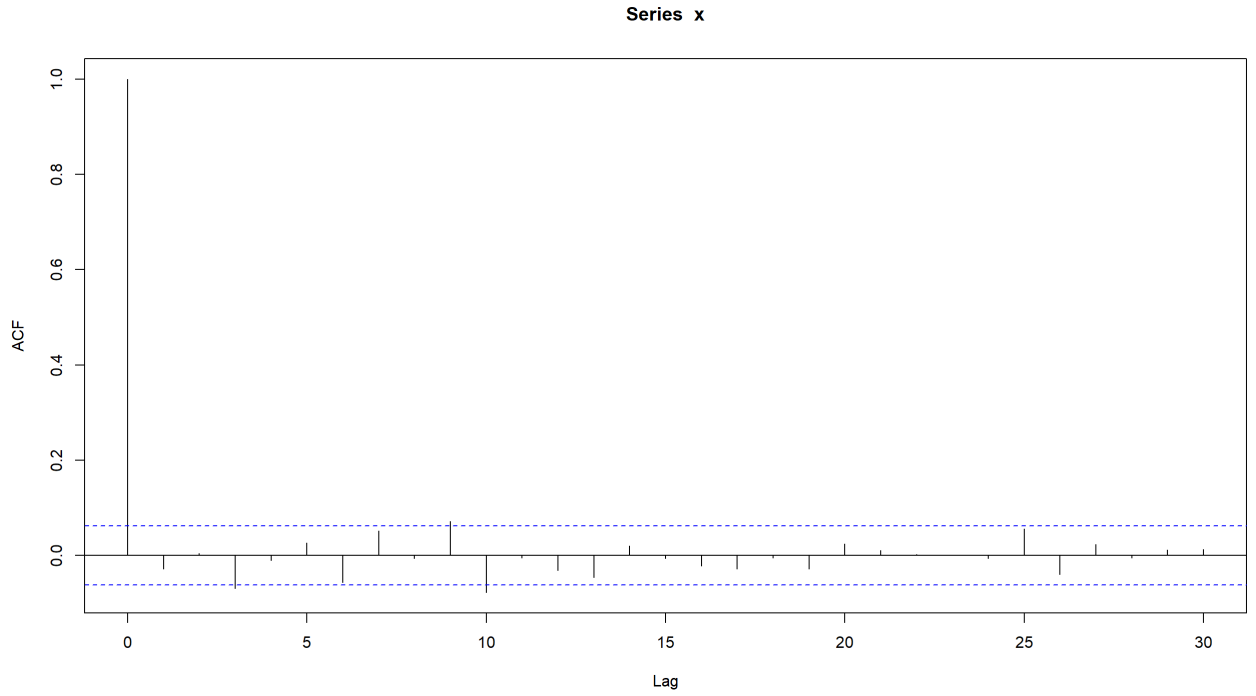
but not exactly the same. The sample correlation of two vectors $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ is defined as

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

The autocovariance function is different in that:

- (a) The mean used, $\bar{x}$, is not the mean of each of the two vectors, $(x_1, \dots, x_{n-h})$ and $(x_{h+1}, \dots, x_n)$ as we would use in the sample correlation, but instead the mean of the whole time series.
 - (b) The denominator n is not the size of the vectors we are correlating ($n - h$). It is defined this way to ensure certain mathematical properties that we will not discuss (positive definiteness of the sample autocovariance matrix).
2. $\hat{\gamma}(h)$ and $\hat{\rho}(h)$ are estimators of $\gamma(h)$ and $\rho(h)$ respectively.
 3. The sample autocovariance and sample autocorrelation can be computed in R as follows using the `acf` command.

```
sigmaSquared = 4
x = rnorm(1000, 0, sigmaSquared)
plot(x, type="l")
acf(x) # the default is autocorrelation
acf(x, type="covariance")
```



As expected, for white noise all numbers except for lag $h = 0$ are close to zero.

4. Both charts above are very similar because they are simply rescaled by $\hat{\gamma}(0)$.

Theorem 6.1. Under general circumstances, if $X_1, \dots, X_n$ is a white noise process, then the sampled autocorrelation, $\rho_X(h)$ can be approximated by a normal distribution with zero mean and variance

$1/n$.

Therefore for a white noise process, 95% of the time the ACF should be in the confidence interval $\pm 1.96/\sqrt{n}$. These are the two blue lines displayed by the `acf` function in R. Remember that 1 out of 20 times (5%) you will see autocorrelations bigger than this, by chance, even if there is no real autocorrelation.

6.6 Moving Average process of order 1

Another example of stationary process can be obtained by mixing past values of white noise.

Definition 6.5. Let Z_t for $t \in \mathbb{Z}$ be independent $N(0, \sigma^2)$ random variables, in other words, white noise. Define X_t as follows

$$X_t = Z_t + \theta Z_{t-1}$$

Then X_t is defined to be *moving average process of order 1*, $MA(1)$, with parameters θ and σ^2 .

Proposition 6.3. A moving average process of order 1 is stationary.

Proof. Let X_t be a moving average process of order 1 with parameters θ and σ^2 .

Let us verify that X_t is weakly stationary³. To see this we just need to confirm that the mean, $\mu(t) = \mathbb{E}(X_t)$ is constant, and that the autocovariance function $\gamma_X(t_1, t_2)$ only depends on the lag $h = t_2 - t_1$.

- The mean of X_t is

$$\mu(t) = \mathbb{E}(X_t) = \mathbb{E}(Z_t + \theta Z_{t-1}) = 0$$

as the Z_t have mean zero.

- From the definition of X_t the autocovariance function is

$$\begin{aligned} \gamma_X(t_1, t_2) &= \text{Cov}(X_{t_1}, X_{t_2}) = \\ &= \text{Cov}(Z_{t_1} + \theta Z_{t_1-1}, Z_{t_2} + \theta Z_{t_2-1}) \\ &= \text{Cov}(Z_{t_1}, Z_{t_2}) + \text{Cov}(Z_{t_1}, \theta Z_{t_2-1}) + \text{Cov}(\theta Z_{t_1-1}, Z_{t_2}) + \text{Cov}(\theta Z_{t_1-1}, \theta Z_{t_2-1}) \\ &= \text{Cov}(Z_{t_1}, Z_{t_2}) + \theta \text{Cov}(Z_{t_1}, Z_{t_2-1}) + \theta \text{Cov}(Z_{t_1-1}, Z_{t_2}) + \theta^2 \text{Cov}(Z_{t_1-1}, Z_{t_2-1}) \end{aligned}$$

But since all Z_t are independent, this will be zero unless some of subindices of Z_t coincide:

- $t_1 = t_2$, or

³You can also verify that it is strictly stationary.

- $t_1 = t_2 - 1$, or
- $t_1 - 1 = t_2$, or
- $t_1 - 1 = t_2 - 1$, or

Therefore the only non-zero autocovariances are:

$$\begin{aligned}
\gamma_X(t_1, t_1) &= \text{Cov}(Z_{t_1}, Z_{t_1}) + \theta \text{Cov}(Z_{t_1}, Z_{t_1-1}) \theta \text{Cov}(Z_{t_1-1}, Z_{t_1}) \theta^2 \text{Cov}(Z_{t_1-1}, Z_{t_1-1}) \\
&= \sigma^2 + \theta^2 \sigma^2 = (1 + \theta^2) \sigma^2 \\
\gamma_X(t_1, t_1 + 1) &= \text{Cov}(Z_{t_1}, Z_{t_1+1}) + \theta \text{Cov}(Z_{t_1}, Z_{t_1}) \theta \text{Cov}(Z_{t_1-1}, Z_{t_1+1}) \theta^2 \text{Cov}(Z_{t_1-1}, Z_{t_1}) \\
&= \theta \sigma^2 \\
\gamma_X(t_1, t_1 - 1) &= \text{Cov}(Z_{t_1}, Z_{t_1-1}) + \theta \text{Cov}(Z_{t_1}, Z_{t_1-2}) \theta \text{Cov}(Z_{t_1-1}, Z_{t_1-1}) \theta^2 \text{Cov}(Z_{t_1-1}, Z_{t_1-2}) \\
&= \theta \sigma^2
\end{aligned}$$

And these quantities do not depend on t_1, t_2 other than through the lag $h = t_2 - t_1$ so the process is weakly stationary. $\square$

From the last part of the proof of the previous proposition we can deduce the following corollary.

Corollary 6.1. The autocovariance function of a $MA(1)$ process with parameters θ and σ^2 is

$$\gamma_X(h) = \begin{cases} (1 + \theta^2) \sigma^2 & \text{if } h = 0 \\ \theta \sigma^2 & \text{if } h = \pm 1 \\ 0 & \text{otherwise} \end{cases}$$

Its autocorrelation function is

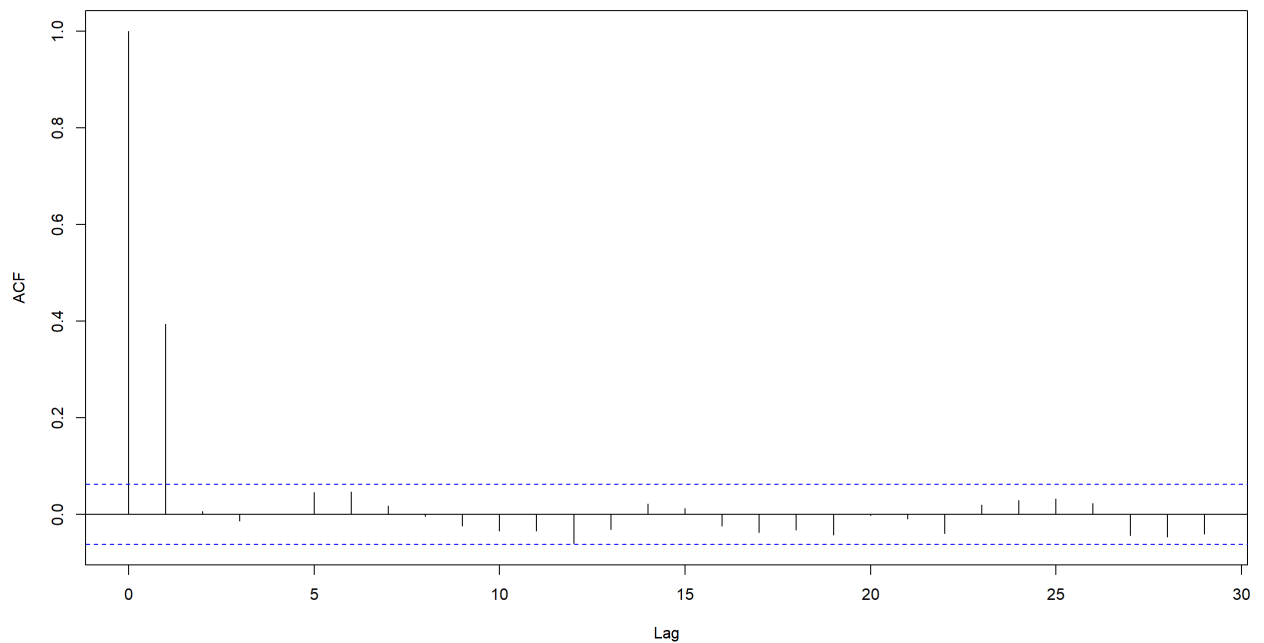
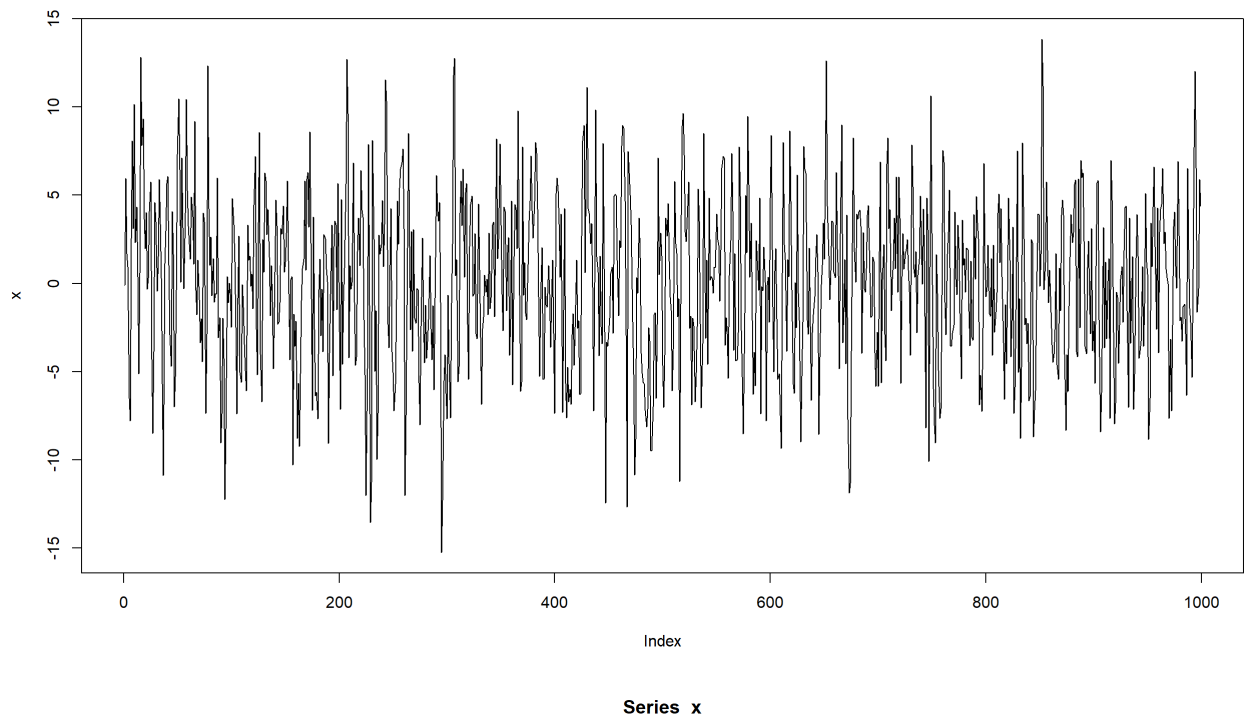
$$\rho_X(h) = \begin{cases} 1 & \text{if } h = 0 \\ \frac{\theta}{1 + \theta^2} & \text{if } h = \pm 1 \\ 0 & \text{otherwise} \end{cases}$$

We can also simulate an $MA(1)$ process as follows

```

sigmaSquared = 4
theta = 0.5
z = rnorm(1000, 0, sigmaSquared)
head(z[-1]) # this is the vector with the first element removed
tail(z[-1]) # this is the vector with the first element removed
head(z[-length(z)]) # this is the vector with the last element removed
tail(z[-length(z)]) # this is the vector with the last element removed
x = z[-length(z)] + theta*z[-1]
plot(x, type="l")
acf(x)

```



Notice that as expected, we can see the autocorrelation at lag 1. Its theoretical value is $\rho_X(1) = \theta/(1 + \theta^2) = 0.5/(1 + 0.25) = 0.4$, the sampled value $\hat{\rho}(1)$ is close. All other autocorrelations are not statistically significant.

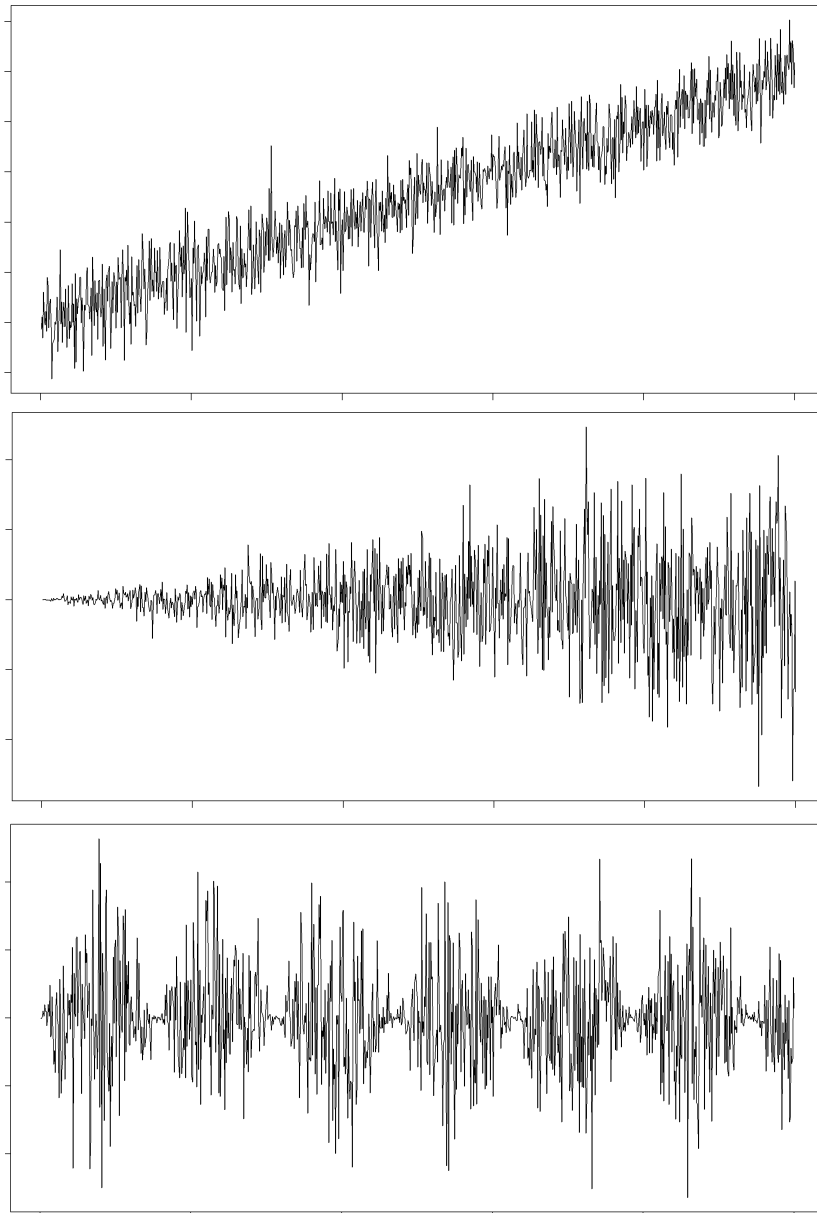
When we see a time series with a similar autocorrelation function we will consider modelling it as

an $MA(1)$ process.

6.7 Non-stationary processes

Sometimes it helps to understand a notion by considering counterexamples.

The following are charts of non-stationary time series.

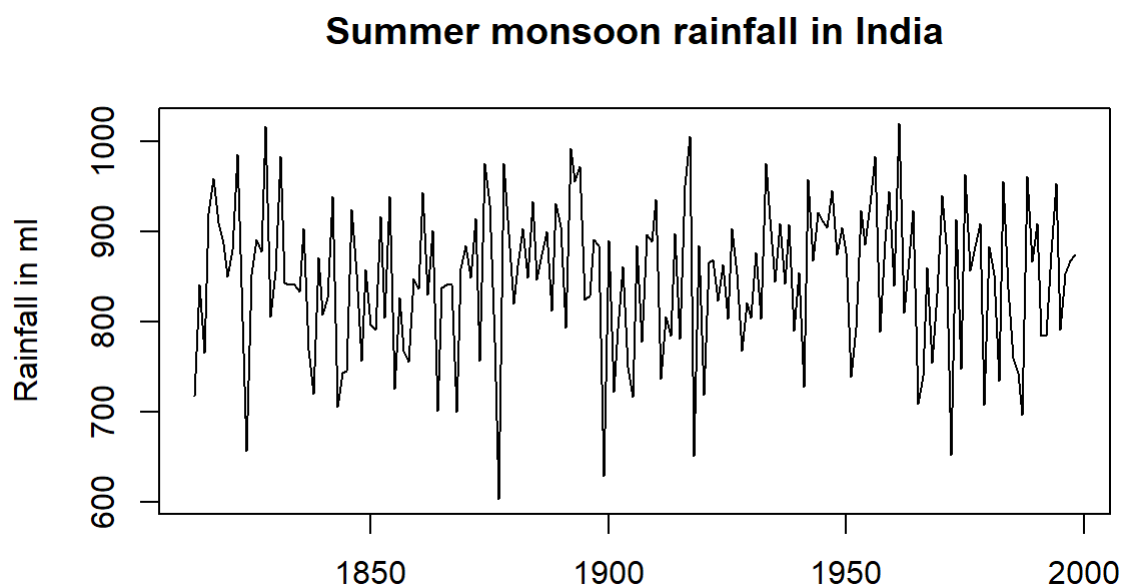


6.8 Statistical tests for stationarity

As well as visual inspection there are statistical hypothesis tests to establish whether a time series is stationary.

For example, the following chart of Monsoon rainfall in India seems stationary as there is no trend and the variance seems to be uniform.

```
monsoon = read.csv("IndianMonsoon.csv")
# convert into a ts object
monsoon = ts(monsoon$RainFall, frequency=1, start=1813)
plot(monsoon, main="Summer monsoon rainfall in India", ylab="Rainfall in ml")
```



The following methods allow us to assess this quantitatively.

6.8.1 The Augmented Dickey-Fuller test

The [Augmented Dickey-Fuller](#) hypothesis test, or ADF, considers the following null and alternative hypothesis:

$$\begin{aligned} H_0 : & \quad \text{the time series is not stationary} \\ H_1 : & \quad \text{the time series is stationary} \end{aligned}$$

We will not get into the details; Dickey-Fuller construct a certain statistic out of the time series such that if this statistic is large H_0 is unlikely. The size of the statistic is, as usual, measured by a p -value. If the p -value is smaller than 5% we reject the null hypothesis.

In R this is computed as follows:

```

> tseries::adf.test(monsoon) # we reject H0

Augmented Dickey-Fuller Test

data: monsoon
Dickey-Fuller = -5.0142, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary

Warning message:
In tseries::adf.test(monsoon) : p-value smaller than printed p-value

```

So in the this case we reject H_0 which supports our intuition that the time series is stationary.

6.8.2 Kwiatkowski-Phillips-Schmidt-Shin test

The [Kwiatkowski-Phillips-Schmidt-Shin](#) test, or KPSS test, is similar to the ADF test but in this case the hypothesis are the other way around:

$$\begin{aligned}
 H_0 : & \quad \text{the time series is stationary} \\
 H_1 : & \quad \text{the time series is not stationary}
 \end{aligned}$$

In R this is computed as follows:

```

> feasts::unitroot_kpss(monsoon)
      kpss_stat kpss_pvalue
0.09777156    0.10000000

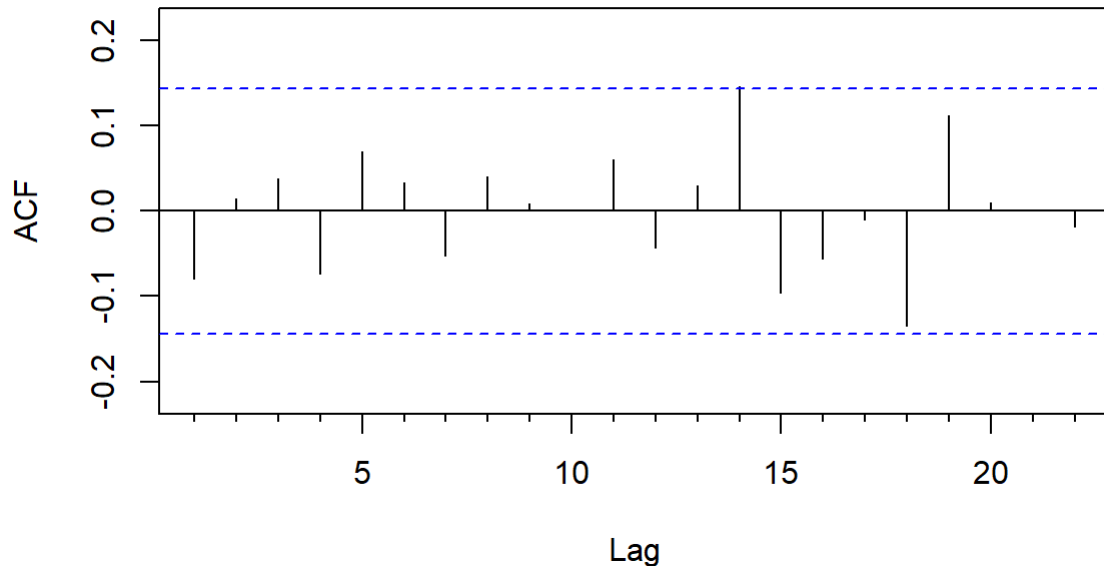
```

Since the p -value is not smaller than 5% we fail to reject H_0 which again supports the intuition that the time series is stationary.

6.8.3 Ljung-Box test for white noise

In the case of the monsoon data above, a chart of the ACF suggests that the different time points are independent (as the autocorrelation is not significantly different from zero).

Series monsoon



This chart was created with the R function `forecast::Acf()` which has the slight advantage over the vanilla `acf()` of hiding the unit bar for lag 0, $\rho(0) = 1$ which zooms into the autocorrelations for lags greater than zero.

To confirm whether a time series is white noise we use the [Ljung-Box](#) test (based on the earlier Box-Pierce test) which considers the following hypothesis

H_0 : white noise, no correlations

H_1 : some non-trivial correlation, not white noise

As usual, we compute a statistic and derive a p -value. If the p -value is below 5% we reject H_0 .

In R this is computed as follows:

```
> Box.test(monsoon, type="Ljung-Box")

Box-Ljung test

data:  monsoon
X-squared = 1.1876, df = 1, p-value = 0.2758
```

So we fail to reject the hypothesis of white noise.

Chapter 7

Moving Average and Autoregressive models

In this chapter we will be dealing with probability. We will describe some models (random processes) such as *moving average* or *autoregressive*, and find their autocovariance functions. This is equivalent to your first steps in probability in Year 1, where you learn about certain named random variables (e.g. exponential, binomial, ...) and find their expectation and variance.

When we later deal with real life time series, after extracting the trend and seasonal components, we will look at the sample autocovariance¹ function and see if it is similar to that of one of our models. This will allow us to model the time series and forecast values. This is similar to Year 1 statistics, where you take data such as arrival times of tubes, and then choose a distribution and parameters that best describe the data.

7.1 Moving average models

Recall that we defined MA(1) models as the following combination of past values of white noise Z_t :

$$X_t = Z_t + \theta Z_{t-1}$$

One can do the same with more past values of white noise as follows.

Definition 7.1. A *moving average* process of order q , MA(q), is defined as:

$$X_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \cdots + \theta_q Z_{t-q}. \quad (7.1)$$

¹You might see the expression *theoretical autocovariance* to refer to the autocovariance of the random process, this is to emphasize the distinction with *sample autocovariance* which will depend of our sampled data and is technically a random value. The same applies to mean and variance, the expressions *theoretical* mean and variance (also called *population* mean and variance) refer to those of a probability distribution as opposed to the sample mean and variance which come from a sample of the data.

where Z_t is white noise with variance σ^2 and $\theta_1, \dots, \theta_q$ are real numbers.

For example, an MA(2) process is defined as

$$X_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}.$$

for values θ_1 and θ_2 .

Equation (7.1) can be written more compactly using the backshift operator, B :

$$\begin{aligned} X_t &= Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \dots + \theta_q Z_{t-q} \\ &= Z_t + \theta_1 B Z_t + \theta_2 B^2 Z_t + \dots + \theta_q B^q Z_t \\ &= (1 + \theta_1 B + \theta_2 B^2 + \dots + \theta_q B^q) Z_t \end{aligned}$$

If we define the polynomial $\theta(z) = 1 + \theta_1 z + \theta_2 z^2 + \dots + \theta_q z^q$, then the MA(q) process can be written as

$$X_t = \theta(B) Z_t$$

7.2 Simulating MA(q) models in R

An important part of understanding maths is being able to look at examples. We can do this in R.

Recall in the case of $q = 1$ covered in the previous chapter, we did this as follows.

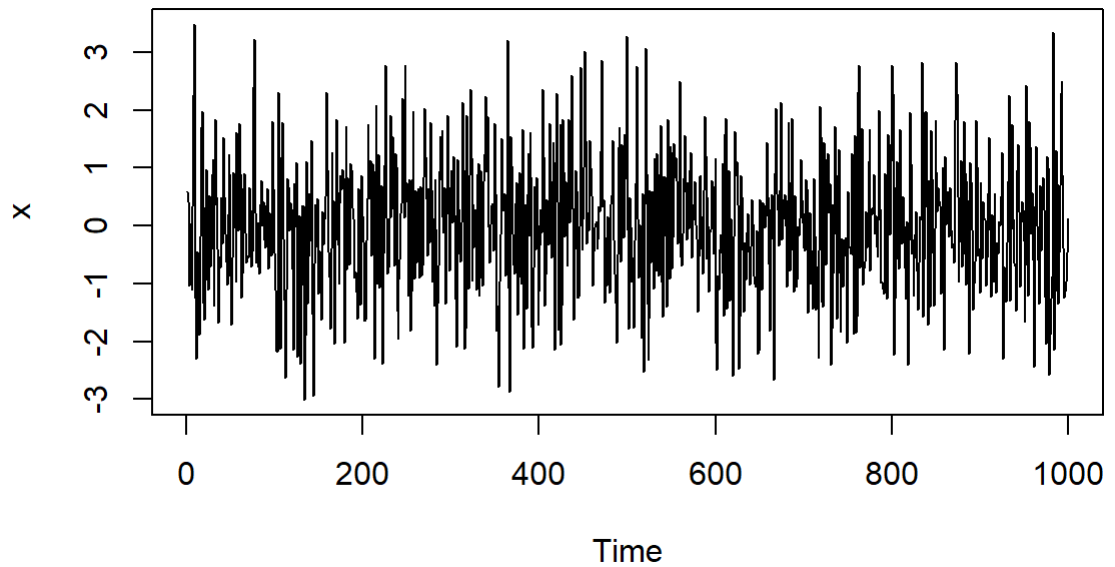
```
sigmaSquared = 4
theta = 0.5
z = rnorm(1000, 0, sigmaSquared)
head(z[-1]) # this is the vector with the first element removed
tail(z[-1]) # this is the vector with the first element removed
head(z[-length(z)]) # this is the vector with the last element removed
tail(z[-length(z)]) # this is the vector with the last element removed
x = z[-length(z)] + theta*z[-1]
plot(x, type="l")
```

It is valuable to understand what is going on in the lines above so make sure you play with it. For example change `n` to 20 and display the shortened vectors `z[-1]` and `z[-length(z)]` in full) until it is very clear that it is implementing $X_t = Z_t + \theta Z_{t-1}$.

However for higher q the code above will become a too complicated. Instead we can use the `arima.sim` function² to simulate the MA(2) process $X_t = Z_t + 0.5Z_{t-1} - 0.3Z_{t-2}$ as follows

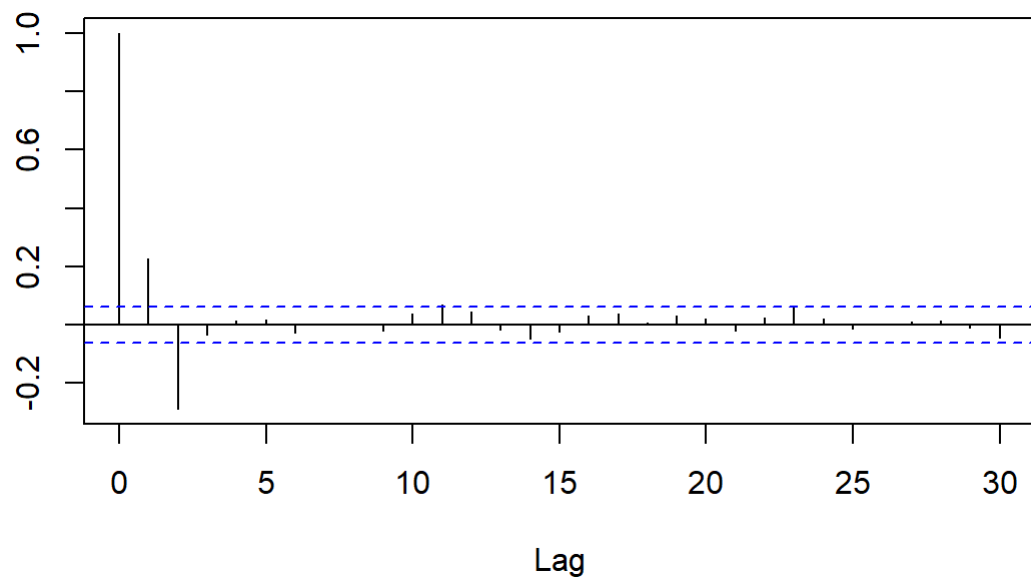
```
# MA(2) model with coefficients theta1=0.5, theta2=-0.3
x = arima.sim(n = 1000, model=list(ma = c(0.5, -0.3)))
plot(x)
```

²The `arima.sim` function resides in the library `stats` which is part of *base R* and loaded by default, so you do not need to load it. As we have seen in the tutorials, this is useful knowledge in the case another library overrides `arima.sim`, in this case you can use `stats::arima.sim`.



As you can see from the chart, it is difficult to visually distinguish any specific features. For this we can use the autocorrelation function as follows

```
acf(x)
```



What might these values of the autocorrelation function be?

With MA(1) the autocorrelation function was $\rho(0) = 1$, $\rho(1) = \theta/(1 + \theta^2)$, and zero for lags greater than 1. For the MA(2) ACF displayed above we see nonzero values for $\hat{\rho}(0) = 1$, $\hat{\rho}(1)$ and $\hat{\rho}(2)$.

How can we calculate the ACF and ACVF of a MA(q) process? Is it the case that they are both zero for lags greater than q as suggested in the chart above?

From the formulæ for MA(1) we expect the ACF and ACVF to be functions involving quadratic polynomials of θ_1 and θ_2 .

7.3 ACF of MA models

Proposition 7.1. The MA(q) process defined by $X_t = \theta(B)Z_t$, with $\theta(z) = 1 + \theta_1 z + \theta_2 z^2 + \dots + \theta_q z^q$ is weakly stationary and its autocovariance function is

$$\gamma(h) = \sigma^2 \left(\sum_{i=0}^{q-|h|} \theta_i \theta_{i+h} \right)$$

for $|h| \in \{0, 1, \dots, q\}$ and zero otherwise. σ^2 is the variance of the white noise, Z_t , and for ease of notation we define $\theta_0 = 1$.

Remarks 7.1. 1. The process above has zero expectation:

$$\mathbb{E}(X_t) = \mathbb{E}(Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q}) = \mathbb{E}(Z_t) + \theta_1 \mathbb{E}(Z_{t-1}) + \dots + \theta_q \mathbb{E}(Z_{t-q}) = 0$$

As $Z_t \sim N(0, \sigma^2)$.

2. Recall that for a stationary process

$$\gamma(h) = \text{Cov}(X_t, X_{t+h}) = \text{Cov}(X_{t-h}, X_t) = \text{Cov}(X_t, X_{t-h}) = \gamma(-h)$$

We therefore normally only focus and display the ACF and ACVF for positive lags, h . Also when we perform calculations we might switch between the two equivalent expressions $\text{Cov}(X_t, X_{t-h})$ and $\text{Cov}(X_t, X_{t+h})$.

3. Note that the ACF and ACVF of a MA(q) process are zero for lags h greater than q . This can be useful to help identify real world time series that might be modelled as a MA(q) process.
4. A process that has zero ACVF for lags greater than q can be proved to be a MA(q) process.
5. It is easy to see that an MA(q) process is also strictly stationary though we do not prove this here.

Corollary 7.1. The autocorrelation function of the MA(q) process defined by $\theta(z)$ is

$$\rho(h) = \frac{\sum_{i=0}^{q-|h|} \theta_i \theta_{i+|h|}}{1 + \theta_1^2 + \theta_2^2 + \dots + \theta_q^2}$$

for $|h| \in \{0, 1, \dots, q\}$ and zero otherwise.

Proof of Corollary. Just divide the expression for $\gamma(h)$ in the proposition by $\gamma(0)$. □

Partial proof of Proposition. $\gamma(h)$ is the covariance between

$$X_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \cdots + \theta_q Z_{t-q}$$

and

$$X_{t-h} = Z_{t-h} + \theta_1 Z_{t-1-h} + \theta_2 Z_{t-2-h} + \cdots + \theta_q Z_{t-q-h}$$

Let us do this in a couple of examples to illustrate the general behaviour. The basic ingredients are:

- Cov is bilinear, so we can write $\text{Cov}(X_t, X_{t-h})$ as a sum of all the constituent summands $\text{Cov}(Z_i, Z_j)$ with coefficients given by the θ s.
- As Z_t are all independent (this is the definition of white noise) $\text{Cov}(Z_i, Z_j)$ is zero if $i \neq j$ and is equal to σ^2 (the variance of the white noise) if $i = j$.

With this, let us compute $\gamma(h)$ for some values:

1. $h = 0$. We line up vertically the Z_t s that have non-zero covariance:

$$\begin{aligned} \gamma(0) = \text{Cov}(X_t, X_t) &= \text{Cov}(Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \cdots + \theta_q Z_{t-q}, \\ &\quad Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \cdots + \theta_q Z_{t-q}) \end{aligned}$$

Therefore

$$\gamma(0) = \sigma^2 + \sigma^2 \theta_1^2 + \sigma^2 \theta_2^2 + \cdots + \sigma^2 \theta_q^2 = \sigma^2 (1 + \theta_1^2 + \theta_2^2 + \cdots + \theta_q^2)$$

Note there are $q + 1$ terms in this sum.

2. $h = 1$. Again, let us line up vertically the Z_t s with non-zero covariance

$$\begin{aligned} \gamma(1) = \text{Cov}(X_t, X_{t-1}) &= \text{Cov}(Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \theta_3 Z_{t-3} + \cdots + \theta_q Z_{t-q}, \\ &\quad Z_{t-1} + \theta_1 Z_{t-2} + \theta_2 Z_{t-3} + \cdots + \theta_{q-1} Z_{t-q} + \theta_q Z_{t-q-1}) \end{aligned}$$

Therefore

$$\gamma(1) = \sigma^2 \theta_1 + \sigma^2 \theta_2 \theta_1 + \cdots + \sigma^2 \theta_q \theta_{q-1} = \sigma^2 (\theta_1 + \theta_1 \theta_2 + \theta_2 \theta_3 + \cdots + \theta_q \theta_{q-1}) = \sigma^2 \sum_{i=0}^{q-1} \theta_i \theta_{i+1}$$

Where we use the convention $\theta_0 = 1$. There are q terms in this sum.

3. In general $\gamma(h)$ will be σ^2 times the sum of all θ_i times θ_{i+h} . And this sum will have $q + 1 - h$ terms.

$$\gamma(h) = \sigma^2 \sum_{i=0}^{q-h} \theta_i \theta_{i+h}.$$

The calculations above showed that $\text{Cov}(X_t, X_{t-h})$ is independent of t , this, together with the fact that $\mathbb{E}(X_t) = 0$ imply that X_t is weakly stationary. $\square$

Examples 7.1. 1. For the case $q = 1$ the formulae above yield

$$\gamma(0) = (1 + \theta_1^2)\sigma^2 \quad \text{and} \quad \gamma(1) = \theta_1\sigma^2$$

as was seen in the previous chapter.

2. For a MA(2) process, defined by $\theta(z) = 1 + \theta_1z + \theta_2z^2$ the formulae above yield

$$\gamma(0) = (1 + \theta_1^2 + \theta_2^2)\sigma^2 \quad \gamma(1) = (\theta_1 + \theta_1\theta_2)\sigma^2 \quad \text{and} \quad \gamma(2) = \theta_2\sigma^2$$

The autocorrelation function is

$$\rho(0) = 1 \quad \rho(1) = \frac{\theta_1 + \theta_1\theta_2}{1 + \theta_1^2 + \theta_2^2} \quad \text{and} \quad \rho(2) = \frac{\theta_2}{1 + \theta_1^2 + \theta_2^2}$$

3. In the R example of a MA(2) process in Section 7.2, with $\theta_1 = 0.5$ and $\theta_2 = -0.3$ we get

$$\begin{aligned} \rho(1) &= \frac{\theta_1 + \theta_1\theta_2}{1 + \theta_1^2 + \theta_2^2} = \frac{0.5 + 0.5(-0.3)}{1 + 0.5^2 + (-0.3)^2} = 0.261194 \\ \rho(2) &= \frac{\theta_2}{1 + \theta_1^2 + \theta_2^2} = \frac{-0.3}{1 + 0.5^2 + (-0.3)^2} = -0.2238806 \end{aligned}$$

which are both close to the numbers $\hat{\rho}(1)$, and $\hat{\rho}(2)$ displayed in the ACF chart in Section 7.2.

7.4 The AR(1) model

Another family of models, assumes that X_t linearly depends on X_{t-1} plus some noise term:

$$X_t = \phi X_{t-1} + Z_t.$$

This is called an *autoregressive* model of order 1, or AR(1).

Note that, unlike the definition of the MA(q) process, this is an *equation* (a recurrence equation) that we need to prove has a solution.

To do so we use a relatively sophisticated theorem.

Theorem 7.1. If $Y_1, Y_2, \dots$, is an infinite sequence of normal random variables $Y_i \sim N(\mu_i, \sigma_i^2)$, such that

$$\sum_{i=1}^{\infty} |\mu_i| < \infty \quad \text{and} \quad \sum_{i=1}^{\infty} \sigma_i^2 < \infty$$

Then the random variable

$$Y = \sum_{i=1}^{\infty} Y_i$$

is well defined, and $Y \sim N(\sum_{i=1}^{\infty} \mu_i, \sum_{i=1}^{\infty} \sigma_i^2)$.

Proposition 7.2. If $|\phi| \neq 1$ then there exists a random process, X_t , verifying

$$X_t = \phi X_{t-1} + Z_t \quad (7.2)$$

Proof. We just need to *unroll* the recurrence equation above:

$$\begin{aligned} X_t &= \phi X_{t-1} + Z_t \\ &= \phi(\phi X_{t-2} + Z_{t-1}) + Z_t = \phi^2 X_{t-2} + \phi Z_{t-1} + Z_t \\ &= \phi^2(\phi X_{t-3} + Z_{t-2}) + \phi Z_{t-1} + Z_t = \phi^3 X_{t-3} + \phi^2 Z_{t-2} + \phi Z_{t-1} + Z_t \\ &\vdots \\ &= Z_t + \phi Z_{t-1} + \phi^2 Z_{t-2} + \phi^3 Z_{t-3} + \cdots + \phi^n Z_{t-n} + \phi^{n+1} X_{t-n-1} \end{aligned}$$

This suggests that a solution to the recurrence might be the infinite sum

$$X_t = Z_t + \phi Z_{t-1} + \phi^2 Z_{t-2} + \phi^3 Z_{t-3} + \cdots \quad (7.3)$$

which by the previous theorem converges if the sum of variances, $\sum_{i=0}^{\infty} \phi^{2i}$, converges. This is so if $|\phi| < 1$.

One can verify that, in the case $|\phi| < 1$, equation (7.3) defines a random process that is a solution to the recurrence equation (7.2).

The case $|\phi| > 1$ can be dealt similarly, by writing X_t in terms of future values of Z_t . Equation (7.2) is equivalent to

$$X_{t-1} = \phi^{-1} X_t - \phi^{-1} Z_t$$

Change time to $t' = t - 1$, which we write simply as t and the recurrence above becomes

$$X_t = \phi^{-1} X_{t+1} - \phi^{-1} Z_{t+1}$$

By proceeding as above:

$$\begin{aligned} X_t &= \phi^{-1} X_{t+1} - \phi^{-1} Z_{t+1} \\ &= \phi^{-1}(\phi^{-1} X_{t+2} - \phi^{-1} Z_{t+2}) - \phi^{-1} Z_{t+1} = \phi^{-2} X_{t+2} - \phi^{-2} Z_{t+2} - \phi^{-1} Z_{t+1} \\ &\vdots \\ &= -\phi^{-1} Z_{t+1} - \phi^{-2} Z_{t+2} - \cdots - \phi^{-n} Z_{t+n} - \phi^{-n-1} X_{t+n} \end{aligned}$$

Which suggests defining X_t as follows

$$X_t = -\phi^{-1} Z_{t+1} - \phi^{-2} Z_{t+2} - \phi^{-3} Z_{t+3} - \cdots$$

This series of random variables converges as the sum of variances is $\sum_{i=1}^{\infty} \phi^{-2i}$ and this converges as $|\phi| > 1$. $\square$

Corollary 7.2. If X_t is the the solution to (7.2) (i.e. X_t is an AR(1) process), then:

- If $|\phi| < 1$ then X_t is independent of $Z_{t+1}, Z_{t+2}, \dots$
- For $|\phi| > 1$, X_t is independent of $Z_t, Z_{t-1}, \dots$

Proof. For simplicity we focus on the case $|\phi| < 1$. In this case

$$X_t = Z_t + \phi Z_{t-1} + \phi^2 Z_{t-2} + \dots$$

And this is independent of future values of Z_t because for all $h > 1$, Z_{t+h} is independent of $Z_t, Z_{t-1}, \dots$ (Z_t is white noise so Z_{t_1} and Z_{t_2} are independent for $t_1 \neq t_2$). $\square$

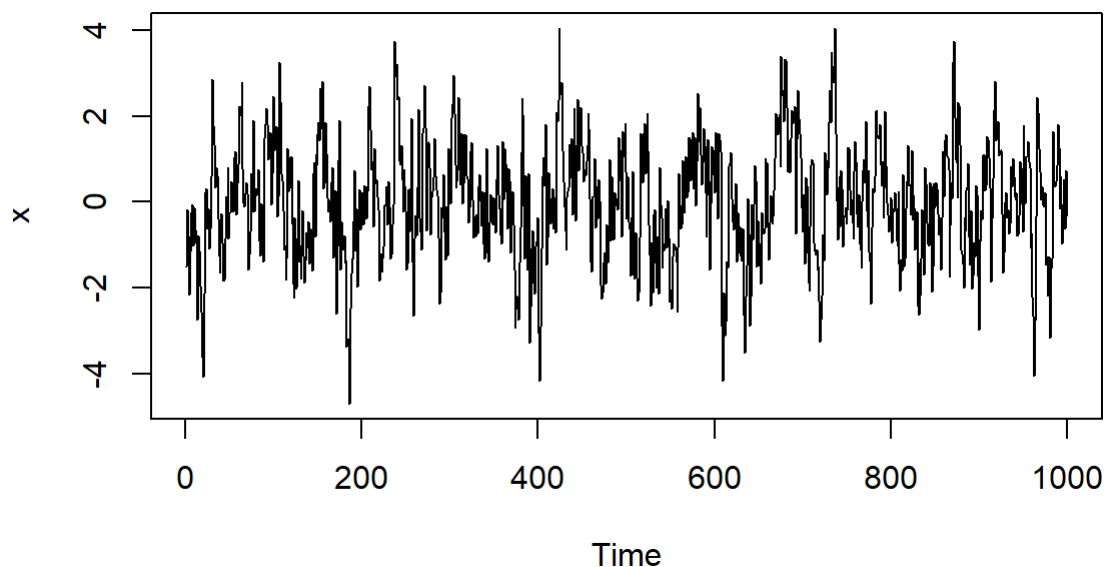
Remarks 7.2. 1. In the case when $|\phi| = 1$ it can be proved that there exists no stationary or weakly stationary random process verifying the recurrence (7.2).

2. In the proof of existence of the AR(1) random process for $|\phi| > 1$ we wrote X_t in terms of future values of the white noise, $Z_{t+1}, Z_{t+2}, \dots$. This is mathematically correct but does not correspond with the intuition that the present should be influenced by the past and not the future. We will later define what a *casual* process is and will see that for $|\phi| > 1$ the AR(1) process is not causal. We will normally restrict our attention to causal processes.

7.5 Simulating AR(1) models in R

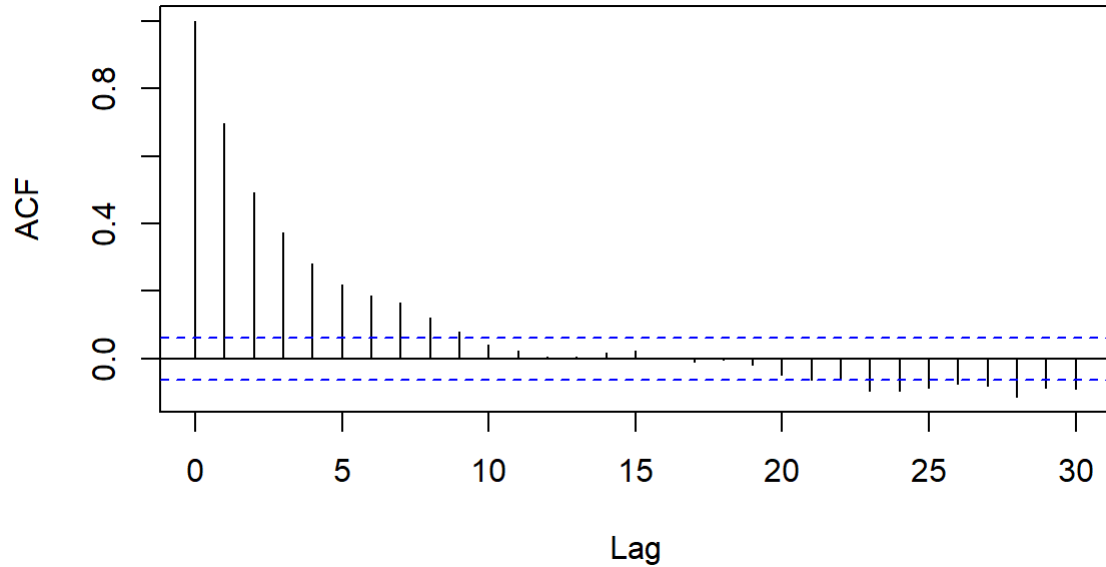
In R we can use the function `arima.sim` to simulate an AR(1) model as follows

```
# AR(1) model with coefficients phi=0.7
phi=0.7
x = arima.sim(n = 1000, model=list(ar = c(phi)))
plot(x)
```



As usual it is not easy to visually identify features of this process. A correlogram helps understand the process.

`acf(x)`



An interesting feature here is that the autocorrelation function of this autoregressive model does not become zero after lag 1 but seems to decay instead.

What are the ACF and ACVF of an autoregressive model?

7.6 The ACF and ACVF of a AR(1) model

Let us calculate the ACF and ACVF of an autoregressive model of order 1.

Proposition 7.3. Let X_t be defined by the AR(1) equation $X_t = \phi X_{t-1} + Z_t$. Then X_t is weakly stationary and

- If $|\phi| < 1$ then $\gamma(h) = \phi^{|h|}\sigma^2/(1 - \phi^2)$ and $\rho(h) = \phi^h$.
- If $|\phi| > 1$ then $\gamma(h) = \phi^{-|h|}\sigma^2/(\phi^2 - 1)$ and $\rho(h) = \phi^{-h}$.

Proof. Let us start with the case $|\phi| < 1$.

We will use the expression found in the proof of the previous proposition

$$X_t = Z_t + \phi Z_{t-1} + \phi^2 Z_{t-2} + \dots$$

The variance of this, by the theorem above, is

$$\text{Var}(X_t) = \sigma^2 + \phi^2\sigma^2 + \phi^4\sigma^2 + \dots = \frac{\sigma^2}{1 - \phi^2}$$

This is independent of t .

The covariance of X_t and X_{t-h} can be calculated in a similar fashion. If we vertically align the terms with non-zero covariance:

$$\begin{aligned} \text{Cov}(X_t, X_{t-h}) = \text{Cov}(Z_t + \phi Z_{t-1} + \phi^2 Z_{t-2} + \dots + \phi^h Z_{t-h} + \phi^{h+1} Z_{t-h-1} + \phi^{h+2} Z_{t-h-2} + \dots \\ Z_{t-h} + \phi \quad Z_{t-h-1} + \phi^2 \quad Z_{t-h-2} + \dots) \end{aligned}$$

we see that

$$\text{Cov}(X_t, X_{t-h}) = \phi^h\sigma^2 + \phi^{h+2}\sigma^2 + \phi^{h+4}\sigma^2 + \dots = \frac{\phi^h\sigma^2}{1 - \phi^2}$$

As these covariances only depend on the lag h , the process is weakly stationary with covariance function as above: $\gamma(h) = \phi^h\sigma^2/(1 - \phi^2)$. The autocorrelation function is $\rho(h) = \gamma(h)/\gamma(0) = \phi^h$.

The case $|\phi| > 1$ is similar. □

Remarks 7.3. 1. As we saw in the correlogram displayed in section, the autocorrelation function, $\rho(h) = \phi^h$, decays to zero. This is different from the $\text{MA}(q)$ processes where all autocorrelations are zero for lags greater than q .

2. It is the case that X_t is also strictly stationary. The proof is not difficult but we will omit it.
3. If we already knew that X_t is weakly stationary, then the calculations above are much simpler.

- In the case $|\phi| < 1$ we also know that X_t is independent of Z_{t+1} and so taking variances on both sides of

$$X_t = \phi X_{t-1} + Z_t$$

we obtain

$$\gamma(0) = \phi^2\gamma(0) + \sigma^2$$

I.e. $\gamma(0) = \sigma^2/(1 - \phi^2)$.

Multiplying (7.2) by X_{t-h}

$$X_t X_{t-h} = \phi X_{t-1} X_{t-h} + Z_t X_{t-h}$$

and taking expectations, we get

$$\gamma(h) = \phi\gamma(h-1)$$

Which together with the starting value $\gamma(0) = \sigma^2/(1 - \phi^2)$ yields

$$\gamma(h) = \frac{\phi^h\sigma^2}{1 - \phi^2}$$

- In the case $|\phi| > 1$ we know that X_t is independent of $Z_t, Z_{t-1}, \dots$ and so taking variances on both sides of

$$X_t - Z_t = \phi X_{t-1}$$

we obtain

$$\gamma(0) + \sigma^2 = \phi^2 \gamma(0)$$

I.e. $\gamma(0) = \sigma^2/(\phi^2 - 1)$.

Multiplying the equation $X_{t-h} - Z_{t-h} = \phi X_{t-h-1}$ by X_t

$$X_t X_{t-h} - X_t Z_{t-h-1} = \phi X_t X_{t-h-1}$$

and taking expectations, we get

$$\gamma(h) = \phi \gamma(h+1)$$

Which together with the starting value $\gamma(0) = \sigma^2/(\phi^2 - 1)$ yields

$$\gamma(h) = \frac{\phi^{-h} \sigma^2}{\phi^2 - 1}$$

Chapter 8

Autoregressive Processes

In this chapter we will extend AR(1) to AR(q) processes.

We will also discuss two important attributes of processes: causality and invertibility processes.

8.1 Causality

Definition 8.1. A *causal process* is defined to be a process X_t that is independent from future values of Z_t .

Remarks 8.1. 1. A MA(q) process is always causal as it is defined by

$$X_t = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}$$

and as all different values of Z_t are independent, so in particular $Z_t, Z_{t-1}, \dots, Z_{t-q}$ are independent of all future values $Z_{t+1}, Z_{t+2}, \dots$

2. Recall the notation introduced in the previous chapter. We define the polynomial $\theta(z) = 1 + \theta_1 z + \theta_2 z^2 + \cdots + \theta_q z^q$, then the MA(q) defining equation can be written as

$$X_t = \theta(B)Z_t$$

Where B is the backshift operator. To see this note that

$$\theta(B)Z_t = (1 + \theta_1 B + \theta_2 B^2 + \cdots + \theta_q B^q)Z_t = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}.$$

3. In the previous chapter we saw that an AR(1) process

$$X_t = \phi X_{t-1} + Z_t$$

with $|\phi| < 1$ can be written as

$$X_t = Z_t + \phi Z_{t-1} + \phi^2 Z_{t-2} + \phi^3 Z_{t-3} + \dots$$

As $Z_t, Z_{t-1}, Z_{t-2}, Z_{t-3}, \dots$ are independent of future values of Z_t we see that such a AR(1) is causal.

4. An AR(1) process

$$X_t = \phi X_{t-1} + Z_t$$

with $|\phi| > 1$ was formally seen to be an expression dependent on future values of Z_t

$$X_t = -\phi^{-1} Z_{t+1} - \phi^{-2} Z_{t+2} - \phi^{-3} Z_{t+3} - \dots$$

Therefore an AR(1) process with $|\phi| > 1$ is not casual.

5. We can write the AR(1) process in terms of the backshift operator as follows:

$$\phi(B)X_t = Z_t$$

where $\phi(z)$ is the polynomial $\phi(z) = 1 - \theta z$ and so the notation above stands for

$$\phi(B)X_t = (1 - \phi B)X_t = X_t - \phi X_{t-1}.$$

In this notation, the condition for causality can be rephrased as: the root of the polynomial $\phi(z)$ (which is $1/\phi$) should have absolute value greater than 1.

Let us summarize these last two examples as a proposition:

Proposition 8.1. A MA(q) process

$$X_t = \theta(B)Z_t$$

is causal. An AR(1) process

$$\phi(B)X_t = Z_t$$

is causal if and only if the zero of $\phi(z)$ is greater than 1 in absolute value.

Proof. See previous two remarks. □

Remarks 8.2. 1. We will mostly focus on causal processes as it is counterintuitive to model a time series on future values.

2. Let us see that there are exactly two AR(1) processes with identical ACVF and ACF, one causal and the other non causal. If by looking at the sample ACF of a time series we identify the features of a AR(1) process, then we will choose the casual process that produces the observed ACF.

Recall that the ACVF and ACF of a AR(1) causal process with $|\phi| < 1$ is

$$\gamma_X(h) = \phi^{|h|} \frac{\sigma^2}{1 - \phi^2} \quad \text{and} \quad \rho_X(h) = \phi^{|h|}$$

We can find a non-causal AR(1) process with the same ACVF and ACF. Namely, define $\phi' = 1/\phi$ and consider the AR(1) process

$$X_t = \phi' X_{t-1} + Z'_t$$

With Z'_t white noise with variance σ'^2 to be defined shortly, then its ACVF and ACF are

$$\gamma_{X'}(h) = \phi'^{-|h|} \frac{\sigma'^2}{\phi'^2 - 1} \quad \text{and} \quad \rho_{X'}(h) = \phi'^{-|h|}$$

These numbers coincide with the ones above if we define $\sigma' = \sigma/\phi$:

$$\gamma_{X'}(h) = \phi'^{-|h|} \frac{\sigma'^2}{\phi'^2 - 1} = \phi^{|h|} \frac{\sigma'^2}{\phi^{-2} - 1} = \phi^{|h|} \frac{\sigma^2}{1 - \phi^2} = \gamma_X(h)$$

The equality $\rho_X(h) = \rho_{X'}(h)$ is easy to derive from this.

8.2 Invertibility

Definition 8.2. An *invertible process*, X_t , is defined to be a process such that we can find the white noise, Z_t , (sometimes called, errors or perturbations in the time series) from present and past values of the time series values X_t .

Remarks 8.3. 1. For example, with an AR(1) process

$$X_t = \phi X_{t-1} + Z_t$$

this is trivial to do: $Z_t = X_t - \phi X_{t-1}$. Therefore an AR(1) process is always invertible.

Recall that we write an AR(1) process more compactly as

$$\boldsymbol{\phi}(B)X_t = Z_t$$

with $\boldsymbol{\phi}(z)$ defined as the polynomial $1 - \phi z$ (note that we use the boldface $\boldsymbol{\phi}$ for the polynomial $\boldsymbol{\phi}(z)$, whereas non-boldface symbol ϕ is for the numerical coefficient).

2. For a MA(1) process

$$X_t = Z_t + \theta Z_{t-1}$$

Invertibility is not obvious, as if we write

$$Z_t = X_t - \theta Z_{t-1}$$

then this is a recurrence equation for Z_t (now we assume X_t is known, and Z_t is unknown). If we unroll this recurrence:

$$\begin{aligned}
Z_t &= X_t - \theta Z_{t-1} \\
&= X_t - \theta(X_{t-1} - \theta Z_{t-2}) = X_t - \theta X_{t-1} - \theta Z_{t-2} \\
&\vdots \\
&= X_t - \theta X_{t-1} - \theta^2 X_{t-2} - \cdots - \theta^n X_{t-n} - \theta^{n-1} Z_{t-n}
\end{aligned}$$

We see that this converges, and describes Z_t in terms of X_t only if $|\theta| < 1$.

In terms of the polynomial $\boldsymbol{\theta}(z) = 1 + \theta z$ defining the MA(1) process we see that the process is invertible if and only if the root of $\boldsymbol{\theta}(z)$ (which is simply $-1/\theta$) is greater than 1 in absolute value.

3. Given an MA(1) process

$$X_t = Z_t + \theta Z_{t-1}$$

recall that its ACVF and ACF are

$$\gamma_X(h) = \begin{cases} (1 + \theta^2)\sigma^2 & \text{if } h = 0 \\ \theta\sigma^2 & \text{if } |h| = 1 \\ 0 & \text{otherwise} \end{cases} \quad \rho_X(h) = \begin{cases} 1 & \text{if } h = 0 \\ \frac{\theta}{1+\theta^2} & \text{if } |h| = 1 \\ 0 & \text{otherwise} \end{cases}$$

And that this process is invertible iff $|\theta| < 1$.

It is possible to find a non-invertible MA(1) process with the same ACVF and ACF. Just take $\theta' = 1/\theta$ and define the process

$$X'_t = Z'_t + \theta' Z'_{t-1}$$

where Z'_t is white noise with variance $\sigma'^2 = \sigma^2\theta^2$. Then the ACVF of X'_t is

$$\gamma_{X'}(h) = \begin{cases} (1 + \theta'^2)\sigma'^2 = (1 + \frac{1}{\theta^2})\sigma^2\theta^2 = (1 + \theta^2)\sigma^2 & \text{if } h = 0 \\ \theta'\sigma'^2 = \frac{1}{\theta}\sigma^2\theta^2 = \theta\sigma^2 & \text{if } |h| = 1 \\ 0 & \text{otherwise} \end{cases}$$

which is the same as the ACVF of X_t .

When we encounter a real world time series with an autocovariance or autocorrelation like the one of a MA(1) process, we could choose to model it with either of the two processes with the same ACF. We will choose the invertible one.

4. It can be seen in general that a MA(q) process

$$X_t = \boldsymbol{\theta}(B)Z_t$$

defined by the polynomial $\boldsymbol{\theta}(z) = 1 + \theta_1 z + \theta_2 z^2 + \cdots + \theta_q z^q$, is invertible if and only if all the roots of $\boldsymbol{\theta}(z)$ (which could be real or imaginary) have absolute value greater than 1.

Proposition 8.2. An AR(1) process

$$\phi(B)X_t = Z_t$$

is always invertible. A MA(q) process

$$X_t = \theta(B)Z_t$$

is invertible if and only if all the roots of the polynomial $\theta(z)$ are greater than 1 in absolute value.

Partial proof. See previous remarks. □

8.3 Autoregressive models

Recall that an AR(1) process is defined as the unique solution to the following recurrence equation

$$X_t = \phi X_{t-1} + Z_t$$

where Z_t is white noise with variance σ^2 .

This can be extended to more terms.

Definition 8.3. A process, X_t , satisfying the recurrence equation

$$\phi(X_t) = Z_t$$

with $\phi(z) = 1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_p z^p$ a polynomial of degree p and Z_t white noise, is called an autoregressive process of order p , AR(p).

Proposition 8.3. A stationary autoregressive process as above exists if and only if all the roots of $\phi(z)$ have absolute value different from 1.

Remarks 8.4. 1. Without reference to the backshift operator, the equation in the definition can be written as

$$X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2} - \cdots - \phi_p X_{t-p} = Z_t$$

or equivalently

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + Z_t.$$

It is called autoregressive because this equation describes X_t as a linear regression of $X_{t-1}, \dots, X_{t-p}$.

2. The proposition generalises what we saw for the case AR(1). Here there was a single coefficient, ϕ , and the condition was that $|\phi| \neq 1$.

Proposition 8.4. A stationary autoregressive process

$$\phi(B)X_t = Z_t$$

is always invertible.

It is causal if and only if all the roots of $\phi(z)$ are greater than one in absolute value.

Proof. Invertibility is trivial as by definition of $\text{AR}(p)$, Z_t can be expressed in terms of past values of X_t :

$$Z_t = \phi(B)X_t = X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2} - \cdots - \phi_p X_{t-p}.$$

We have already seen that in the case $p = 1$, X_t is casual if and only if $|\phi| < 1$.

For $p > 1$ the proof is essentially the same but more fiddly. It requires complex analysis to guarantee that $1/\phi(z)$ can be expanded in a power series with square summable coefficients. We will not cover this in the course. $\square$

Remark 8.1. When an $\text{AR}(p)$ process is causal, we often rewrite

$$\phi(B)X_t = Z_t$$

as

$$X_t = \frac{1}{\phi(B)}Z_t$$

We will not go into great details here. Basically, if $\phi(z)$ is a polynomial, we can expand the function $1/\phi(z)$ as a power series around the origin

$$\frac{1}{\phi(z)} = a_0 + a_1 z + a_2 z^2 + \cdots$$

If $\phi(z)$ does not have zeros of absolute value lower than 1, then the coefficients of this expansion decay fast and we can then define $\frac{1}{\phi(B)}Z_t$ to be

$$a_0 Z_t + a_1 Z_{t-1} + a_2 Z_{t-2} + \cdots$$

In the case $p = 1$, $\phi(z) = 1 - \phi z$, this is precisely what we did to prove the existence of X_t , in the case $|\phi| < 1$:

$$\begin{aligned} \frac{1}{1 - \phi B} Z_t &= (1 + \phi B + \phi^2 B^2 + \phi^3 B^3 + \cdots) Z_t \\ &= Z_t + \phi Z_{t-1} + \phi^2 Z_{t-2} + \phi^3 Z_{t-3} + \cdots \end{aligned}$$

8.4 ACVF and ACF of a causal $\text{AR}(p)$ process

A way of identifying the best model for our data is to compare its *sample autocorrelation* function with the autocorrelation function of our models. Therefore it would be useful to know what these functions are for a $\text{AR}(p)$ model.

For simplicity we will restrict our attention to causal $\text{AR}(p)$ processes.

Examples 8.1. 1. In the case $p = 1$ we have

$$X_t = \phi X_{t-1} + Z_t$$

and we saw then that

$$\gamma(h) = \phi^{|h|} \frac{\sigma^2}{1 - \phi^2} \quad \text{and} \quad \rho(h) = \phi^{|h|}$$

We just write these here as the formulæ for higher p should be similar.

2. Let us take the case $p = 2$ then

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t \tag{8.1}$$

As we assume we are in the causal case, Z_t is independent of X_{t-1} and X_{t-2} .

Let us calculate $\gamma(h)$

- The case $h = 0$ is just the variance of X_t . As the expectations are all zero, we can multiply (8.1) by X_t and take expectations

$$\mathbb{E}(X_t X_t) = \phi_1 \mathbb{E}(X_t X_{t-1}) + \phi_2 \mathbb{E}(X_t X_{t-2}) + \mathbb{E}(X_t Z_t)$$

The last term can be computed by multiplying equation (8.1) by Z_t and taking expectations: $\mathbb{E}(X_t Z_t) = \phi_1 \mathbb{E}(X_{t-1} Z_t) + \phi_2 \mathbb{E}(X_{t-2} Z_t) + \mathbb{E}(Z_t^2) = \sigma^2$. Therefore:

$$\gamma(0) = \phi_1 \gamma(-1) + \phi_2 \gamma(-2) + \sigma^2$$

which looks like a recurrence equation.

- The case $h > 0$ can be computed similarly. Multiply (8.1) on both sides by X_{t-h} and then take expectations:

$$\mathbb{E}(X_{t-h} X_t) = \phi_1 \mathbb{E}(X_{t-h} X_{t-1}) + \phi_2 \mathbb{E}(X_{t-h} X_{t-2}) + \mathbb{E}(X_{t-h} Z_t)$$

This results in:

$$\gamma(h) = \phi_1 \gamma(h-1) + \phi_2 \gamma(h-2) \tag{8.2}$$

Equation (8.2) is a 2-step linear recurrence. To solve it we need to know $\gamma(0)$ and $\gamma(1)$ and then recursively apply (8.2). How can we get $\gamma(0)$ and $\gamma(1)$? Using the fact that γ is even ($\gamma(-h) = \gamma(h)$) the first 3 recurrence equations are:

$$\begin{aligned} \gamma(0) &= \phi_1 \gamma(1) + \phi_2 \gamma(2) + \sigma^2 \\ \gamma(1) &= \phi_1 \gamma(0) + \phi_2 \gamma(1) \\ \gamma(2) &= \phi_1 \gamma(1) + \phi_2 \gamma(0) \end{aligned}$$

But this is just a set of 3 linear equations on the 3 unknowns $\gamma(0)$, $\gamma(1)$, and $\gamma(2)$ (the other variables, ϕ_1 , ϕ_2 and σ are known).

So to summarize: we find $\gamma(0)$, $\gamma(1)$ and $\gamma(2)$ by solving the linear equations above and then use the recurrence to find all other $\gamma(h)$.

3. If we divide the recurrence equations above by $\gamma(0)$ we obtain an equivalent recurrence for the autocorrelation functions

$$\rho(h) = \phi_1\rho(h-1) + \phi_2\rho(h-2)$$

Now the first starting value is $\rho(0) = 1$. The next two can be found by dividing the linear system by $\gamma(0)$:

$$\begin{aligned}\rho(1) &= \phi_1 + \phi_2\rho(1) \\ \rho(2) &= \phi_1\rho(1) + \phi_2\end{aligned}$$

From which we can easily derive $\rho(1) = \phi_1/(1-\phi_2)$. The other values result from the recurrence above.

4. Recurrences like this are not entirely trivial to solve. We will see how this is done later. The solution is that if r_1 and r_2 are the roots of the polynomial $\phi(z)$ then both $\gamma(h)$ and $\rho(h)$ have the form $\alpha r_1^h + \beta r_2^h$ for constants α and β that depend on the initial values.

The derivation in the previous example can be used to prove the following.

Proposition 8.5. If X_t is the solution to a causal $AR(p)$ process equation

$$\phi(B)X_t = Z_t$$

with $\phi(z) = 1 - \phi_1z - \dots - \phi_pz^p$, then the autocovariance function satisfies the recurrence

$$\begin{aligned}\gamma(h) &= \phi_1\gamma(h-1) + \dots + \phi_p\gamma(h-p) & \text{if } h > 0 \\ \gamma(0) &= \phi_1\gamma(-1) + \dots + \phi_p\gamma(-p) + \sigma^2\end{aligned}$$

The recurrence requires p initial values. These can be found by writing the recurrence for $h = 0, \dots, p$, using that γ is even, and solving the resulting linear system:

$$\begin{aligned}\gamma(0) &= \phi_1\gamma(1) + \phi_2\gamma(2) + \dots + \phi_p\gamma(p) + \sigma^2 \\ \gamma(1) &= \phi_1\gamma(0) + \phi_2\gamma(1) + \dots + \phi_p\gamma(p-1) \\ &\vdots \\ \gamma(p) &= \phi_1\gamma(p-1) + \phi_2\gamma(p-2) + \dots + \phi_p\gamma(0)\end{aligned}$$

(there are $p+1$ linear equations in the $p+1$ unknowns $\gamma(0), \dots, \gamma(p)$)

Proof. The proof is essentially the same as the case $p = 2$ covered in the example above. □

Corollary 8.1. The autocorrelation function of a $AR(p)$ causal process satisfies the recurrence

$$\rho(h) = \phi_1\rho(h-1) + \dots + \phi_p\rho(h-p) \quad \text{if } h > 0$$

The initial p values for the recurrence are $\rho(0) = 1$ and the solutions to the $p-1$ linear equation

$$\begin{aligned}\rho(1) &= \phi_1 + \phi_2\rho(1) + \dots + \phi_p\rho(p-1) \\ &\vdots \\ \rho(p-1) &= \phi_1\rho(p-2) + \phi_2\rho(p-3) + \dots + \phi_p\rho(1)\end{aligned}$$

8.5 Linear recurrence equations (optional)

This section is flagged as optional as we do not intend to grade you on the fine details of the derivations below. But this is an important and interesting subject and will be used below, so we suggest you do invest some time trying to understand the contents.

8.5.1 Baby case: order 1 linear recurrences

This case is only of interest as it tells us what to expect when we move to higher order.

An order 1 recurrence equation is an equation of the form

$$\begin{aligned}a_0 &= \alpha \\ a_n &= \lambda a_{n-1}\end{aligned}$$

where λ and α are two given numbers. This recurrence equation is trivial to solve as

$$a_n = \lambda a_{n-1} = \lambda^2 a_{n-2} = \cdots = \lambda^n a_0 = \alpha \lambda^n$$

So the solution is a multiple of the power function λ^n . The coefficient α is given by the initial value a_0 and the number λ is the zero of the *characteristic* polynomial $p(z) = z - \lambda$.

This is what will happen for higher orders, but there will be more λ s and more α s. The λ s will be the zeroes of a polynomial $p(z)$ constructed from the coefficients of the recurrence, and the α s will be found from the initial solutions.

8.5.2 Motivation: the Fibonacci sequence

Recall the Fibonacci sequence, defined by

$$\begin{aligned}a_0 &= 1 \\ a_1 &= 1 \\ a_n &= a_{n-1} + a_{n-2}\end{aligned}$$

Can we find a closed form for a_n ?

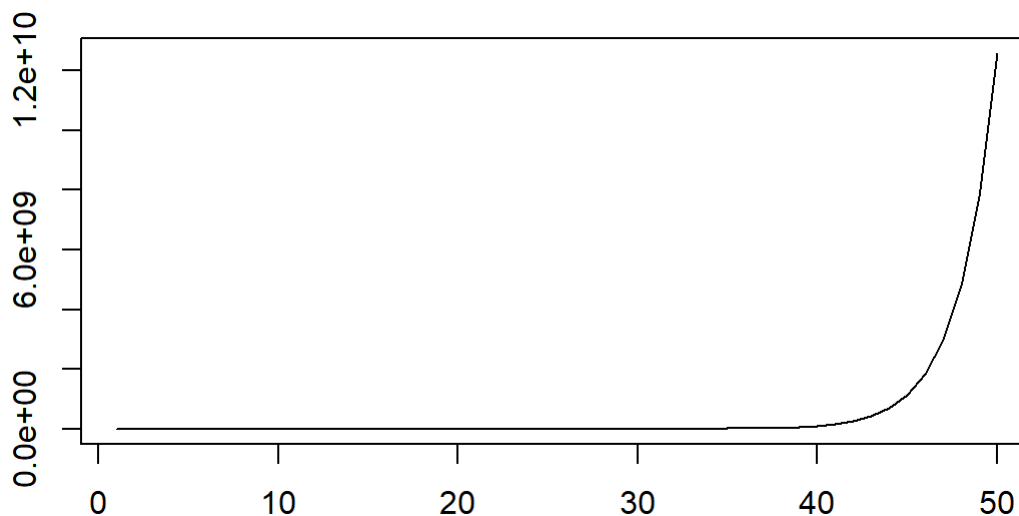
As usual, it is useful to get an idea of what numbers we are dealing with. We can plot a_n using R as follows

```
# You can change the 50 below to other numbers
# Try it!
n = 50
f = c(1,1)
```

```

for(i in 2:n){
  f = c(f, f[i-1]+f[i-2])
}
plot(1:n,f, type="l")

```



The numbers grow exponentially, a_{50} is a 10 digit number. So we expect any closed form for a_n to contain some power.

To find a closed form solution we perform a trick to end up with a single step recurrence. Change the single number a_n to a vector as follows:

$$\begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_n \\ a_{n-1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^2 \begin{pmatrix} a_{n-1} \\ a_{n-2} \end{pmatrix} = \dots = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n \begin{pmatrix} a_1 \\ a_0 \end{pmatrix}$$

So if we can find $F = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n$ we are done. To do this, we just need to diagonalise F . The eigenvalues of this matrix are the zeroes of the characteristic polynomial:

$$p(z) = \begin{vmatrix} 1-z & 1 \\ 1 & -z \end{vmatrix} = z^2 - z - 1$$

which are¹ $\lambda_1 = (1 + \sqrt{5})/2$, and $\lambda_2 = (1 - \sqrt{5})/2$. If A is the matrix with columns the eigenvalues of F , then

$$\begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix} = A \begin{pmatrix} \left(\frac{1+\sqrt{5}}{2}\right)^n & 0 \\ 0 & \left(\frac{1-\sqrt{5}}{2}\right)^n \end{pmatrix} A^{-1} \begin{pmatrix} a_1 \\ a_0 \end{pmatrix}$$

¹These numbers are called *the golden ratio* and according to the classics, define the perfect aesthetic proportions. One day I observed the EURGBP exchange rate was exactly the GBPEUR rate minus one. What were these exchange rates?

You can save yourself finding the eigenvectors and operating the matrices above by realising that it will result in an expression of the form

$$a_n = \alpha \left(\frac{1 + \sqrt{5}}{2} \right)^n + \beta \left(\frac{1 - \sqrt{5}}{2} \right)^n$$

and α and β can be found by imposing the initial values $a_0 = 1$ and $a_1 = 1$.

Homework 8.1. Find α and β by equating the value of the expression above to the known values $a_0 = 1$ and $a_1 = 1$. Then compare the values you obtain to the ones calculate above in R. You can plot them alongside, or calculate and plot the difference. The difference might not be zero because of rounding errors.

8.5.3 Linear order 2 recurrences

A linear recurrence of order 2 is something similar to the Fibonacci sequence:

$$a_n = Aa_{n-1} + Ba_{n-2}$$

where a_0 and a_1 are some known initial values, and A, B are numbers defining the recurrence.

In order to find the solution of this recurrence, following the ideas above, we find the eigenvalues of the matrix

$$\begin{pmatrix} A & B \\ 1 & 0 \end{pmatrix}.$$

These are the zeroes of the characteristic polynomial

$$p(z) = \begin{vmatrix} A - z & B \\ 1 & -z \end{vmatrix} = z^2 - Az - B$$

The zeroes of this polynomial are $\lambda_1 = (A + \sqrt{A^2 + 4B})/2$ and $\lambda_2 = (A - \sqrt{A^2 + 4B})/2$. The solutions of the recurrence are

$$a_n = \alpha \lambda_1^n + \beta \lambda_2^n$$

where α and β are found by imposing the initial values we are given for a_0 and a_1 .

$$\begin{cases} a_0 &= \alpha + \beta \\ a_1 &= \alpha \lambda_1 + \beta \lambda_2 \end{cases}$$

Multiplying the first equation by λ_1 and subtracting from the second

$$a_1 - \lambda_1 a_0 = \beta \lambda_2 - \beta \lambda_1$$

I.e.

$$\begin{aligned} \beta &= \frac{a_1 - \lambda_1 a_0}{\lambda_2 - \lambda_1} \\ \alpha &= a_0 - \beta = a_0 - \frac{a_1 - \lambda_1 a_0}{\lambda_2 - \lambda_1} = \frac{a_0 \lambda_2 - a_1}{\lambda_2 - \lambda_1} \end{aligned}$$

Putting it all together:

$$\begin{aligned}
a_n &= \frac{a_0\lambda_2 - a_1}{\lambda_2 - \lambda_1} \left(\frac{A + \sqrt{A^2 + 4B}}{2} \right)^n + \frac{a_1 - \lambda_1 a_0}{\lambda_2 - \lambda_1} \left(\frac{A - \sqrt{A^2 + 4B}}{2} \right)^n \\
&= \frac{a_1 - a_0 \frac{A - \sqrt{A^2 + 4B}}{2}}{\sqrt{A^2 + 4B}} \left(\frac{A + \sqrt{A^2 + 4B}}{2} \right)^n + \frac{a_0 \frac{A + \sqrt{A^2 + 4B}}{2} - a_1}{\sqrt{A^2 + 4B}} \left(\frac{A - \sqrt{A^2 + 4B}}{2} \right)^n
\end{aligned}$$

We will not require this level of explicit detail but it is a nice exercise in algebraic manipulation to be able to find these expressions. You can check that with $n = 0$ and $n = 1$ the formula above gives back a_0 and a_1 .

Homework 8.2. Pick your favourite numbers for A , B , a_0 and a_1 . Emulate the R code above to calculate and plot the values a_n . Then implement the formula we have found for a_n and compare it with these values.

8.5.4 The case of repeated or imaginary roots

Repeated roots

The discussion above is valid only in the case that the roots of the characteristic polynomial are different. Otherwise you cannot diagonalise the matrix. In that case one uses something called the Jordan decomposition which has the form

$$\begin{pmatrix} A & B \\ 1 & 0 \end{pmatrix} = S \begin{pmatrix} \lambda & 0 \\ 1 & \lambda \end{pmatrix} S^{-1}$$

Where S is a matrix with a certain basis (the Jordan basis) as column vectors. Proceeding as above, since we have

$$\begin{aligned}
\begin{pmatrix} \lambda & 0 \\ 1 & \lambda \end{pmatrix}^2 &= \begin{pmatrix} \lambda & 0 \\ 1 & \lambda \end{pmatrix} \begin{pmatrix} \lambda & 0 \\ 1 & \lambda \end{pmatrix} = \begin{pmatrix} \lambda^2 & 0 \\ 2\lambda & \lambda^2 \end{pmatrix} \\
\begin{pmatrix} \lambda & 0 \\ 1 & \lambda \end{pmatrix}^3 &= \begin{pmatrix} \lambda & 0 \\ 1 & \lambda \end{pmatrix} \begin{pmatrix} \lambda^2 & 0 \\ 2\lambda & \lambda^2 \end{pmatrix} = \begin{pmatrix} \lambda^3 & 0 \\ 3\lambda^2 & \lambda^3 \end{pmatrix} \\
&\vdots \\
\begin{pmatrix} \lambda & 0 \\ 1 & \lambda \end{pmatrix}^n &= \begin{pmatrix} \lambda & 0 \\ 1 & \lambda \end{pmatrix} \begin{pmatrix} \lambda^{n-1} & 0 \\ (n-1)\lambda^{n-2} & \lambda^{n-1} \end{pmatrix} = \begin{pmatrix} \lambda^n & 0 \\ n\lambda^{n-1} & \lambda^n \end{pmatrix}
\end{aligned}$$

the solution of the recurrence will have the form

$$a_n = \alpha \lambda^n + \beta n \lambda^{n-1}$$

where α and β will be found by imposing the initial values a_0 , a_1 .

Complex roots

In the case s_1 and s_2 are complex, the formula above is still valid, but as a_n is a real number, it is preferable to have a formula only involving real numbers. This will also manifest certain oscillatory features.

In the expression

$$a_n = \alpha\lambda_1^n + \beta\lambda_2^n$$

λ_1 , λ_2 , α and β are now complex numbers. But since λ_1 and λ_2 are roots of a real quadratic equation $\lambda_2 = \overline{\lambda_1}$. And since a_n is real, we must have $\beta = \overline{\alpha}$.

$$a_n = \alpha\lambda_1^n + \overline{\alpha}\overline{\lambda_1}^n$$

If we write λ_1 in polar coordinates, $\lambda_1 = re^{i\theta}$, $\overline{\lambda_1} = re^{-i\theta}$ and $\alpha = c + di$ then:

$$\begin{aligned} a_n &= r^n ((c + di)e^{in\theta} + (c - di)e^{-in\theta}) \\ &= r^n ((c + di)(\cos(n\theta) + i\sin(n\theta)) + (c - di)(\cos(n\theta) - i\sin(n\theta))) \\ &= r^n (2c\cos(n\theta) - 2d\sin(n\theta)) \end{aligned}$$

So a_n has the form

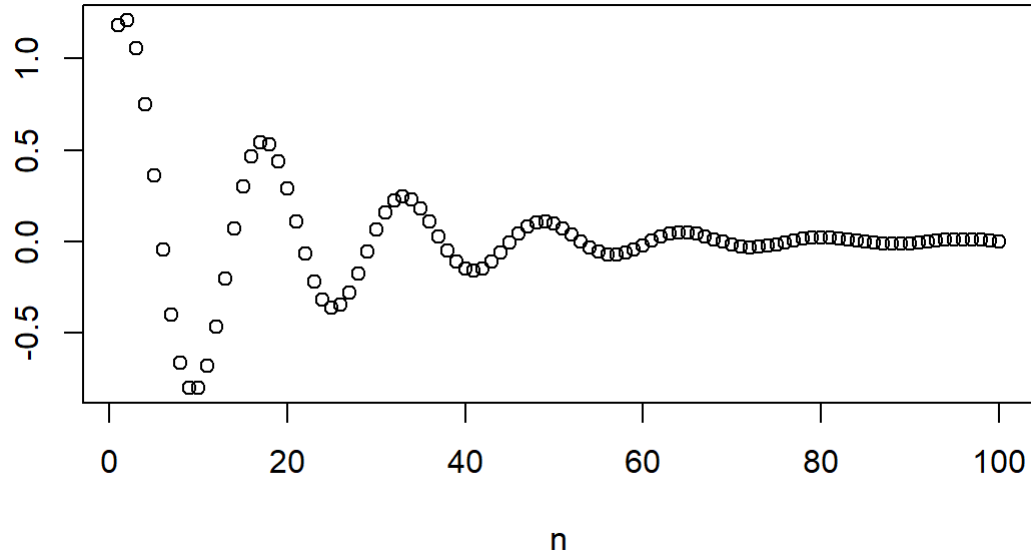
$$a_n = r^n (\delta \cos(n\theta) + \gamma \sin(n\theta))$$

For some coefficients δ and γ that can be found by imposing the equation above for the known values of a_0 and a_1 . Recall that r and θ are the modulus and argument of the complex solutions of $z^2 - Az - B$.

Remark 8.2. An important observation in this case is that the values of a_n will increase or decrease in size depending on whether $r > 1$ or $r < 1$. And that, they will also oscillate because of the periodic sin and cos. Below is an example in R.

```
# You can change the numbers below
gamma = 1
delta = 1
# If r<1 the sequence will decrease
# If r> it will increase in size
r = 0.95
theta = 0.4

n=1:100
an = r^n*(delta*r*cos(n*theta)+gamma*r*sin(n*theta))
plot(n, an)
```



8.6 The ACVF and ACF of a causal $\text{AR}(p)$ process II

8.6.1 Linear recurrence equations of order p

The following summarizes what you need to know about linear recurrence equations.

1. Definition and notation

A linear recurrence equation is an equation of the form

$$a_n = \phi_1 a_{n-1} + \phi_2 a_{n-2} + \cdots + \phi_p a_{n-p}$$

where the initial values $a_0, \dots, a_{p-1}$ are known, as well as the coefficients $\phi_1, \dots, \phi_p$.

2. Objective of analysis below

The aim is to come up with an explicit formula for a_n that does not involve a recurrence.

3. The characteristic polynomial

The coefficients of the recurrence are codified in a polynomial called the *characteristic* polynomial

$$p(z) = z^n - \phi_1 z^{n-1} - \phi_2 z^{n-2} - \cdots - \phi_p$$

4. The characteristic roots

The zeroes of this polynomial are called the characteristic roots and we will note them: $\lambda_1, \dots, \lambda_p$.

5. Solution to the recurrence equation

In the case these roots are all different, the solution to the recurrence is

$$a_n = \alpha_1 \lambda_1^n + \dots + \alpha_p \lambda_p^n$$

where $\alpha_1, \dots, \alpha_p$ are numbers found by imposing that the equation is valid for the known initial values $a_0, \dots, a_{p-1}$.

6. Remarks

From the expression above we can discern the following cases:

- If all the roots are smaller than one in absolute value and real, then a_n will monotonically decay to zero as n grows.
- If all the roots are smaller than one in absolute value and some are complex, then a_n will decay to zero oscillating as n grows.
- If one of the roots is greater than 1 in absolute value then a_n will exponentially grow to infinity.

6. Example

To solve the recurrence

$$a_n = \frac{5}{2}a_{n-1} - a_{n-2}, \quad a_0 = 1, \quad a_1 = -1$$

We write the characteristic polynomial, $p(z) = z^2 - 2.5z + 1$. Using the known solution for quadratic equations, its roots are seen to be $\lambda_1 = 2$ and $\lambda_2 = 1/2$, therefore the solution to the recurrence is

$$a_n = \alpha_1 2^n + \alpha_2 \left(\frac{1}{2}\right)^n$$

To find α_1 and α_2 we impose the solution above with the known values of a_0 and a_1

$$\begin{aligned} a_0 = 1 &= \alpha_1 2^0 + \alpha_2 \left(\frac{1}{2}\right)^0 \\ a_1 = -1 &= \alpha_1 2^1 + \alpha_2 \left(\frac{1}{2}\right)^1 \end{aligned}$$

This results in the linear system

$$\left. \begin{array}{rcl} \alpha_1 & + & \alpha_2 = 1 \\ 2\alpha_1 & + & \frac{1}{2}\alpha_2 = -1 \end{array} \right\}$$

whose solution is $\alpha_1 = -1$ and $\alpha_2 = 2$. Thus the solution of the recurrence is:

$$a_n = -2^n + 2 \left(\frac{1}{2} \right)^n.$$

As a check, let us calculate a_2 and a_3 . From the recurrence:

$$\begin{aligned} a_2 &= \frac{5}{2}a_1 - a_0 = -\frac{5}{2} - 1 = -\frac{7}{2}, \\ a_3 &= \frac{5}{2}a_2 - a_1 = -\frac{5}{2} \cdot \frac{7}{2} + 1 = -\frac{31}{4}. \end{aligned}$$

According to the explicit solution

$$\begin{aligned} a_2 &= -2^2 + 2 \left(\frac{1}{2} \right)^2 = -4 + \frac{1}{2} = -\frac{7}{2}, \\ a_3 &= -2^3 + 2 \left(\frac{1}{2} \right)^3 = -8 + \frac{1}{4} = -\frac{31}{4}. \end{aligned}$$

8.6.2 The case $p = 2$

1. The recurrence equation

The autocorrelation function of the AR(2) process

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

is the solution of the recurrence

$$\rho(h) = \phi_1 \rho(h-1) + \phi_2 \rho(h-2)$$

2. The initial values

As discussed above the initial values are $\rho(0) = 1$ and $\rho(1) = \phi_1/(1 - \phi_2)$. This last value results from writing the recurrence for $h = 1$:

$$\rho(1) = \phi_1 \rho(0) + \phi_2 \rho(-1)$$

Using that $\rho(-1) = \rho(1)$.

3. The characteristic polynomial

The characteristic polynomial is

$$p(z) = z^2 - \phi_1 z - \phi_2$$

which is different from

$$\phi(z) = 1 - \phi_1 z - \phi_2 z^2$$

You can see that they are related by $p(z) = z^2 \phi(1/z)$.

4. The characteristic roots

As $p(z) = z^2 \phi(1/z)$, the roots of p are the inverses of the roots of ϕ . If we note the roots of $p(z)$ by λ_1 and λ_2 and the roots of $\phi(z)$ by r_1 and r_2 , we must have $\lambda_1 = 1/r_1$ and $\lambda_2 = 1/r_2$.

For the purposes of working with AR(2) processes we prefer the polynomial $\phi(z)$ to $p(z)$.

So, we solve the quadratic equation $\phi(z) = 0$ and find its roots r_1 and r_2 . For a causal process these roots are of absolute value greater than one.

Remember: The numbers r_i are the roots of the polynomial $\phi(z)$. They are the inverses of the characteristic roots λ_i that appear in the discussion of linear recurrence equations.

5. The autocorrelation function

The autocorrelation function will have the form

$$\rho(h) = \alpha_1 r_1^{-h} + \alpha_2 r_2^{-h}$$

(the negative sign of the exponents is there because $\lambda_i = r_i^{-1}$)

6. Finding the coefficients

α_1 and α_2 are the solutions of the linear equations

$$\left. \begin{array}{rcl} \alpha_1 & + & \alpha_2 = 1 \\ \alpha_1 r_1 & + & \alpha_2 r_2 = \rho(1) \end{array} \right\}$$

where $\rho(1) = \phi_1/(1 - \phi_2)$.

7. The complex case

In the case in which r_1 and r_2 are complex numbers, the solution a_n will oscillate between positive and negative values and decay to zero exponentially in absolute value.

8. Example

Consider the AR(2) process

$$X_n = \frac{5}{6}X_{n-1} - \frac{1}{6}X_{n-2} + Z_n$$

To find its ACF we write its recurrence equation

$$\rho(h) = \frac{5}{6}\rho(h-1) - \frac{1}{6}\rho(h-2)$$

To solve this recurrence equation we write need to find the roots of the polynomial $\phi(z) = 1 - \frac{5}{6}z + \frac{1}{6}z^2$. Its roots are $r_1 = 2$ and $r_2 = 3$ so the solution to the recurrence will be

$$\rho(h) = \alpha_1 2^{-h} + \alpha_2 3^{-h}.$$

To find α_1 and α_2 we impose the values of $\rho(0) = 1$ and $\rho(1)$. Recall from the previous section that the value for $\rho(1)$ can be derived by writing the first terms of the recurrence and using ρ is an even function of the lag:

$$\begin{aligned}\rho(0) &= \frac{5}{6}\rho(1) - \frac{1}{6}\rho(2) \\ \rho(1) &= \frac{5}{6} - \frac{1}{6}\rho(1) \\ \rho(2) &= \frac{5}{6} - \frac{1}{6}\rho(0)\end{aligned}$$

From the second line we instantly see that $\rho(1) = 5/7$. So we find α_1 and α_2 by solving the linear system:

$$\left. \begin{aligned}\alpha_1 2^0 + \alpha_2 3^0 &= \alpha_1 + \alpha_2 = \rho(0) = 1 \\ \alpha_1 2^{-1} + \alpha_2 3^{-1} &= \frac{1}{2}\alpha_1 + \frac{1}{3}\alpha_2 = \rho(1) = \frac{5}{7}\end{aligned} \right\}$$

This yields $\alpha_1 = 32/5$ and $\alpha_2 = -27/5$ and so the ACF is

$$\rho(h) = \frac{32}{5}2^{-h} - \frac{27}{5}3^{-h}.$$

8.6.3 The ACF function of a AR(p) process

The ideas above extend without material changes.

The recurrence equation

Write down the recurrence equation for the ACF. If the process is defined by

$$X_n = \phi_1 X_{n-1} + \cdots + \phi_p X_{n-p} + Z_n$$

Then the recurrence equation is

$$\rho(h) = \phi_1 \rho(h-1) + \cdots + \phi_p \rho(h-p)$$

The characteristic roots

Find the roots of the polynomial $\phi(z)$ and call them $r_1, \dots, r_p$. These are the inverses of the characteristic roots and will therefore be used exponentiated to negative powers.

The ACF

The ACF will take the form:

$$\rho(h) = \alpha_1 r_1^{-h} + \dots + \alpha_p r_p^{-h}$$

for some unknown coefficients $\alpha_1, \dots, \alpha_p$.

Finding the coefficients

The coefficients $\alpha_1, \dots, \alpha_p$ can be found from the initial p values $\rho(0) = 1, \rho(1), \dots, \rho(p-1)$.

To find these initial values we write the first p terms of the recurrence equation and use the property $\rho(-h) = \rho(h)$. This will result in $p-1$ linear equations on the $p-1$ unknowns $\rho(1), \dots, \rho(p-1)$ (we do not write the equation for $\rho(0)$ as it is not needed):

$$\left. \begin{array}{rcl} \phi_1 \rho(-1) + \dots + \phi_p \rho(1-p) & = & \rho(1) \\ \phi_1 \rho(1) + \dots + \phi_p \rho(2-p) & = & \rho(2) \\ & \vdots & \\ \phi_1 \rho(p-3) + \dots + \phi_p \rho(-2) & = & \rho(p-2) \\ \phi_1 \rho(p-2) + \dots + \phi_p \rho(-1) & = & \rho(p-1) \end{array} \right\} \quad (8.3)$$

Chapter 9

The partial autocorrelation function

9.1 Motivation

In the last chapters we have seen that the ACF of MA(q) and AR(p) processes behaves as follows:

- For an MA(q) process, the ACVF and ACF are zero for lags greater than q .
- For a causal AR(p) process, the ACVF and ACF are given by recurrences and can be written as

$$\gamma(h) = \alpha_1 r_1^{-h} + \cdots + \alpha_p r_p^{-h}$$

where $r_1, \dots, r_p$ are the zeroes of the polynomial $\phi(z)$ and they all have absolute value greater than 1. As a consequence $\gamma(n)$ and $\rho(n)$ both decay exponentially.

This means that if we see a real world time series with sample ACF that cuts off after lag q we can model it with a MA(q) process.

AR(p) processes seem harder to identify because $\gamma(h)$ never drops to zero, so we cannot guess what p might be.

At the root cause of this, is the fact that, even for an order 1 AR(1) process

$$X_t = \phi X_{t-1} + Z_t$$

the correlation between X_t and X_{t-2} is non-zero as

$$X_t = \phi^2 X_{t-2} + \phi Z_t + Z_{t-1}$$

has correlation ϕ^2 with X_{t-2} .

However this correlation is in some sense *mediated by* X_{t-1} and not a *first hand* dependency of X_t on X_{t-2} .

To analyse this we consider partial correlation.

9.2 Definition of partial correlation

Recall from Chapter 3 (see also the tutorials) that the partial correlation of X and Y with respect to Z , is the correlation between $X' = X - \alpha Z$ and $Y' = Y - \beta Z$ where α and β are determined so that $\text{Cov}(X', Z) = 0$ and $\text{Cov}(Y', Z) = 0$. This partial correlation is noted $\rho_{XY \cdot Z}$ and it is understood as some form of correlation *discounting* the joint influence of Z .

Example 9.1. Let W_1 , W_2 and W_3 be independent standard normal random variables. And set

$$\begin{aligned} X &= 2W_1 + W_2 + 3W_3 \\ Y &= W_1 - W_2 - 5W_3 \\ Z &= 3W_3 \end{aligned}$$

Then using bilinearity and the fact W_1 , W_2 and W_3 are independent, the covariance between X and Y is

$$\text{Cov}(X, Y) = 2 - 1 - 15 = -14$$

And given the variances of X and Y are $\text{Var}(X) = 2^2 + 1^2 + 3^2 = 14$ and $\text{Var}(Y) = 1^2 + (-1)^2 + (-5)^2 = 27$, we find that the correlation is

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{-14}{\sqrt{14 \cdot 27}} = -0.7200823$$

This did not use Z at all. For partial correlation between X and Y with respect to Z we need to subtract the effect of this variable. By plain sight, as $Z = 3W_3$, we see what needs to be subtracted to be independent of Z :

$$\begin{aligned} X' &= X - Z = 2W_1 + W_2 + 3W_3 - 3W_3 = 2W_1 + W_2 \\ Y' &= Y + \frac{5}{3}Z = W_1 - W_2 - 5W_3 + 5W_3 = W_1 - W_2 \end{aligned}$$

These random variables have no W_3 and so are independent of $Z = 3W_3$. The partial correlation of X and Y with respect to Z is therefore

$$\rho_{XY \cdot Z} = \rho_{X'Y'} = \frac{\text{Cov}(X', Y')}{\sqrt{\text{Var}(X')\text{Var}(Y')}} = \frac{2 - 1}{\sqrt{(2^2 + 1^2)(1^2 + (-1)^2)}} = 0.3162278$$

The definition can be extended to several Z as follows.

Definition 9.1. The partial correlation and covariance of X and Y with respect to $Z_1, \dots, Z_n$ is defined as the correlation and covariance respectively between

$$X' = X - \alpha_1 Z_1 - \dots - \alpha_n Z_n$$

and

$$Y' = Y - \beta_1 Z_1 - \cdots - \beta_n Z_n$$

where $\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n$ are numbers found by demanding that

$$\text{Cov}(X', Z_1) = 0, \dots, \text{Cov}(X', Z_n) = 0$$

$$\text{Cov}(Y', Z_1) = 0, \dots, \text{Cov}(Y', Z_n) = 0.$$

It is denoted by $\rho_{XY \cdot X_1 \dots X_n}$.

9.3 Partial correlation of a random process

Definition 9.2. The partial partial autocorrelation function of a random process X_t is the partial autocorrelation between X_t and X_{t-h} with respect to the intermediate values $X_{t-1}, \dots, X_{t-h+1}$. It will be noted as ρ_{hh} .

Remark 9.1. For the cases $h = 0$ and $h = 1$ there are no intermediate values to condition on, so we have $\rho_{00} = \rho(0) = 1$ and $\rho_{11} = \rho(1)$.

9.4 Partial correlation of an AR(1) process

For the case of the AR(1) process above, we would expect that the partial correlation between X_t and X_{t-2} with respect to X_{t-1} should be zero as X_t depends on X_{t-2} only through its dependency on X_{t-1} .

Let us see.

To find the partial correlation between

$$X_t = Z_t + \phi Z_{t-1} + \phi^2 Z_{t-2} + \cdots$$

and

$$X_{t-2} = Z_{t-2} + \phi Z_{t-3} + \phi^2 Z_{t-4} + \cdots$$

conditional on

$$X_{t-1} = Z_{t-1} + \phi Z_{t-2} + \phi^2 Z_{t-3} + \cdots$$

We need to subtract a certain amount of X_{t-1} from X_t so that it becomes uncorrelated with X_{t-1} . By visual inspection we see that

$$X'_t = X_t - \phi X_{t-1} = Z_t$$

which is independent of X_{t-1} . We do the same with X_{t-2} , again by visual inspection we see that

$$X'_{t-2} = X_{t-2} - \phi^{-1}X_{t-1} = Z_{t-2}$$

The partial correlation between X_t and X_{t-2} with respect to X_{t-1} is, by definition, the correlation

$$\rho_{22} = \text{Corr}(X'_t, X'_{t-2}) = \text{Corr}(Z_t, Z_{t-2}) = 0$$

This can be easily generalised as follows.

Proposition 9.1. For a causal AR(1) process the partial correlation function is:

$$\rho_{hh} = \begin{cases} 1 & \text{if } h = 0 \\ \rho(1) = \phi & \text{if } h = 1 \\ 0 & \text{if } h > 1 \end{cases}$$

Corollary 9.1. The PACF of an AR(1) process is zero for lags, h , larger than 1.

9.5 Partial correlation of an AR(p) process

The results above can be extended to arbitrary AR(p) processes.

Proposition 9.2. The PACF of a AR(p) process is zero for lags, h , greater than p .

9.6 Summary of ACF and PACF of MA(q) and AR(p) processes

The following table summarizes what the ACF and PACF of the moving average and autoregressive models look like

	ACF	PACF
MA(q)	cuts off at q	decays to zero
AR(p)	decays to zero	cuts off at p

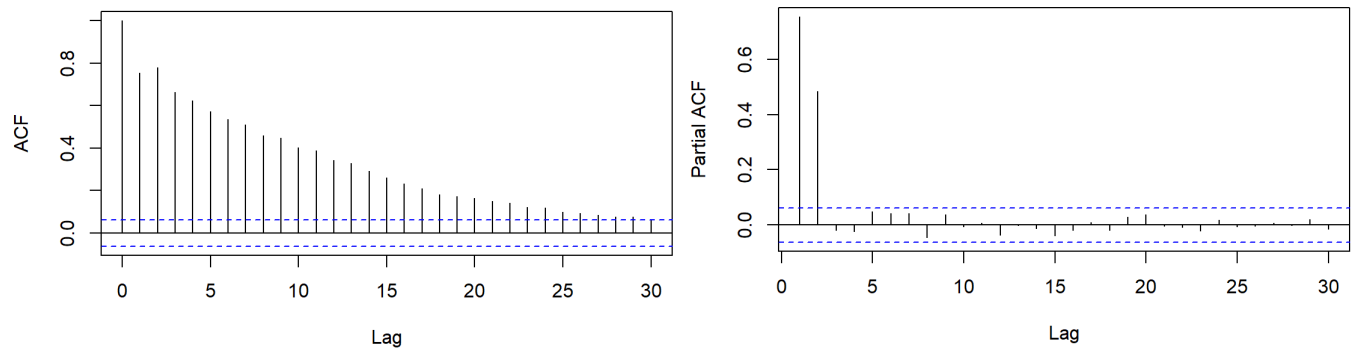
9.7 Sample partial autocorrelation function

In order to identify the best model to fit some real world data we will need to estimate the sample partial correlation of a time series. This uses an algorithm called the Durbin-Lenvinson algorithm.

This algorithm takes in the sample ACF described in Section 6.5 and outputs an estimate for the partial autocorrelation function.

In R this is calculated with the function `pacf`.

```
x = arima.sim(1000, model=list(ar=c(0.4,0.5)))  
acf(x)  
pacf(x)
```



A time series with a PACF as above, zero for lags greater than 2, would be modelled as a $AR(2)$ process.

Chapter 10

Fitting and Forecasting MA(q) and AR(p) models (optional)

10.1 Fitting models

10.1.1 Summary

As explained above once we have extracted the stationary component of our time series¹ we will plot the sample ACF and PACF. Based on these we will choose an appropriate model:

- If the ACF cuts off after lag q then we use a MA(q) model.
- If the PACF cuts off after lag p then we use a AR(p) model.

Once a model has been chosen we need to estimate its parameters ($\omega_1, \dots, \omega_q$ for MA(q) and $\phi_1, \dots, \phi_p$ for AR(p)). This is described as *fitting* the model.

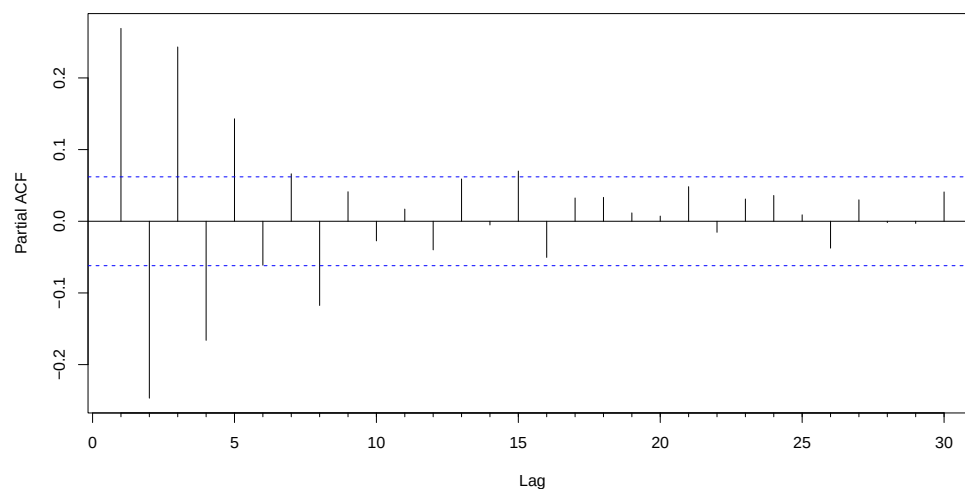
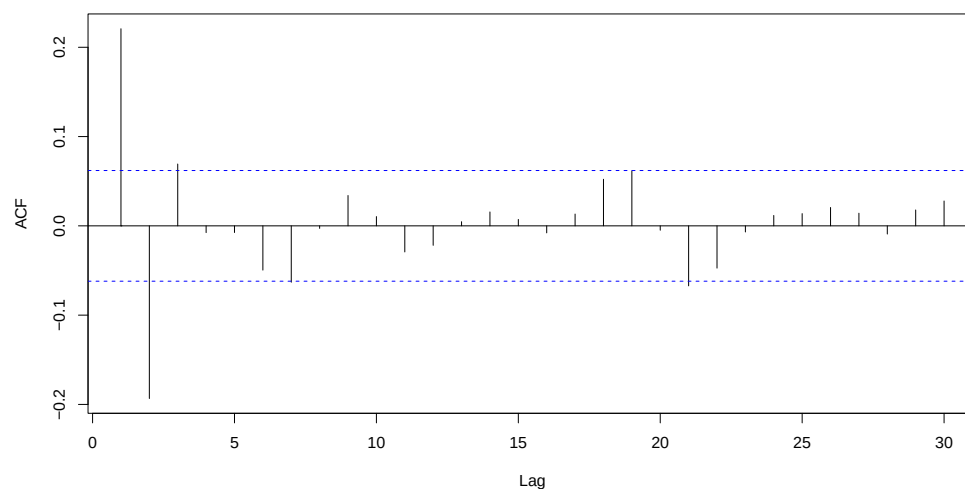
The numerical techniques used in fitting models lie outside of the scope of this course. In R we can use the function `arima` as described in the next sections.

10.1.2 Fitting a MA(q) model

First we identify the most likely model (the instructions `forecast::Acf` and `forecast::Pacf` remove the first correlation number $\rho(0) = 1$ so as to improve readability, the argument `main=""` removes the default title).

¹Recall that this can be accomplished either by subtracting the estimated trend, m_t , and seasonal component s_t , or by differencing the time series so as to remove trend and seasonal effects.

```
forecast::Acf(x, main=" ")
forecast::Pacf(x, main=" ")
```



From this we see that the time series seems to follow a MA(2) process. We fit it as follows. The 3 numbers in the parameter `order=c(0,0,2)` are usually denoted p , d , and q and are employed in models that combine an AR(p), an MA(q) and d -th order differencing (these are called ARMIMA(p, d, q) models). In our case we only have moving average component so $p = 0$ and $d = 0$.

```
> arima(x, order=c(0,0,2))
Call:
arima(x = x, order = c(0, 0, 2))
```

```
Coefficients:
```

```

          ma1      ma2  intercept
0.5188 -0.2710   -0.0129
s.e.  0.0289   0.0286    0.0447

sigma^2 estimated as 1.283:  log likelihood = -1543.98,  aic = 3095.96

```

According to this our times series can be modelled with the random process

$$X_t = Z_t + 0.5188Z_{t-1} - 0.2710Z_{t-2}$$

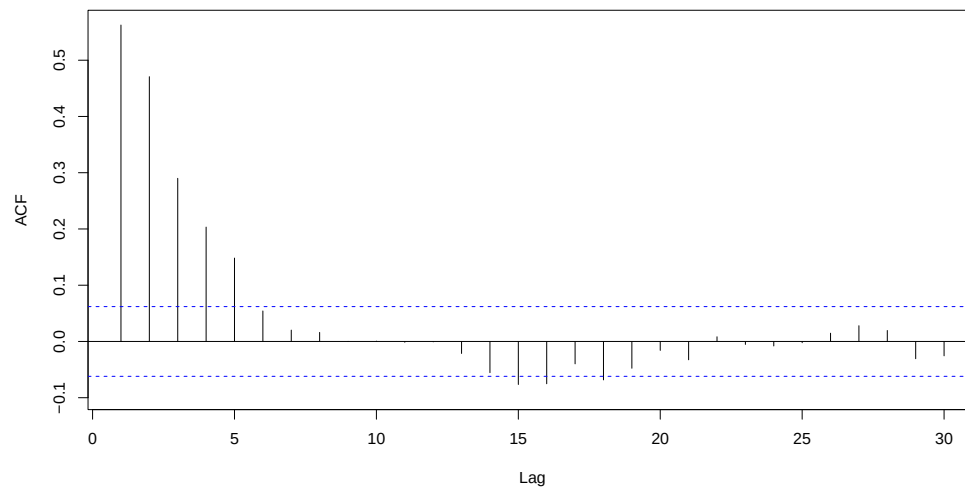
10.1.3 Fitting a $AR(p)$ model

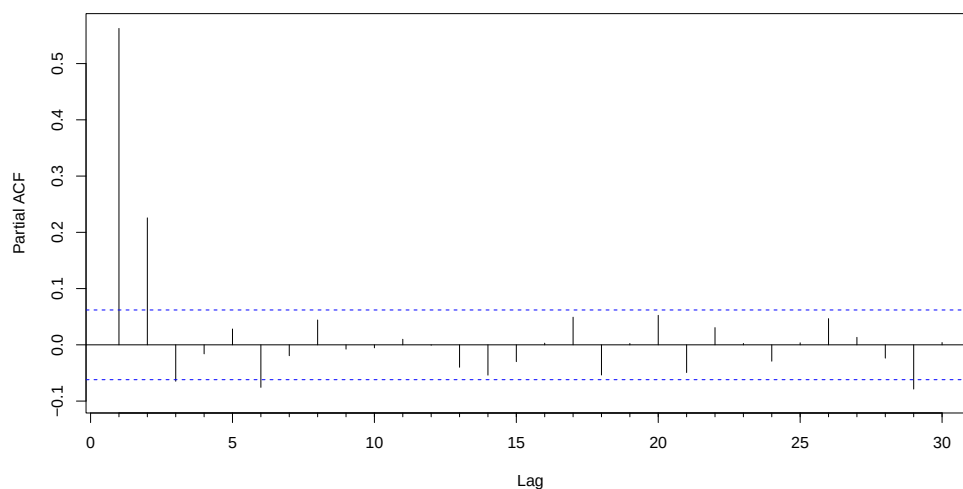
As we did before, given a stationary time series we use `forecast::Acf` and `forecast::Pacf` to identify the suitable model.

```

forecast::Acf(x, main=" ")
forecast::Pacf(x, main=" ")

```





Here an AR(2) model seems appropriate so we fit it using the R `arima` function.

```
> arima(x, order=c(2,0,0))

Call:
arima(x = x, order = c(2, 0, 0))

Coefficients:
          ar1          ar2    intercept
      0.4375    0.2260     -0.0086
s.e.  0.0309    0.0309      0.0914

sigma^2 estimated as 0.951:  log likelihood = -1394.09,  aic = 2796.18
```

According to this our times series can be modelled with the random process

$$X_t = 0.4375X_{t-1} + 0.2260X_{t-2} + Z_t$$

10.2 Forecasting

Once we have chosen a model amongst

- MA(q): $X_t = Z_t + \omega_1 Z_{t-1} + \dots + \omega_p Z_{t-p}$.
- AR(p): $X_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + Z_t$.

and estimated the relevant coefficients, we might wish to use the model in forecasting future values of X_t .

10.3 Forecasting for MA(q) models

10.3.1 One step forecasting

Assume the time series is known for time t and earlier times. Then the value at time $t + 1$ is the random variable

$$X_{t+1} = Z_{t+1} + \omega_1 Z_t + \cdots + \omega_p Z_{t-p+1}$$

Here $Z_t, \dots, Z_{t-p+1}$ are now numbers (not random variables) as we assume we are at time t and they have already been sampled. Only $Z_{t+1} \sim N(0, \sigma^2)$ is random (it is a future unknown value).

The notion of invertibility of the random process is now key, it implies that the $q - 1$ numbers $Z_t, \dots, Z_{t-p+1}$ can be written in terms of $X_t, X_{t-1}, \dots$. We therefore assume the process is invertible.

As $Z_{t+q} \sim N(0, \sigma^2)$ its expected value is zero and so a reasonable estimate for X_{t+1} is

$$\hat{X}_{t+1} = \omega_1 Z_t + \cdots + \omega_p Z_{t-p+1}$$

10.3.2 The general case

In general, the same arguments as above show that for $h \leq p$ we have

$$\hat{X}_{t+h} = \omega_{h-1} Z_t + \cdots + \omega_{h-p} Z_{t-p+1}$$

whereas for $h > p$, $\hat{X}_{t+h} = 0$ (note that for the process X_t expectations are always zero, so that zero here is not an *extreme* value, it is just the point in the middle of the time series).

10.3.3 In R

In R, as usual we can use the function `predict`. For example, to find the next 2 forecast points in the MA(2) process in section 10.1.2 we can run:

```
> model = arima(x, order=c(0,0,2))
> predict(model, 2, se.fit=FALSE)
Time Series:
Start = 1001
End = 1002
Frequency = 1
[1] -0.2005094  0.1665311
```

I.e. $\hat{X}_{t+1} = -0.2005094$ and $\hat{X}_{t+2} = 0.1665311$ where t indicates the time of the last known value of the time series.

10.4 Forecasting for MA(q) models

10.4.1 One step forecasting

Assume the time series is known for time t and earlier times. Then the value at time $t + 1$ is the random variable

$$X_{t+1} = \phi_1 X_t + \cdots + \omega_p X_{t-p+1} + Z_t$$

As $Z_t \sim N(0, \sigma^2)$ its expected value is zero and so a reasonable estimate for X_{t+1}

$$\hat{X}_{t+1} = \phi_1 X_t + \cdots + \omega_p X_{t-p+1}$$

10.4.2 The general case

For lags greater than 1, we can use the same approach as above recursively. So form example

$$\hat{X}_{t+2} = \phi_1 \hat{X}_{t+1} + \phi_2 X_t + \cdots + \omega_p X_{t-p+2}.$$

10.4.3 In R

Again we just use the `predict` function. For example, to find the next 3 forecast points in the AR(2) process in section 10.1.3 we can run:

```
> model = arima(x, order=c(2,0,0))
> predict(model, 2, se.fit=FALSE)
Time Series:
Start = 1001
End = 1002
Frequency = 1
[1] -0.7330219 -0.5698441
```

So that the forecasting yields $\hat{X}_{t+1} = -0.7330219$ and $\hat{X}_{t+2} = -0.5698441$ where t indicates the time of the last known value of the time series.

As the forecast for AR models is so easy we can check these numbers manually. We just implement the formula $\hat{X}_{t+1} = \phi_1 X_t + \phi_2 X_{t-1}$ as follows:

```
> phi1= model$coef[1]
> phi2= model$coef[2]
> phi1*x[length(x)] + phi2*x[length(x)-1]
      ar1
-0.7234847
```

which coincides with the output of the function `predict` above. We have used two items of R syntax:

- The syntax `model$coef` accesses the array of coefficients that are stored inside the model. These coefficients are just $\phi_1 = \text{model\$coef}[1]$ and $\phi_2 = \text{model\$coef}[2]$.
- The instruction `length(x)` returns the length of the time series `x` and so `x[length(x)]` is the last element of our time series and `x[length(x)-1]` is the penultimate, which in the maths above are denoted X_t and X_{t-1} .